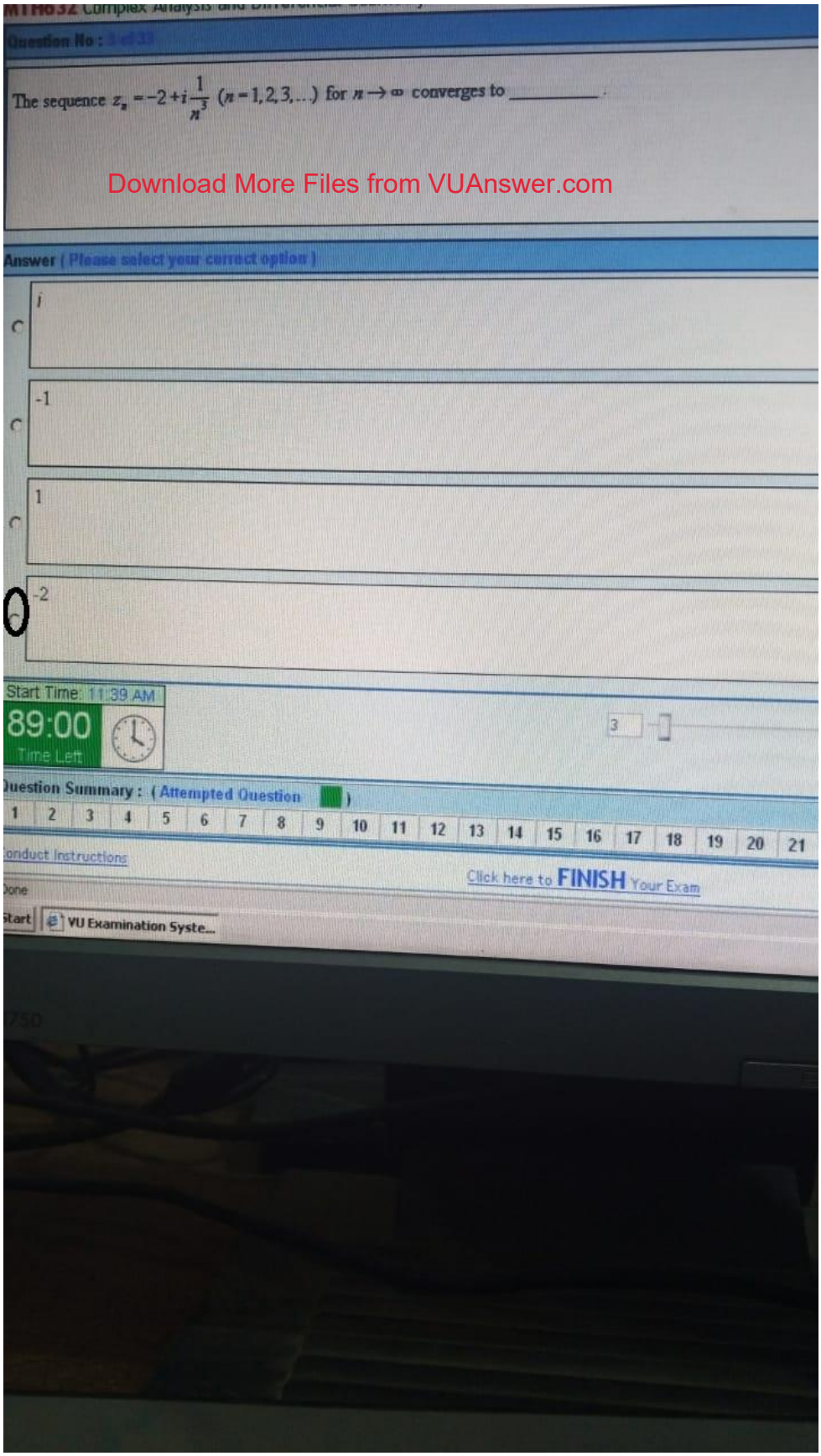
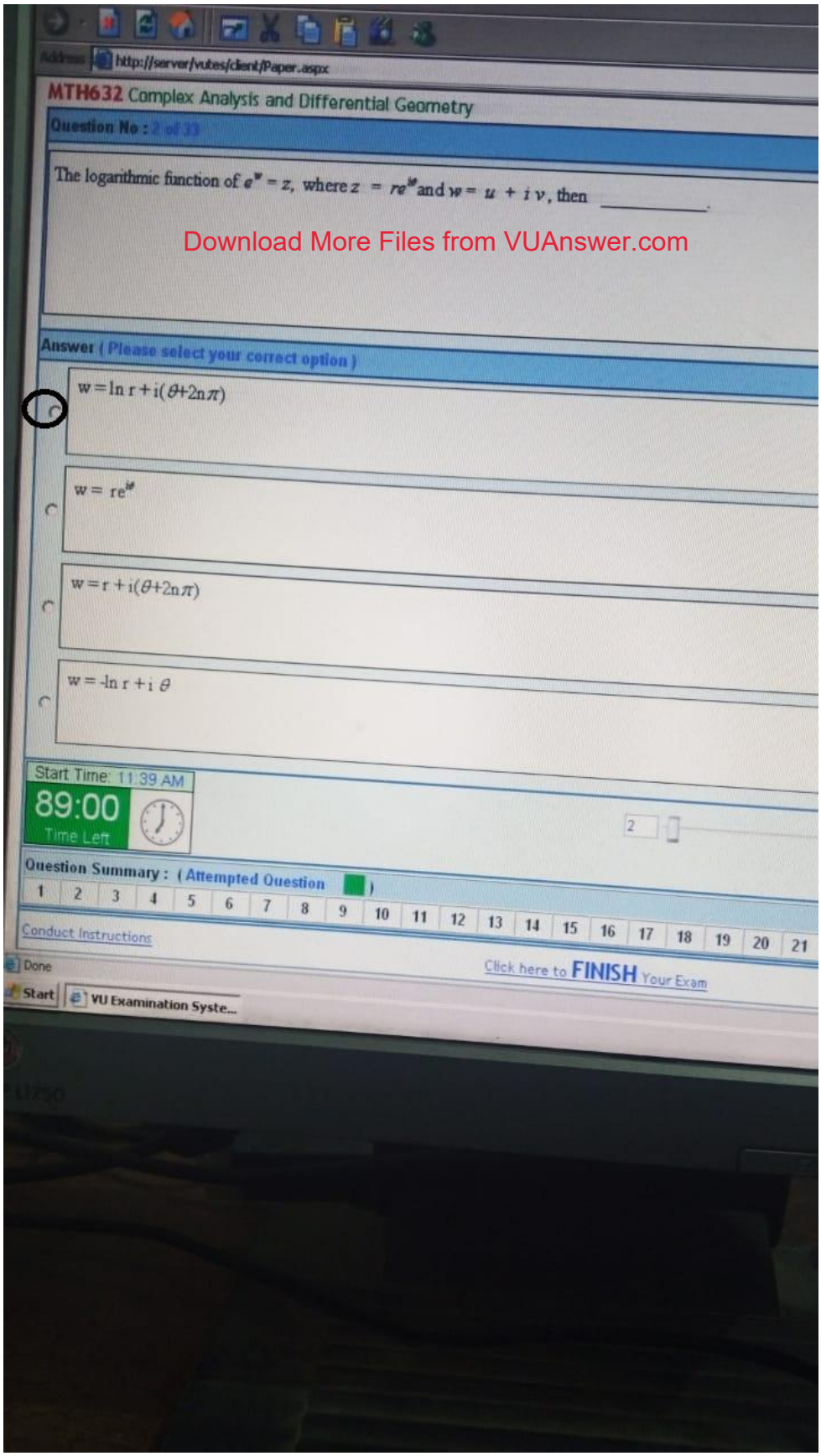
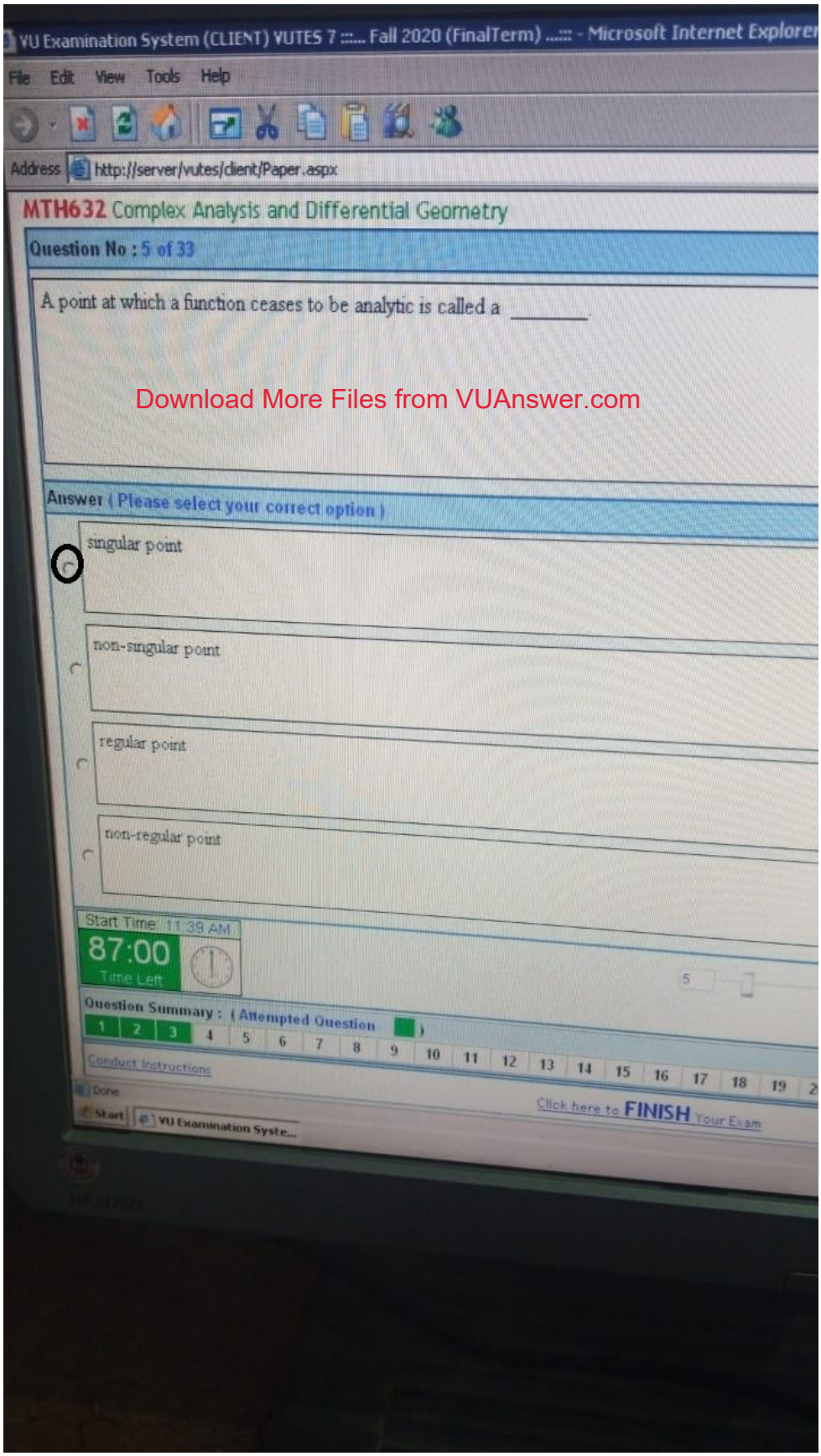
MTH632 FINAL

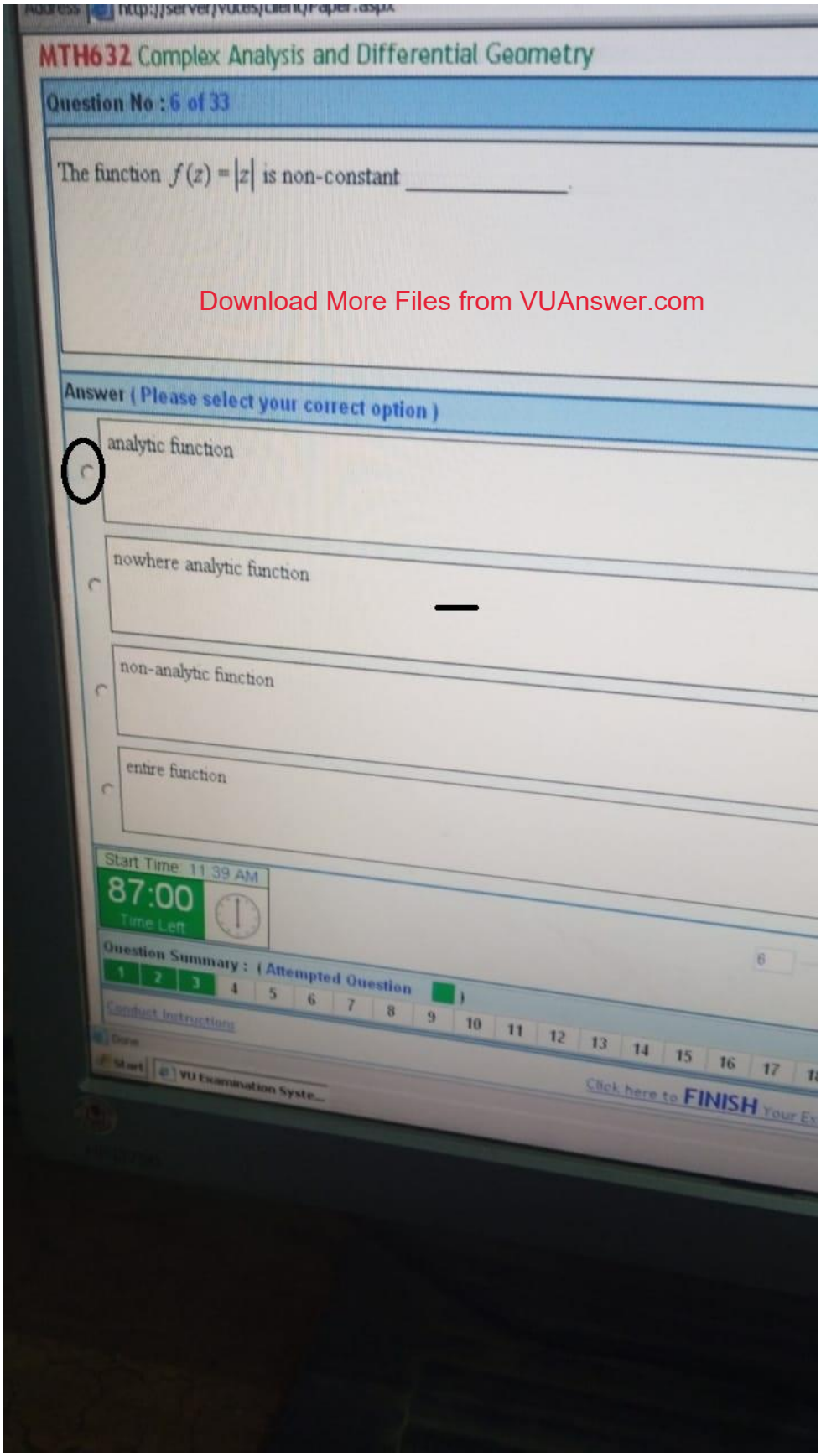
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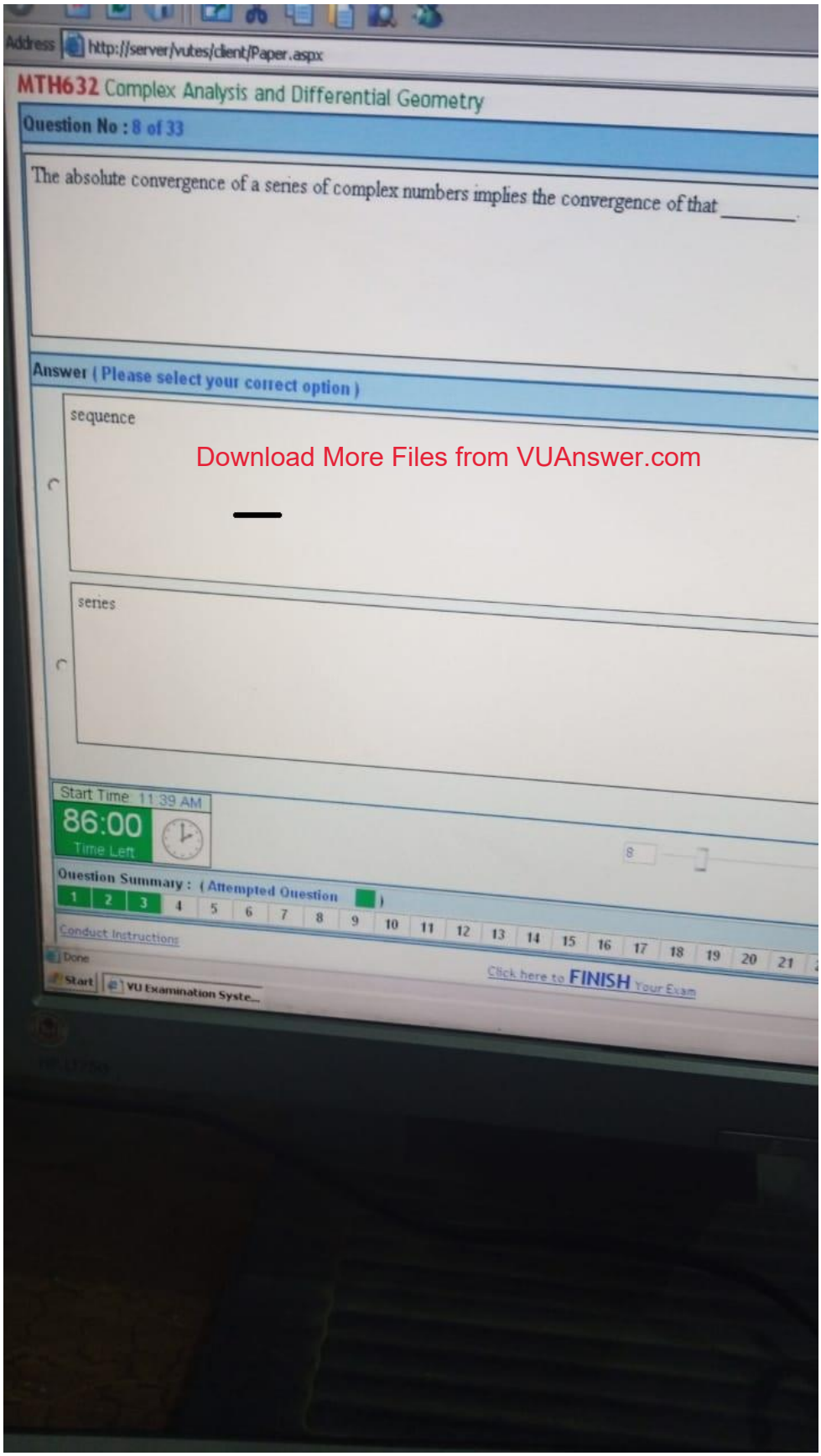
6-3-21

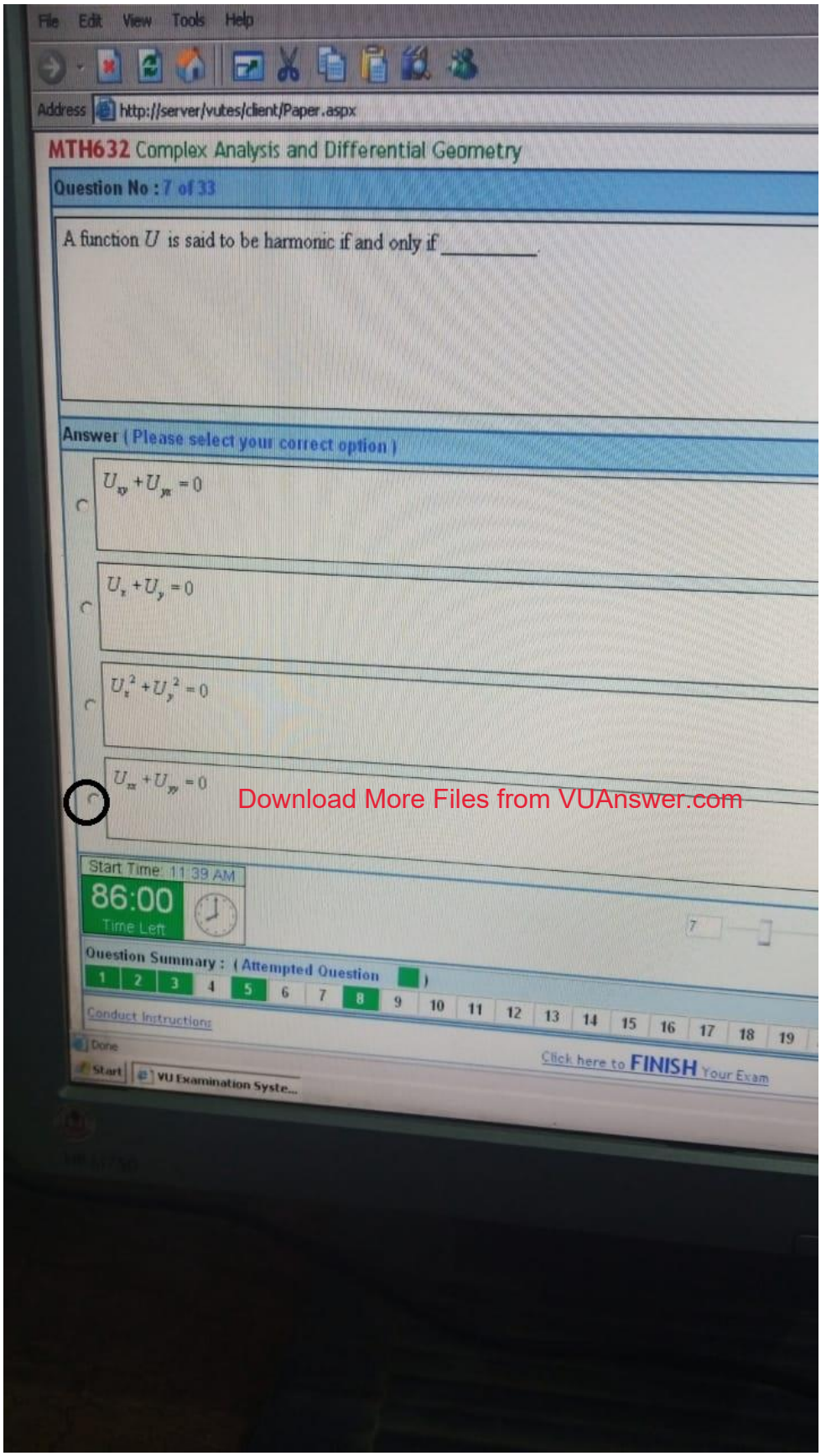

MTH632 Complex Analysis and Differential Geometry
Question No : 3 of 33
The sequence $z_n = -2 + i\frac{1}{n^3}$ $(n = 1, 2, 3, \ldots)$ for $n \to \infty$ converges to _____.
Download More Files from VUAnswer.com
Answer (Please select your correct option)
i
-1
1
-2
Start Time: 11:39 AM
89:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21
Conduct Instructions
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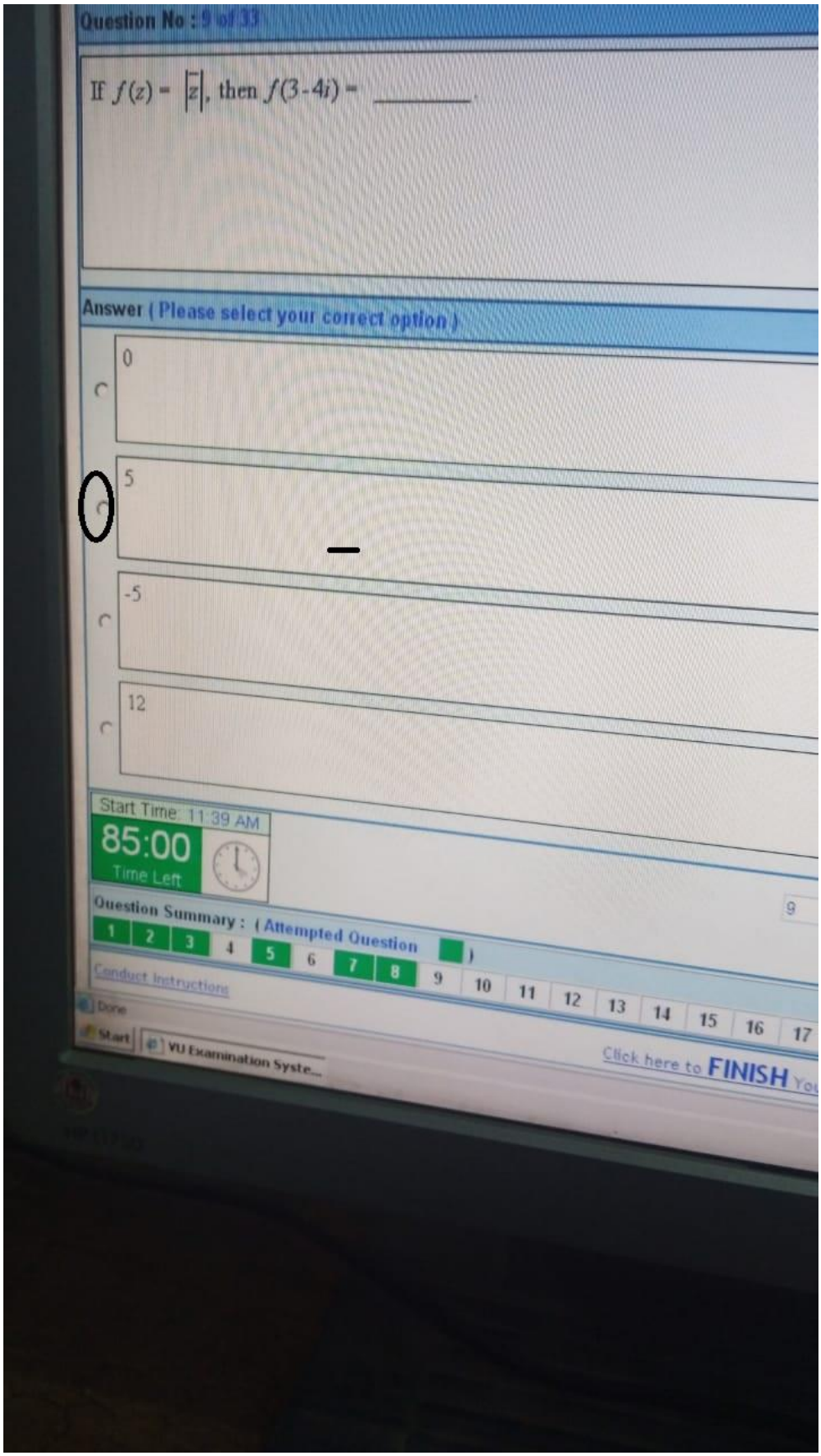

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MTH632 Complex Analysis and Differential Geometry
Question No : 2 of 33
The logarithmic function of $e^w = z$, where $z = re^{i\theta}$ and $w = u + iv$, then _____.
Download More Files from VUAnswer.com
Answer (Please select your correct option)
$w = \ln r + i(\theta + 2n\pi)$
$w = re^{i\theta}$
$w = r + i(\theta + 2n\pi)$
$w = -\ln r + i\theta$
Start Time: 11:39 AM
89:00
Time Left
Question Summary : (Attempted Question)
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Conduct Instructions
Click here to FINISH Your Exam
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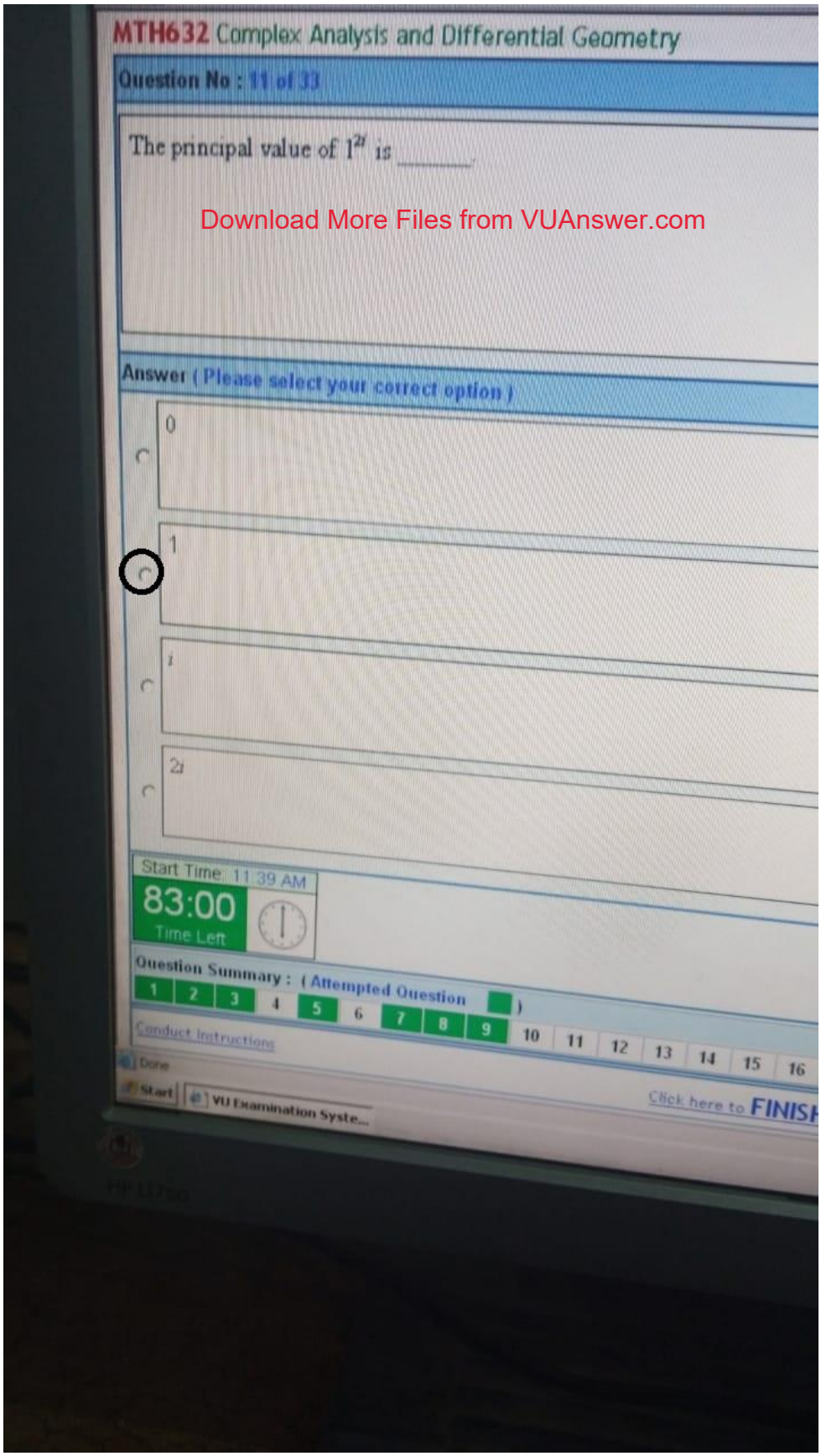

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MTH632 Complex Analysis and Differential Geometry
Question No : 5 of 33
A point at which a function ceases to be analytic is called a _____.
Download More Files from VUAnswer.com
Answer (Please select your correct option)
singular point
non-singular point
regular point
non-regular point
Start Time 11:39 AM
87:00
Time Left
Question Summary : (Attempted Question)
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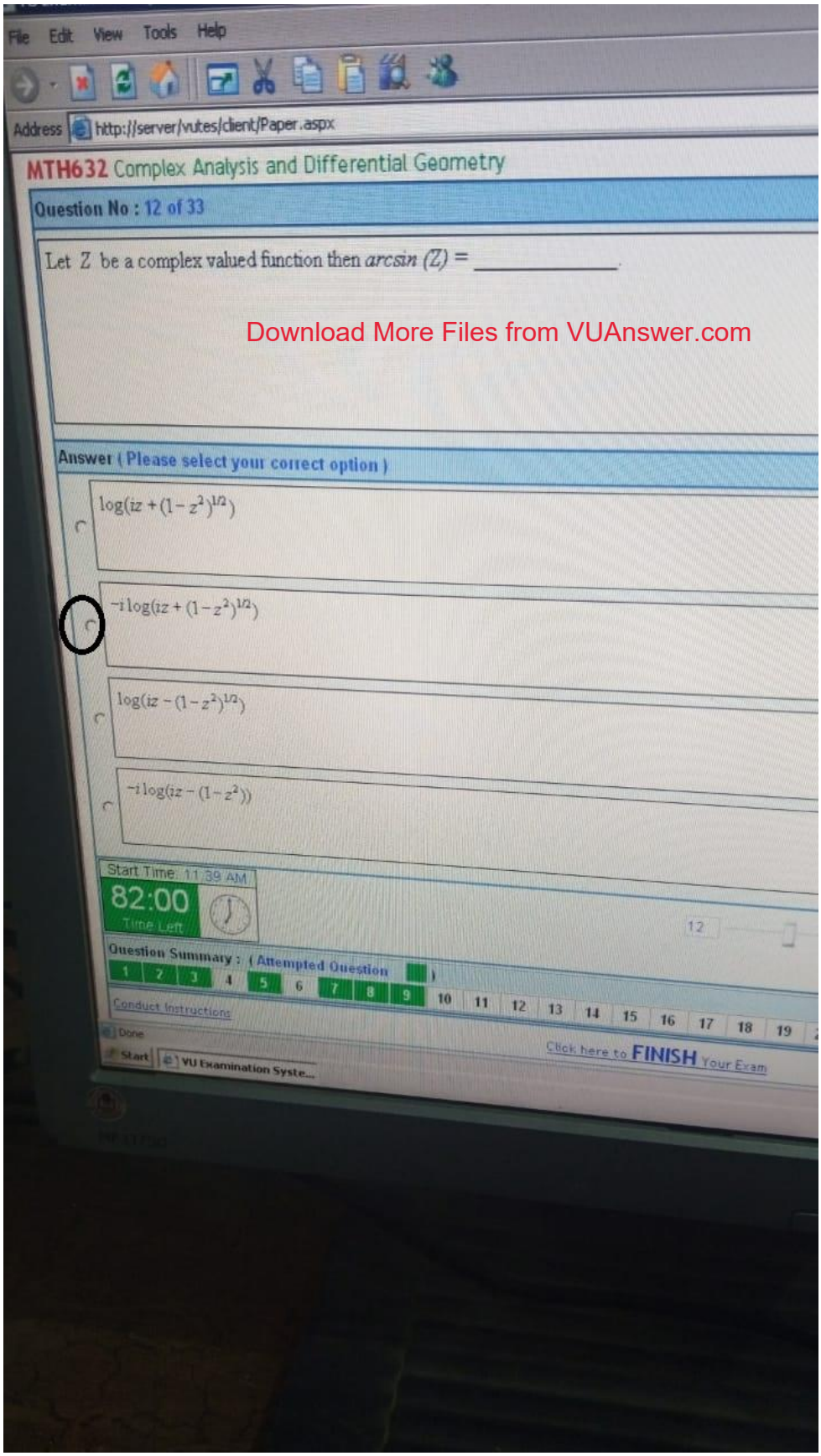

MTH632 Complex Analysis and Differential Geometry
Question No : 6 of 33
The function $f(z) = |z|$ is non-constant _____.
Download More Files from VUAnswer.com
Answer (Please select your correct option)
analytic function
nowhere analytic function
non-analytic function
entire function
Start Time: 11:39 AM
87:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Click here to FINISH

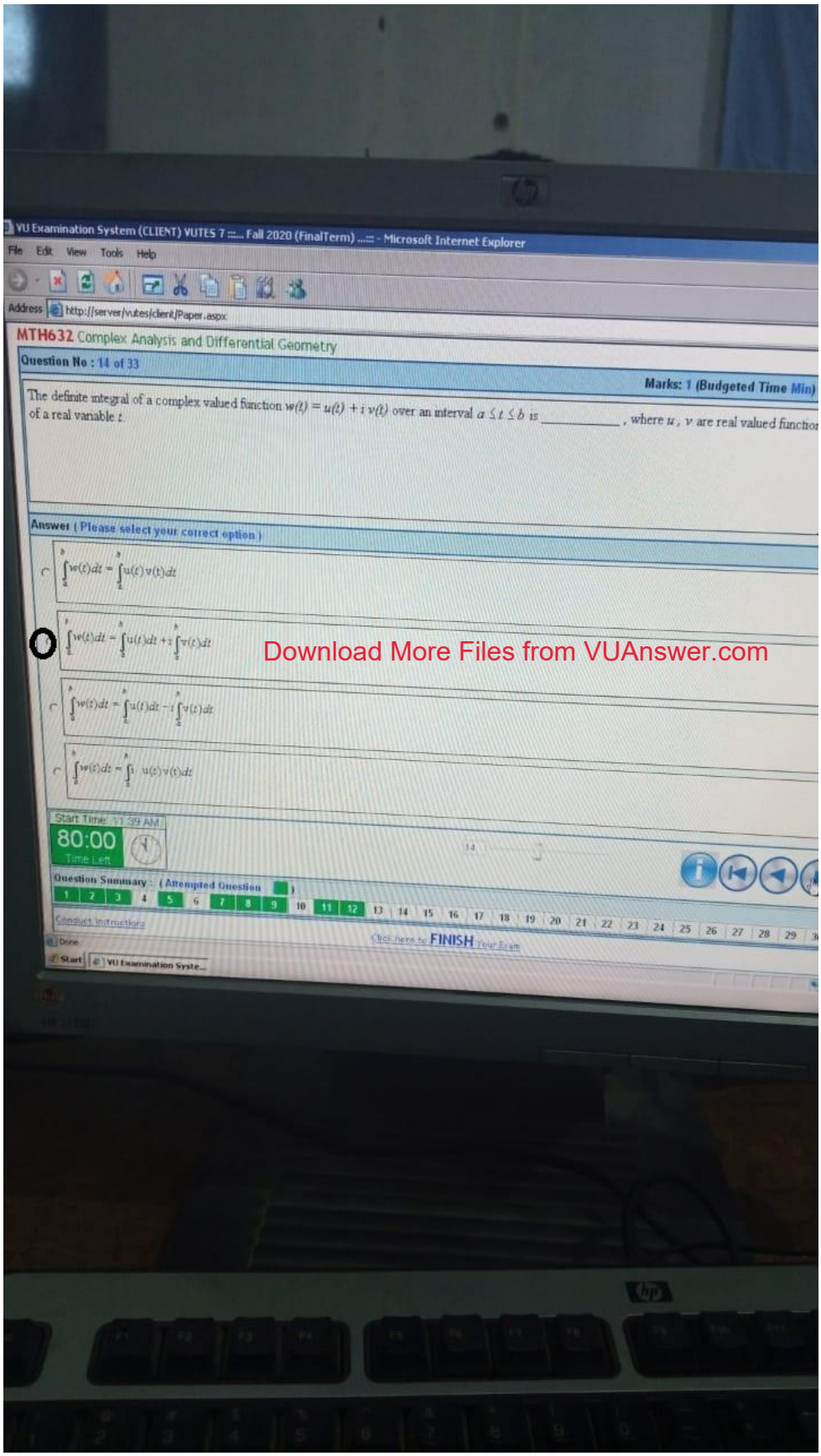

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MTH632 Complex Analysis and Differential Geometry
Question No : 8 of 33
The absolute convergence of a series of complex numbers implies the convergence of that _____.
Answer (Please select your correct option)
sequence
Download More Files from VUAnswer.com
series
Start Time 11:39 AM
86:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21
Conduct Instructions
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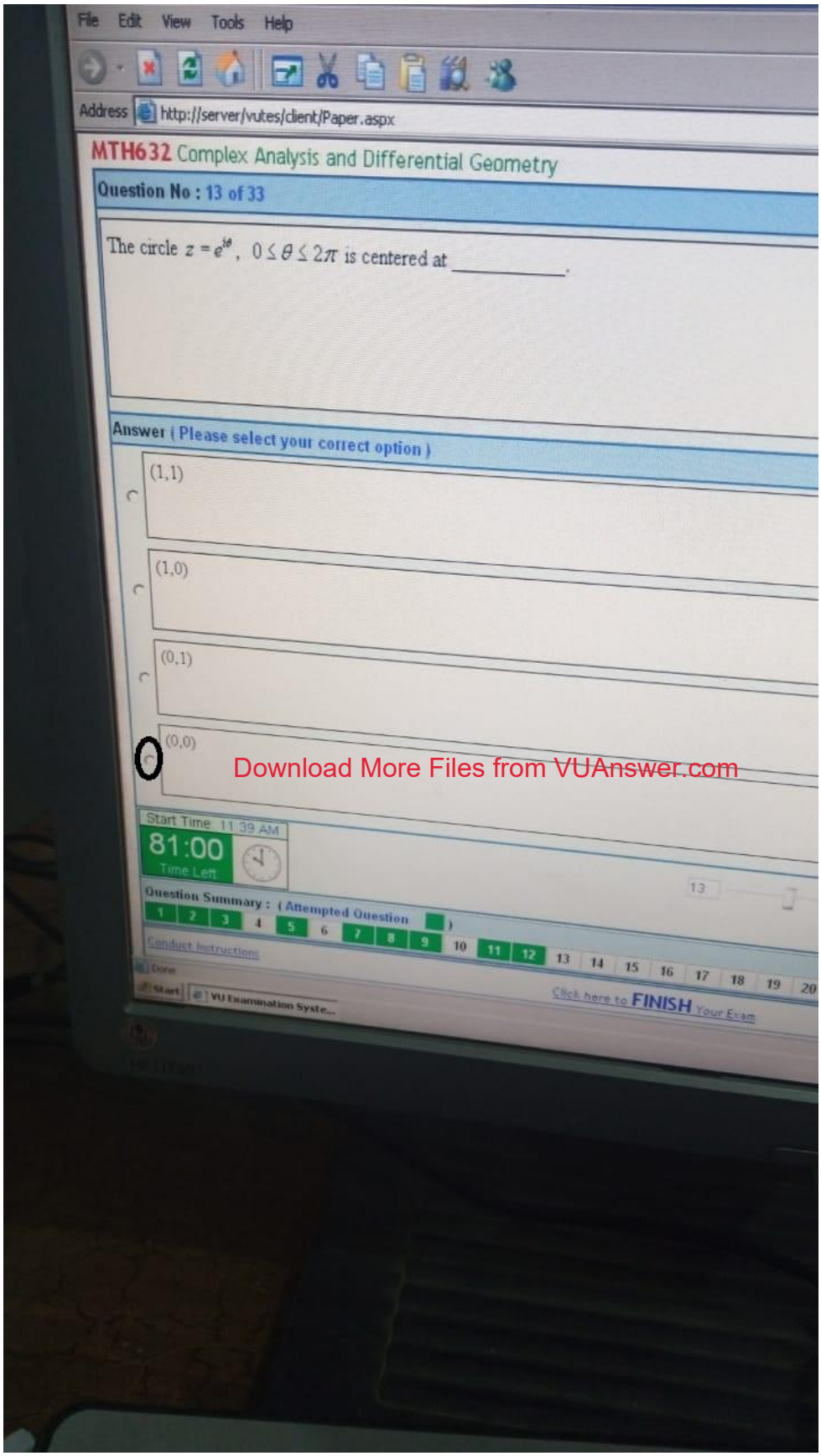

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MTH632 Complex Analysis and Differential Geometry
Question No : 7 of 33
A function U is said to be harmonic if and only if _____.
Answer (Please select your correct option)
$U_{xy} + U_{yx} = 0$
$U_x + U_y = 0$
$U_x^2 + U_y^2 = 0$
$U_{xx} + U_{yy} = 0$
Download More Files from VUAnswer.com
Start Time 11:39 AM
86:00
Time Left
Question Summary : (Attempted Question)
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Conduct Instructions
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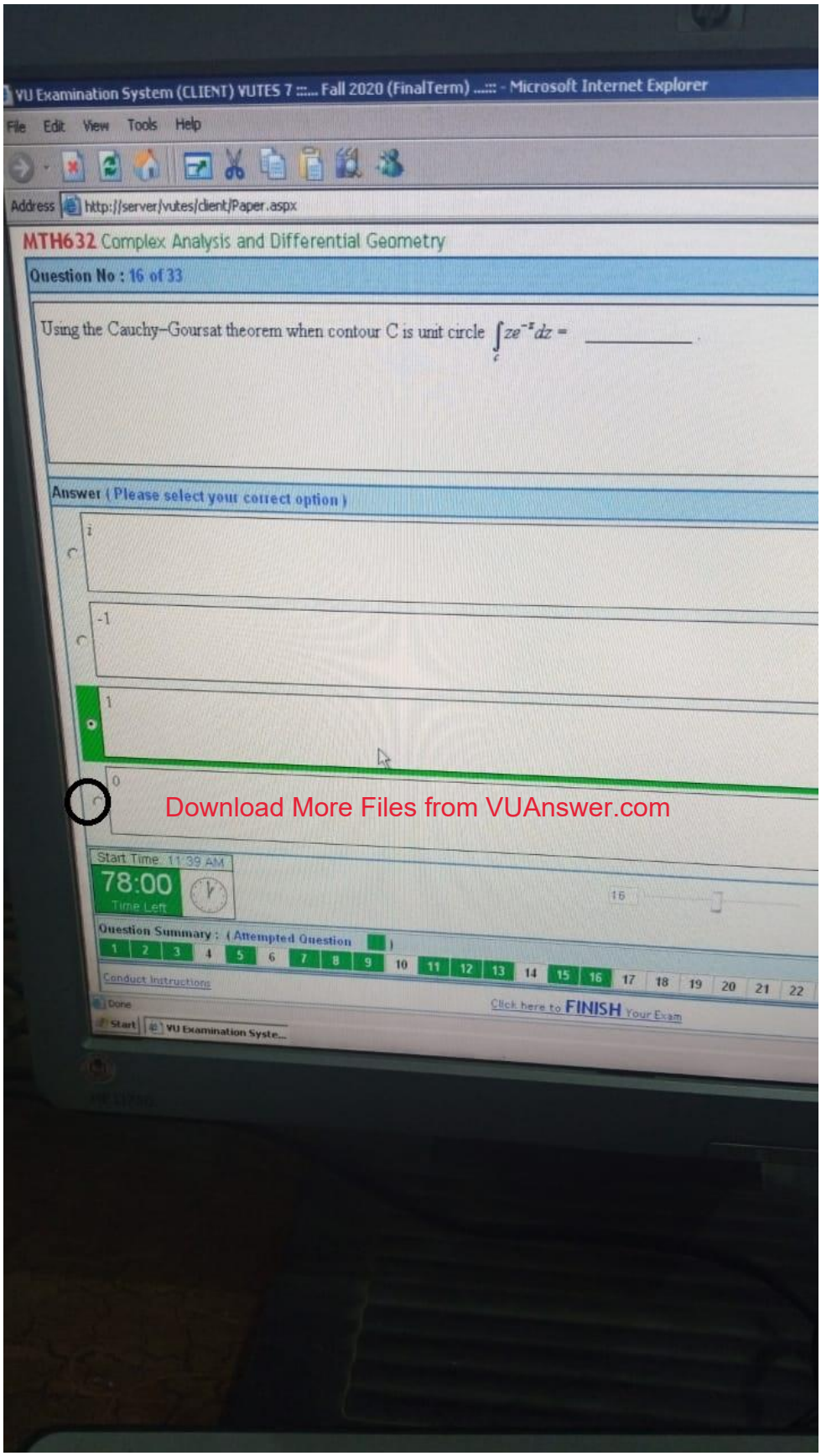

Question No : 9 of 33
If $f(z) = |\bar{z}|$, then $f(3-4i) =$ _____.
Answer (Please select your correct option)
0
5
-5
12
Start Time 11:39 AM
85:00
Time Left
Question Summary : (Attempted Question)
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Conduct Instructions
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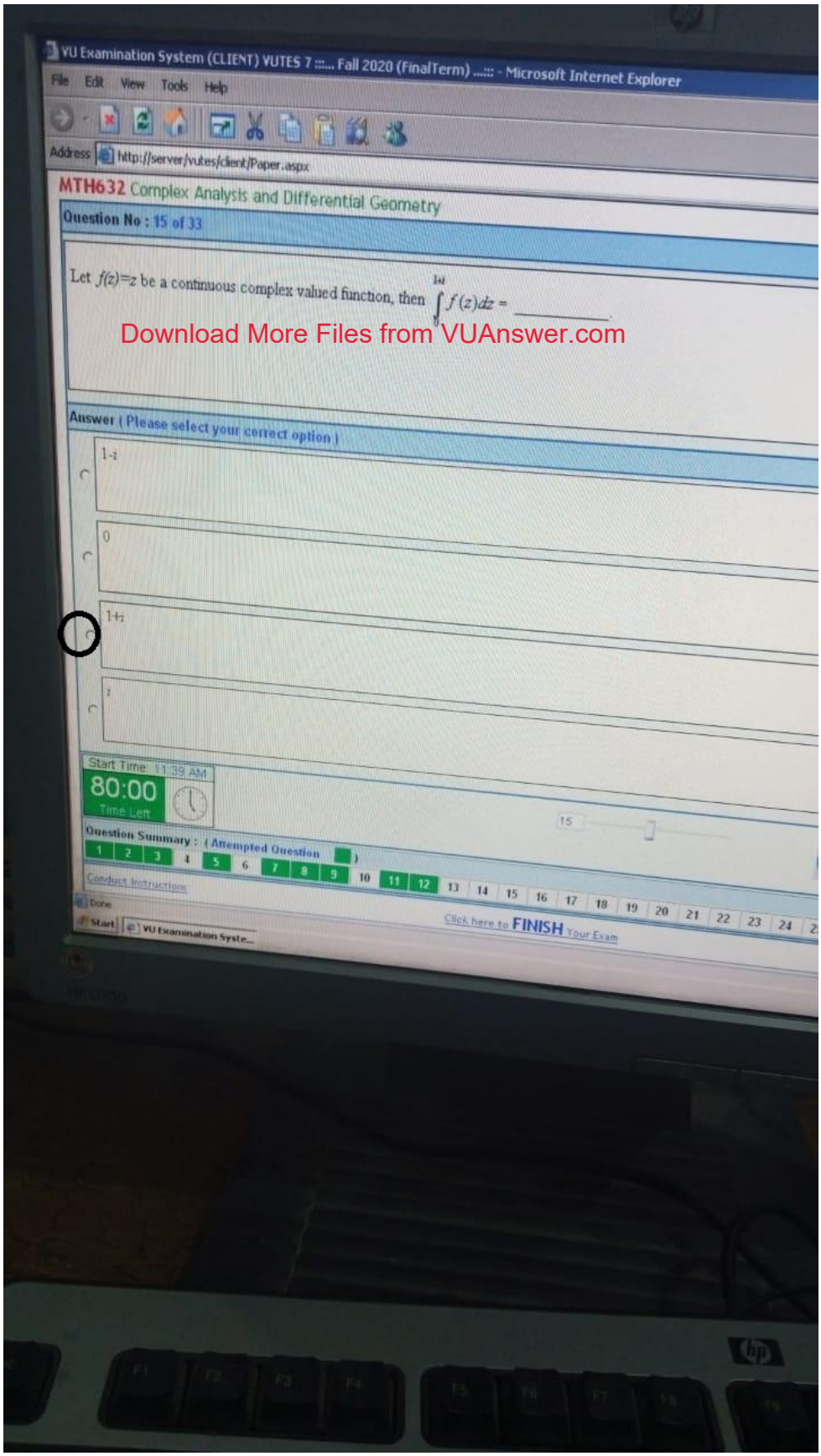

MTH632 Complex Analysis and Differential Geometry
Question No : 11 of 33
The principal value of 1^{2i} is _____.
Download More Files from VUAnswer.com
Answer (Please select your correct option)
0
1
i
2i
Start Time: 11:39 AM
83:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16
Conduct Instructions
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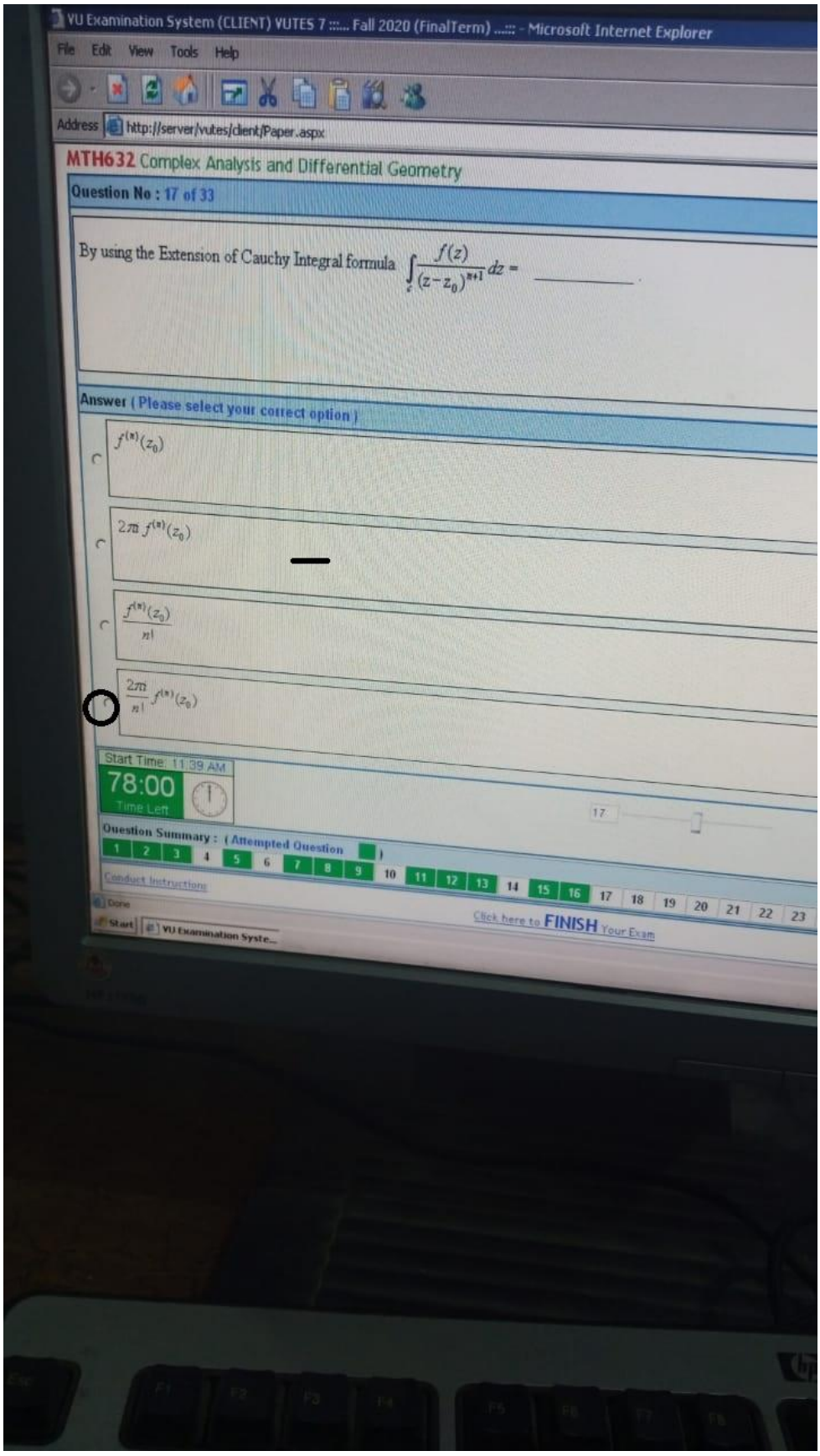

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MTH632 Complex Analysis and Differential Geometry
Question No : 12 of 33
Let Z be a complex valued function then arcsin (Z) = _____.
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Answer (Please select your correct option)
log(iz + (1 − z²)^1/2)
−i log(iz + (1 − z²)^1/2)
log(iz − (1 − z²)^1/2)
−i log(iz − (1 − z²))
Start Time: 11:39 AM
82:00
Time Left
12
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19
Conduct Instructions
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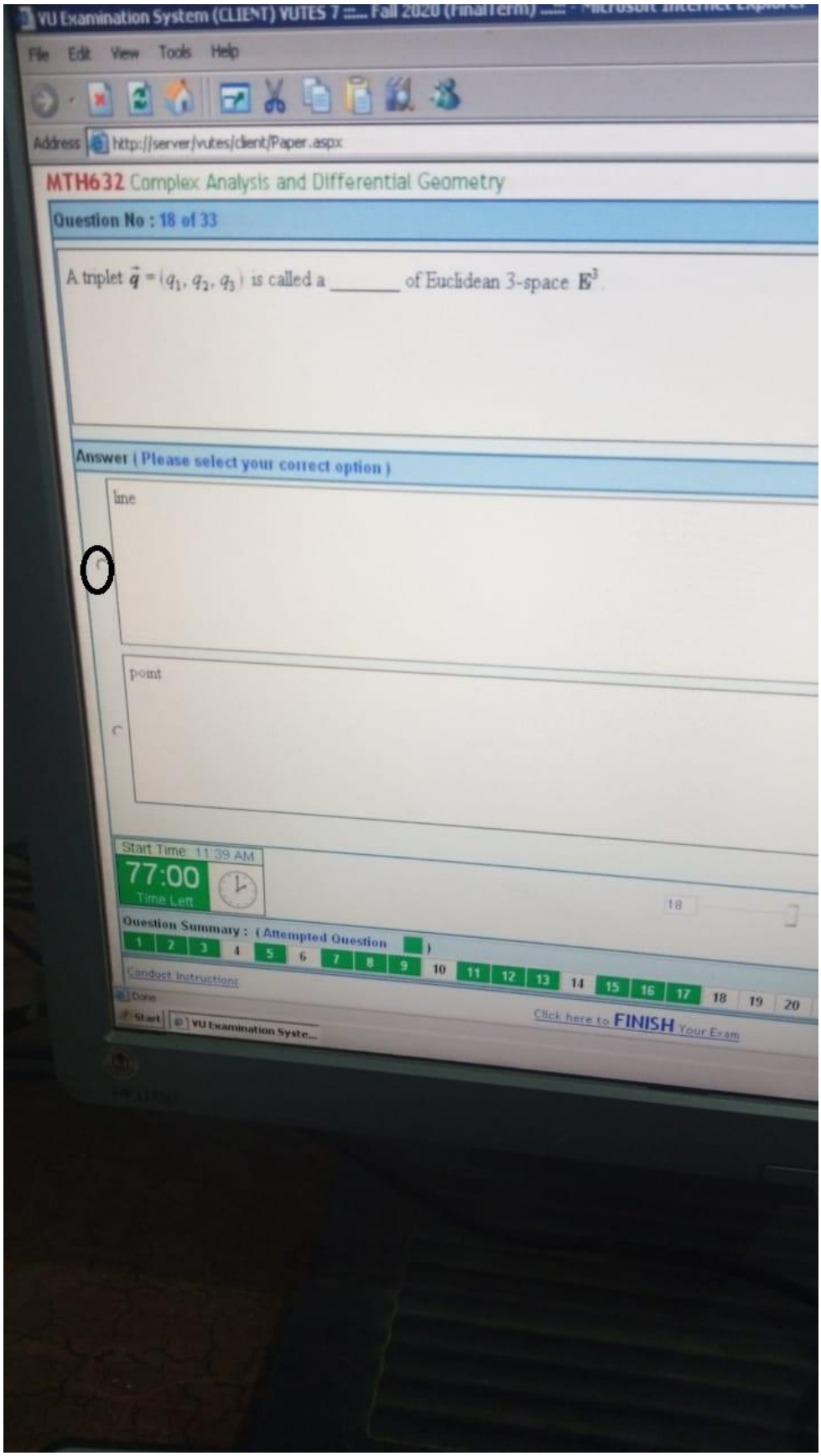

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MTH632 Complex Analysis and Differential Geometry
Question No : 14 of 33
Marks: 1 (Budgeted Time Min)
The definite integral of a complex valued function w(t) = u(t) + i v(t) over an interval a ≤ t ≤ b is _____, where u, v are real valued functio
of a real variable t.
Answer (Please select your correct option)
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80:00
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Question Summary : (Attempted Question)
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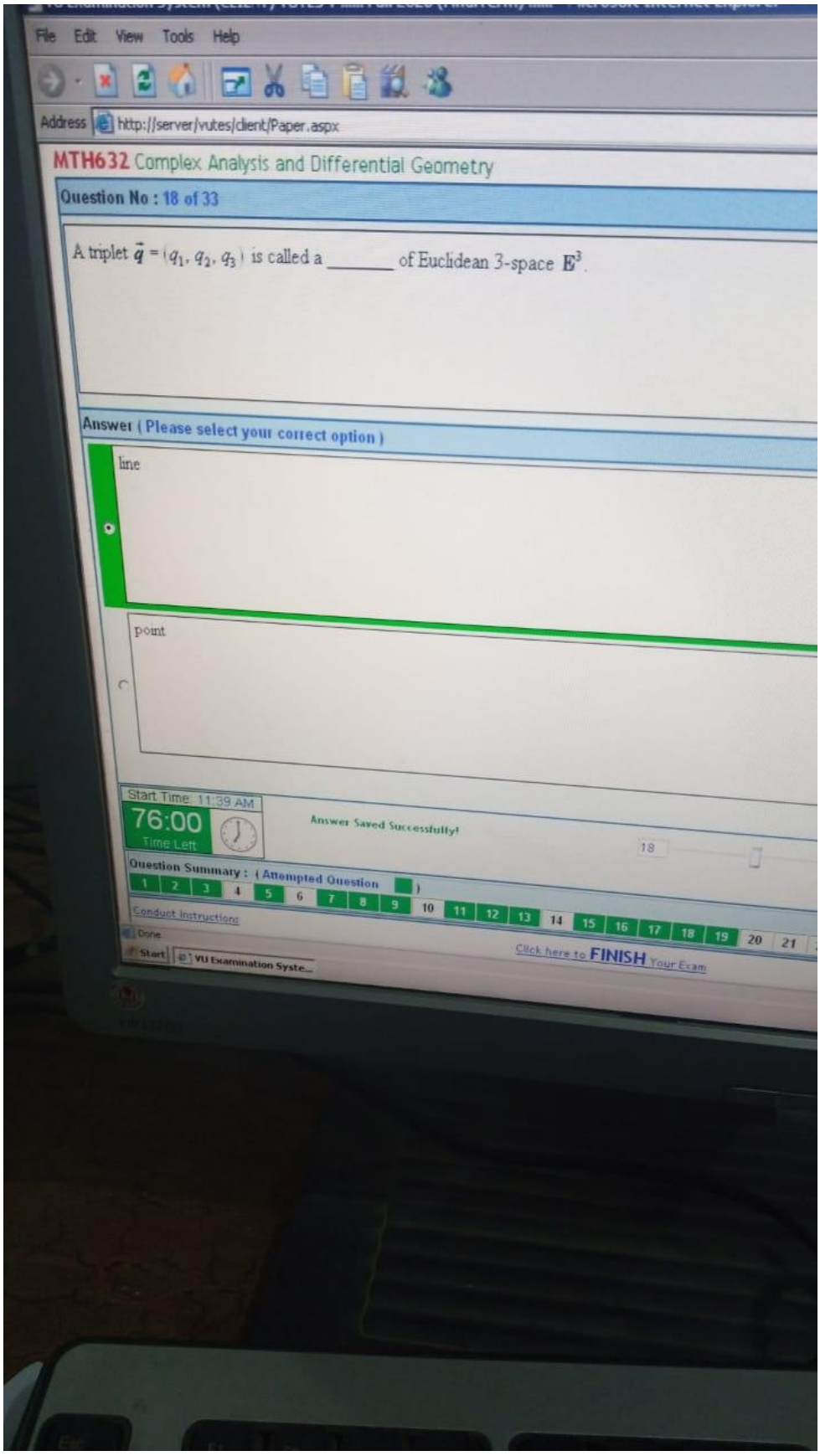

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MTH632 Complex Analysis and Differential Geometry
Question No : 13 of 33
The circle $z = e^{i\theta}$, $0 \le \theta \le 2\pi$ is centered at _____.
Answer (Please select your correct option)
(1,1)
(1,0)
(0,1)
(0,0)
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Start Time 11:39 AM
81:00
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Question Summary : (Attempted Question)
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Conduct Instructions
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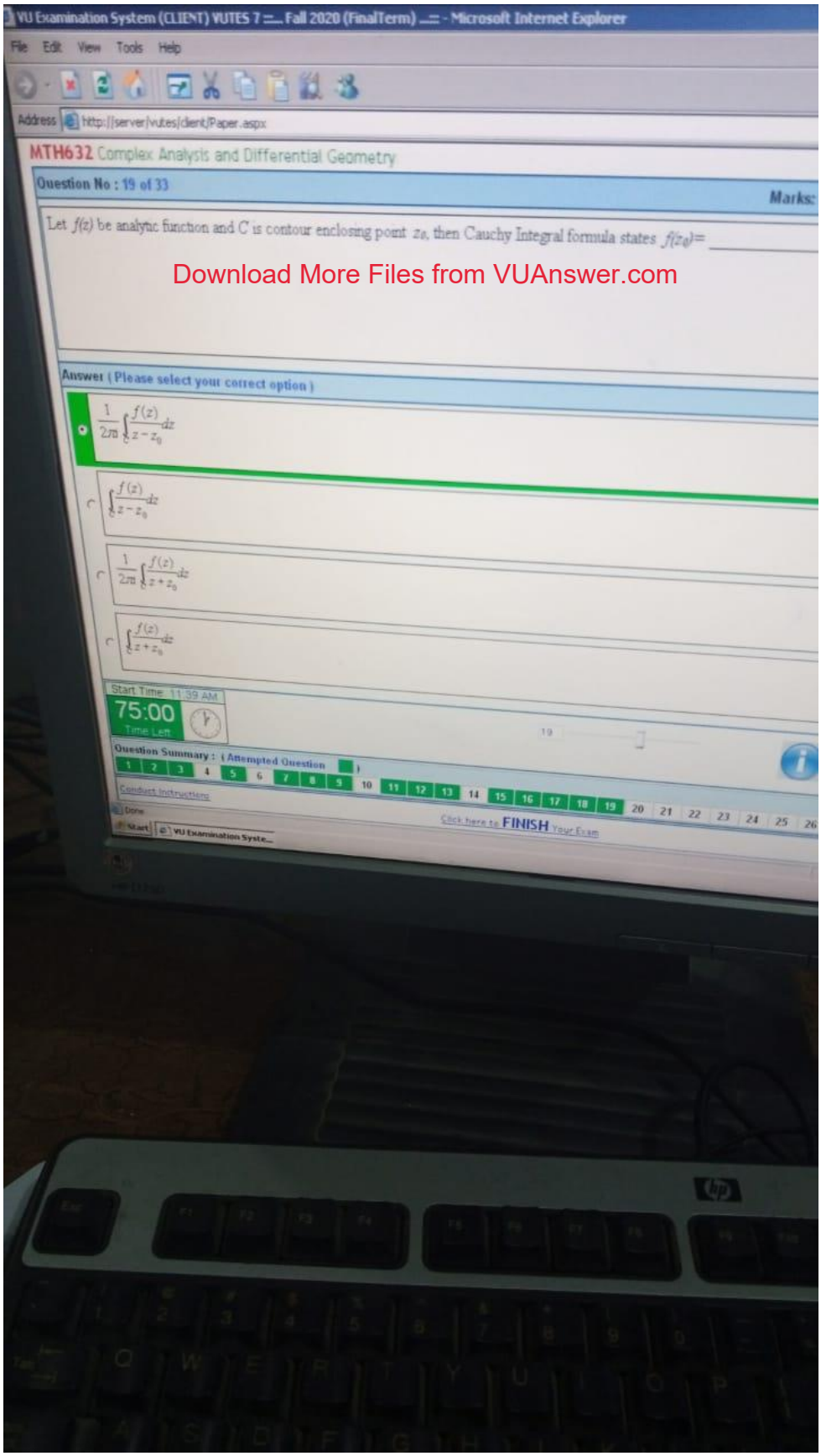

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MTH632 Complex Analysis and Differential Geometry
Question No : 16 of 33
Using the Cauchy–Goursat theorem when contour C is unit circle $\int_C ze^{-z}dz =$ _____ .
Answer (Please select your correct option)
i
-1
1
0
Download More Files from VUAnswer.com
Start Time 11:39 AM
78:00
Time Left
Question Summary : (Attempted Question)
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Conduct Instructions
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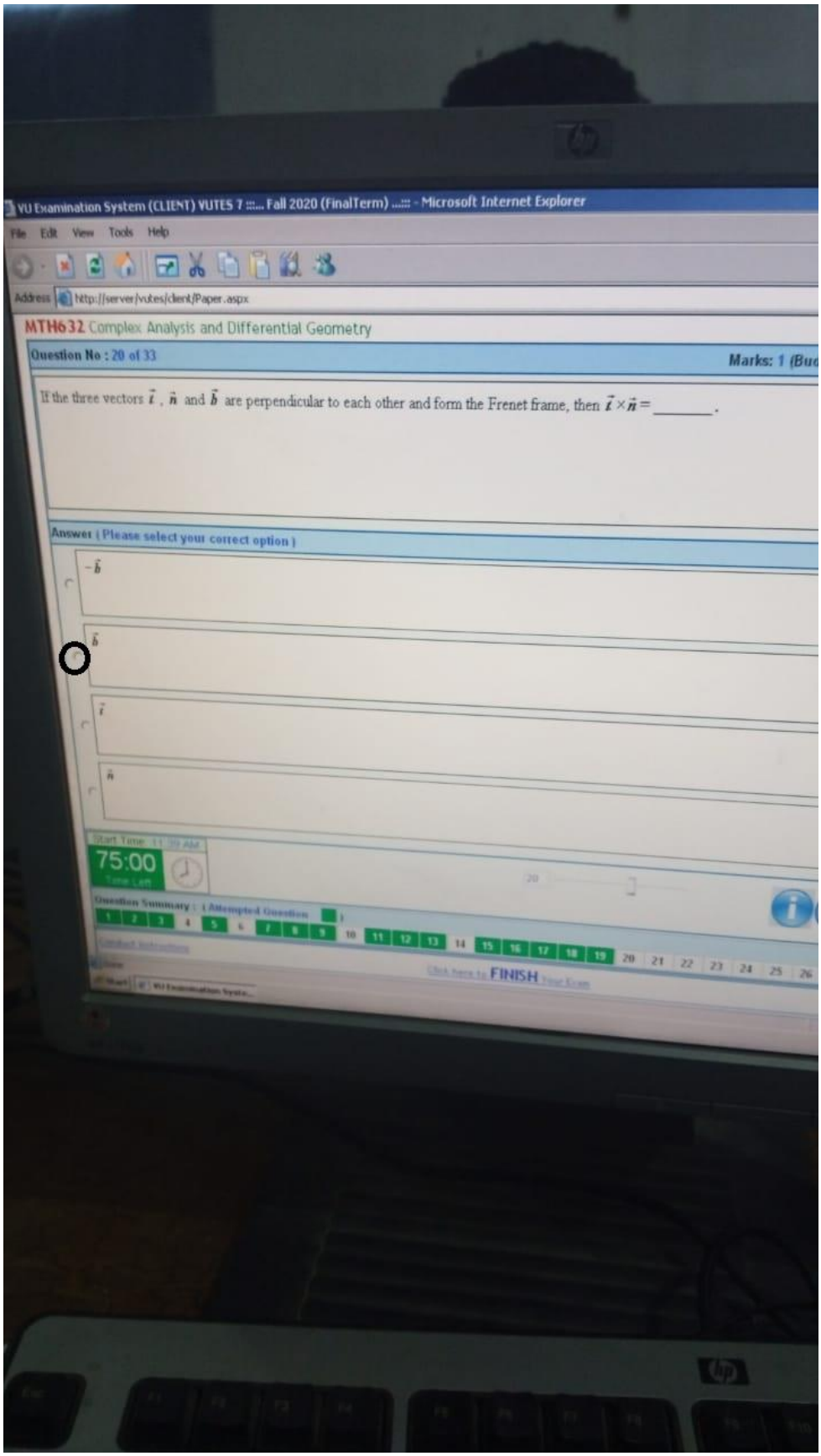

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MTH632 Complex Analysis and Differential Geometry
Question No : 15 of 33
Let f(z)=z be a continuous complex valued function, then $\int_0^{1+i} f(z)dz =$ _____.
Download More Files from VUAnswer.com
Answer (Please select your correct option)
1-i
0
1+i
i
Start Time: 11:39 AM
80:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24
Conduct Instructions
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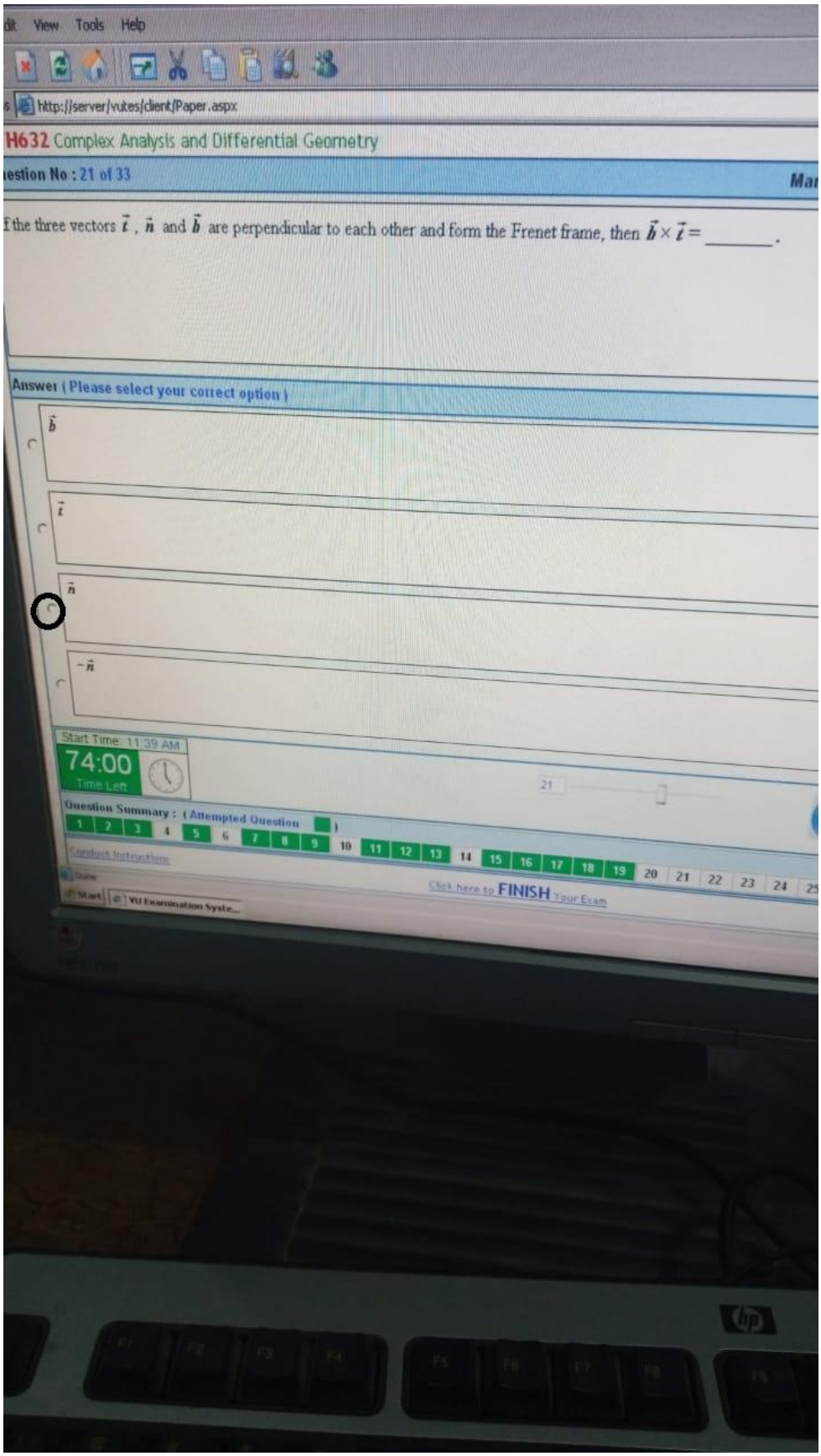

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MTH632 Complex Analysis and Differential Geometry
Question No : 17 of 33
By using the Extension of Cauchy Integral formula $\int_c \frac{f(z)}{(z-z_0)^{n+1}} dz =$ _____.
Answer (Please select your correct option)
$f^{(n)}(z_0)$
$2\pi i f^{(n)}(z_0)$
$\frac{f^{(n)}(z_0)}{n!}$
$\frac{2\pi i}{n!} f^{(n)}(z_0)$
Start Time: 11:39 AM
78:00
Time Left
17
Question Summary : (Attempted Question)
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Conduct Instructions
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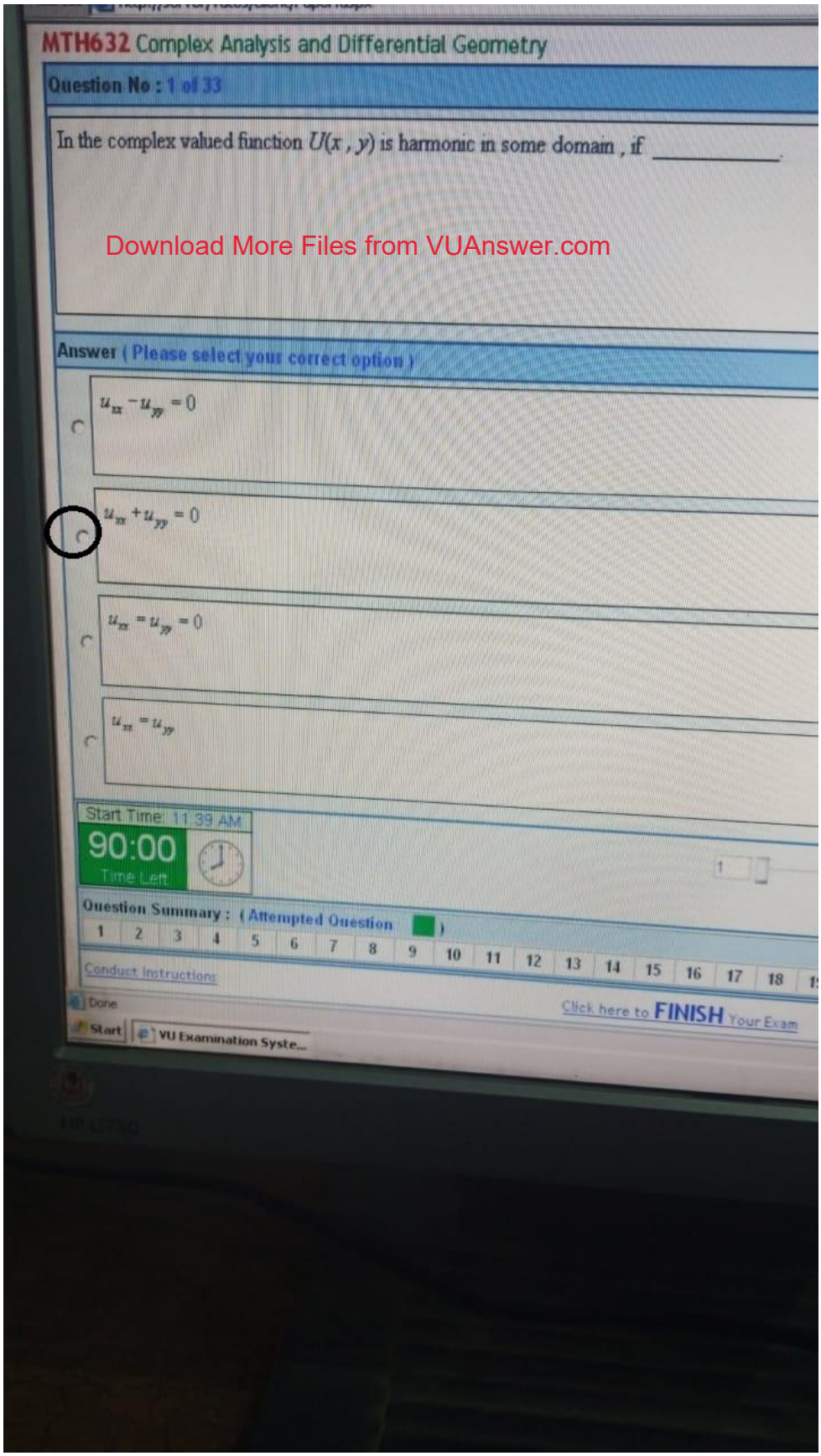

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MTH632 Complex Analysis and Differential Geometry
Question No : 18 of 33
A triplet $\vec{q} = (q_1, q_2, q_3)$ is called a _____ of Euclidean 3-space E^3.
Answer (Please select your correct option)
line
point
Start Time 11:39 AM
77:00
Time Left
18
Question Summary : (Attempted Question)
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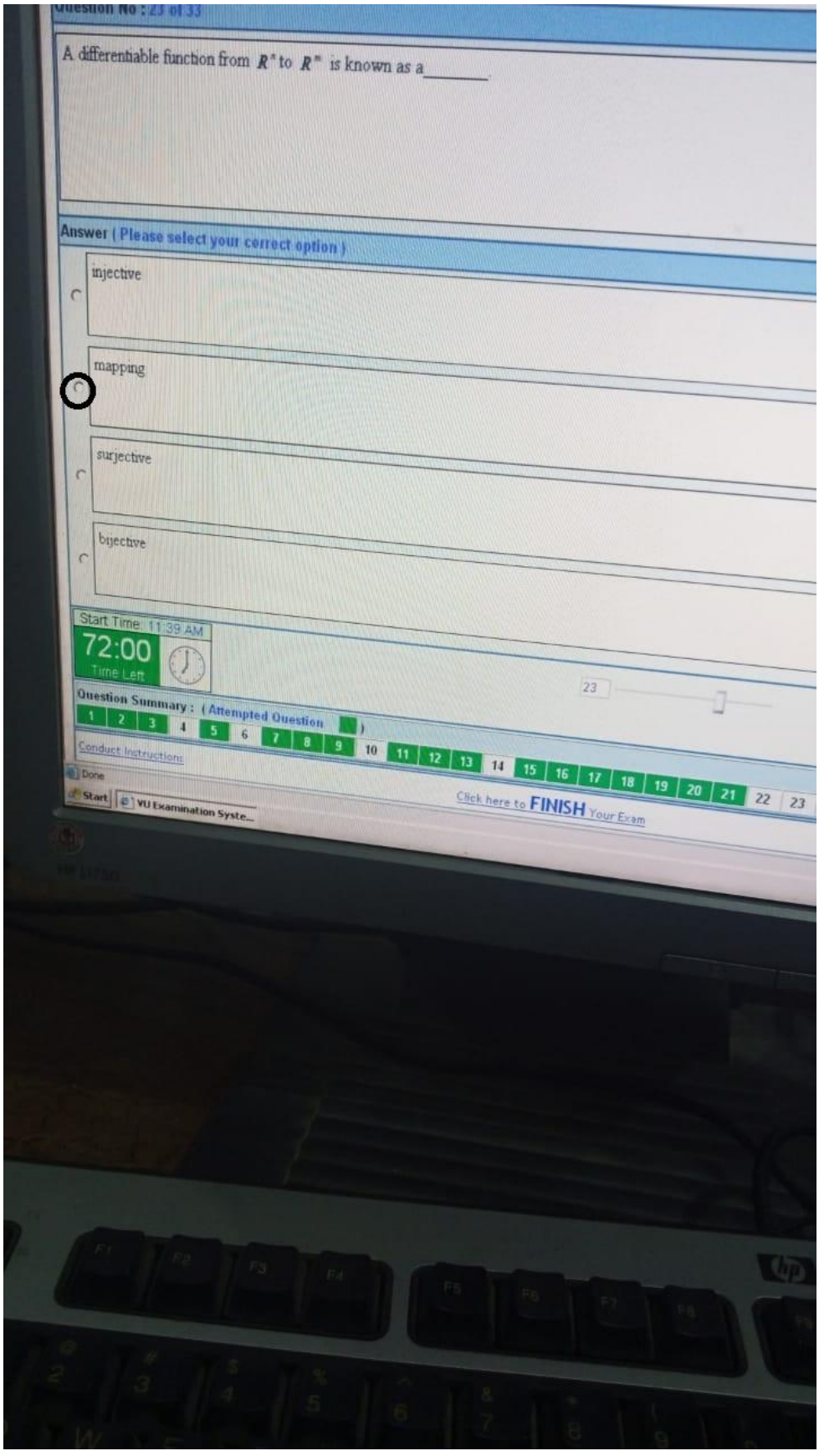

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MTH632 Complex Analysis and Differential Geometry
Question No : 18 of 33
A triplet $\vec{q} = (q_1, q_2, q_3)$ is called a _____ of Euclidean 3-space E^3.
Answer (Please select your correct option)
line
point
Start Time 11:39 AM
76:00
Time Left
Answer Saved Successfully!
Question Summary : (Attempted Question)
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Conduct Instructions
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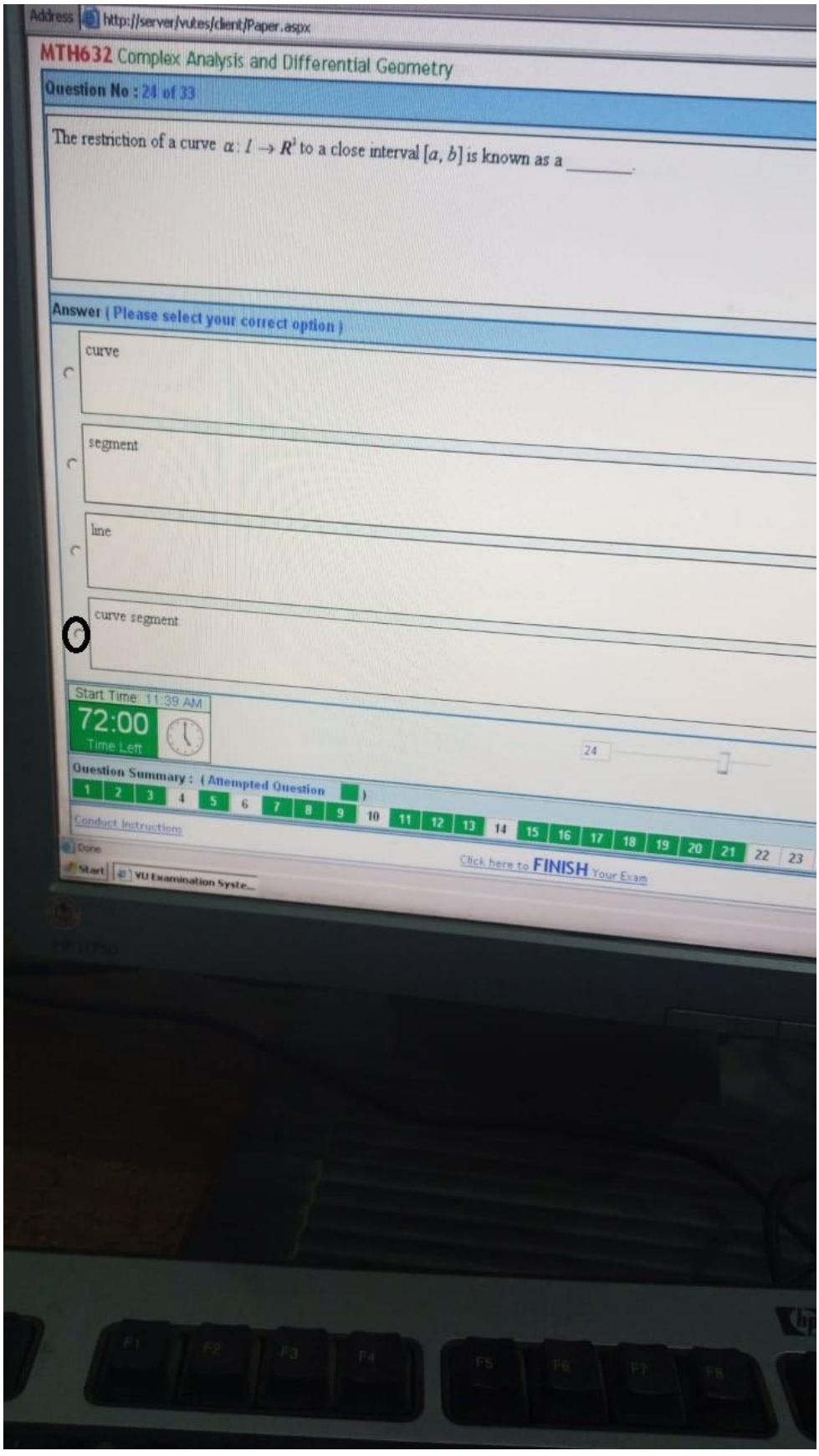

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MTH632 Complex Analysis and Differential Geometry
Question No : 19 of 33
Marks:
Let f(z) be analytic function and C is contour enclosing point z_0, then Cauchy Integral formula states $f(z_0)$=
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Answer (Please select your correct option)
$\frac{1}{2\pi i}\int_C \frac{f(z)}{z-z_0}dz$
$\int_C \frac{f(z)}{z-z_0}dz$
$\frac{1}{2\pi i}\int_C \frac{f(z)}{z+z_0}dz$
$\int_C \frac{f(z)}{z+z_0}dz$
Start Time 11:39 AM
75:00
Time Left
Question Summary : (Attempted Question)
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Conduct Instructions
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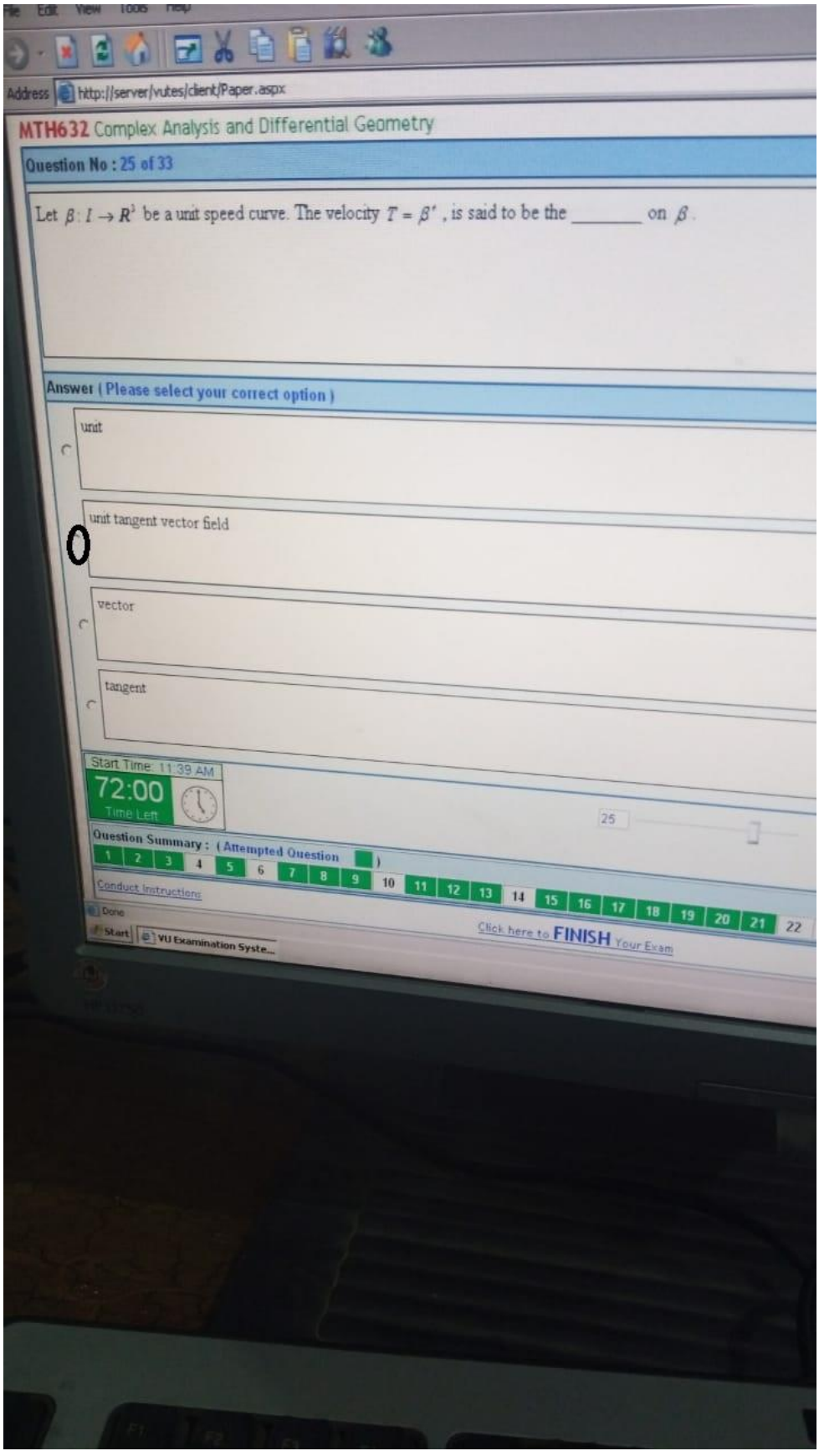

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MTH632 Complex Analysis and Differential Geometry
Question No : 20 of 33
Marks: 1 (Bu
If the three vectors $\vec{t}$, $\vec{n}$ and $\vec{b}$ are perpendicular to each other and form the Frenet frame, then $\vec{t} \times \vec{n} =$ _____.
Answer (Please select your correct option)
$-\vec{b}$
$\vec{b}$
$\vec{t}$
$\vec{n}$
75:00
Question Summary : 1 Attempted Question
FINISH

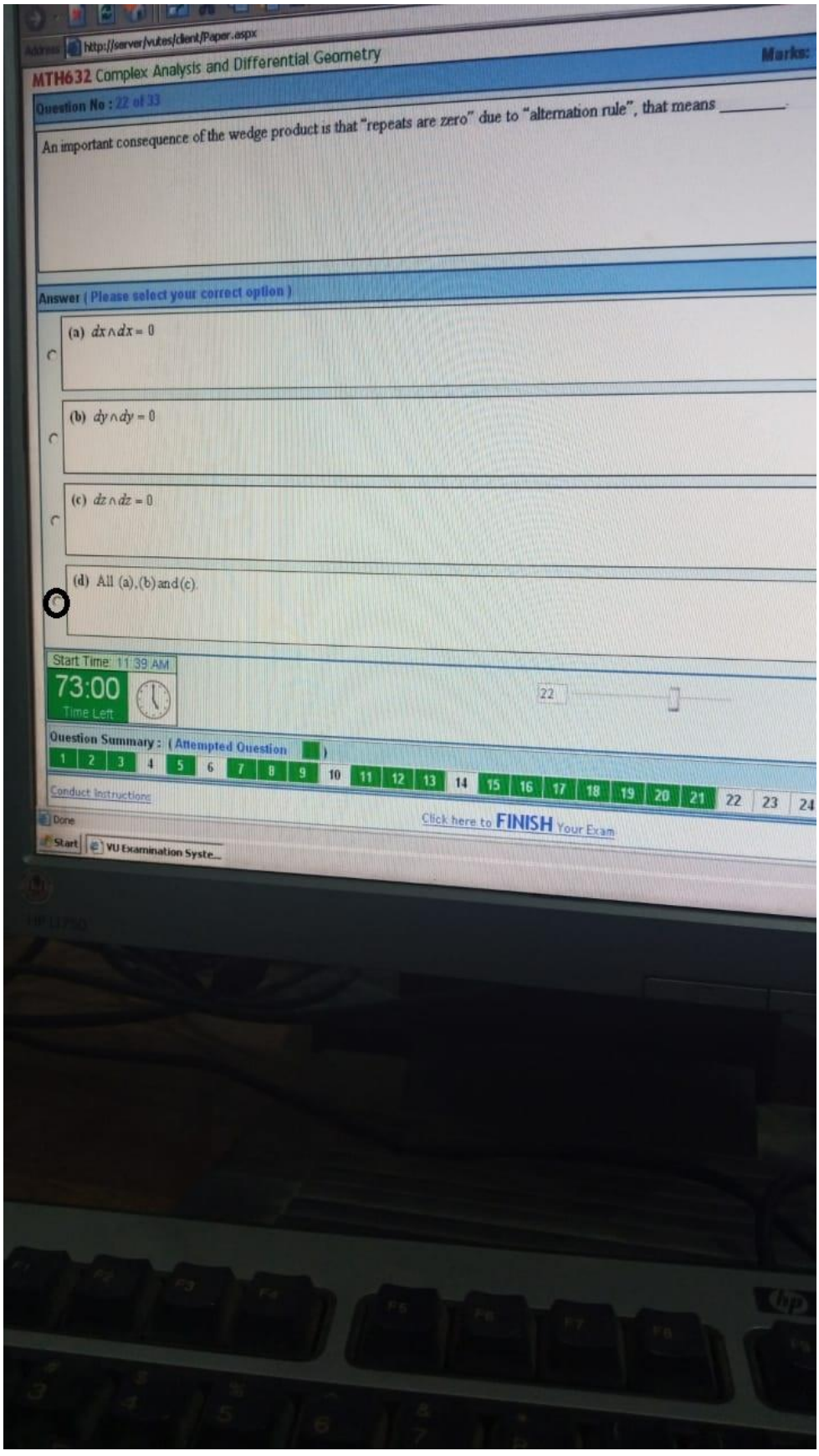

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H632 Complex Analysis and Differential Geometry
estion No : 21 of 33
Mar
f the three vectors $\vec{t}$, $\vec{n}$ and $\vec{b}$ are perpendicular to each other and form the Frenet frame, then $\vec{b} \times \vec{t} =$ _____.
Answer (Please select your correct option)
$\vec{b}$
$\vec{t}$
$\vec{n}$
$-\vec{n}$
Start Time 11:39 AM
74:00
Time Left
21
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25
Conduct Instructions
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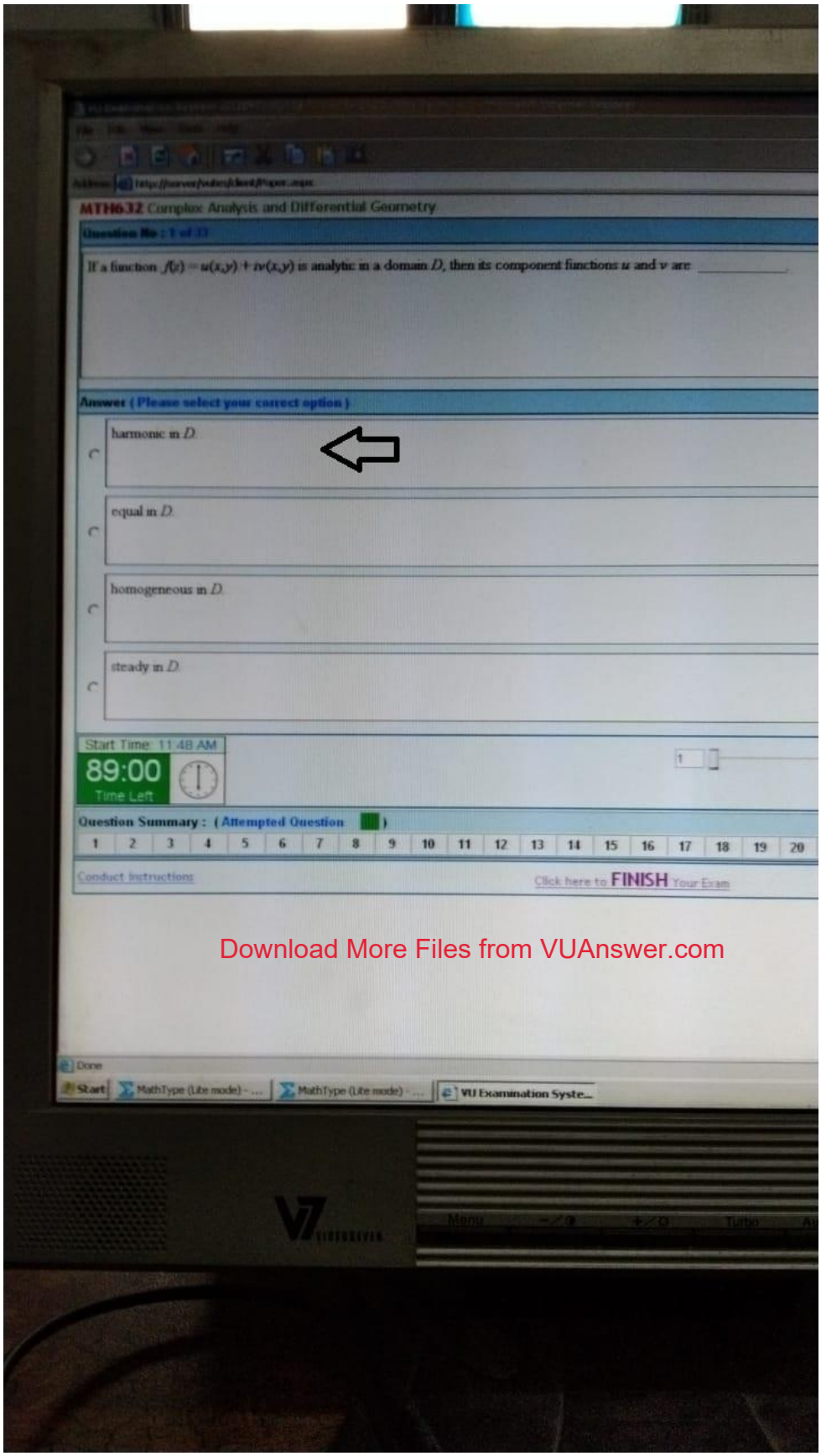

MTH632 Complex Analysis and Differential Geometry
Question No : 1 of 33
In the complex valued function $U(x, y)$ is harmonic in some domain , if _____.
Download More Files from VUAnswer.com
Answer (Please select your correct option)
$u_{xx} - u_{yy} = 0$
$u_{xx} + u_{yy} = 0$
$u_{xx} = u_{yy} = 0$
$u_{xx} = u_{yy}$
Start Time: 11:39 AM
90:00
Time Left
Question Summary : (Attempted Question)
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Conduct Instruction
Done
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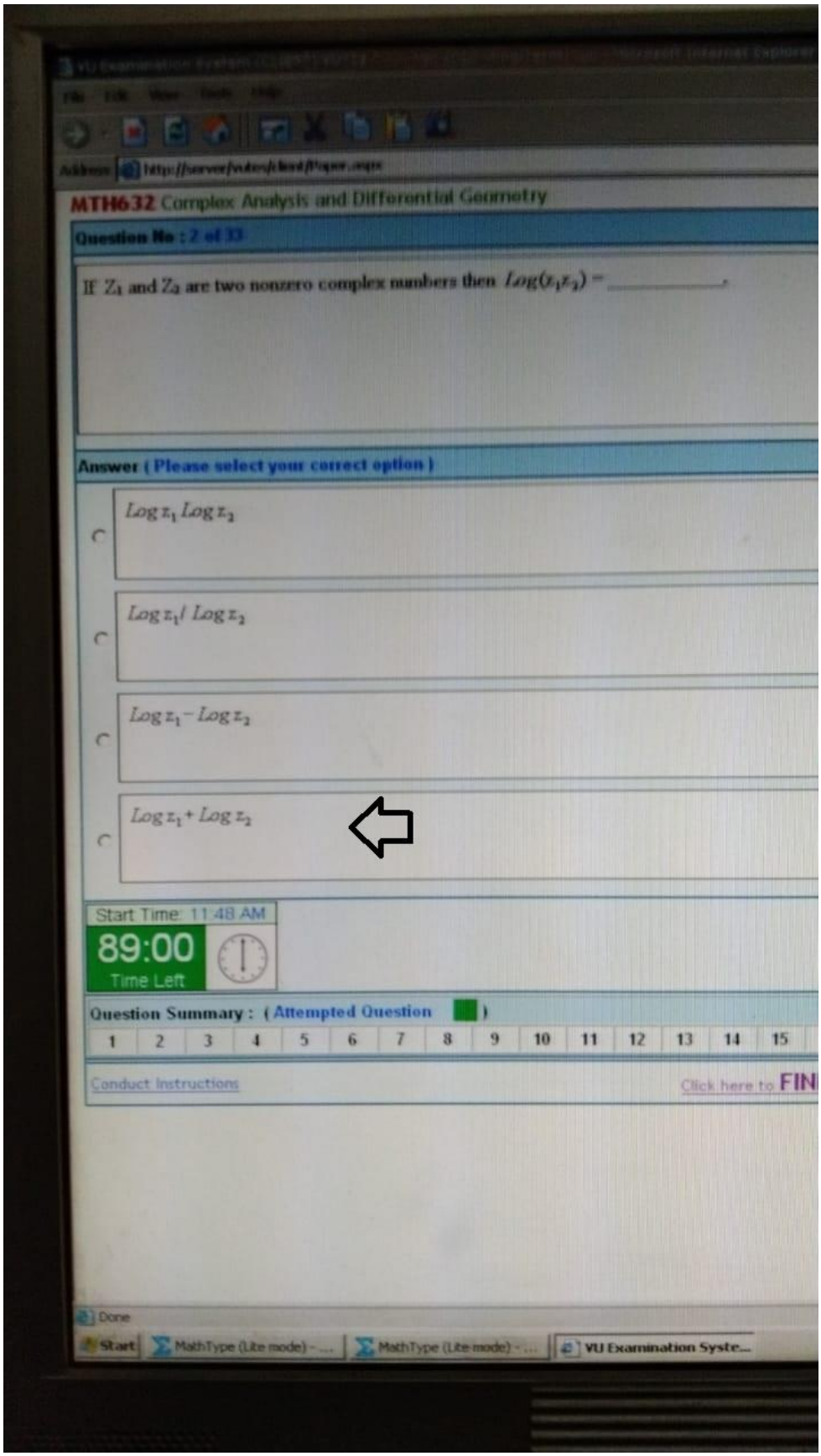

Question No : 23 of 33
A differentiable function from R^n to R^m is known as a_____.
Answer (Please select your correct option)
injective
mapping
surjective
bijective
Start Time 11:39 AM
72:00
Time Left
23
Question Summary : (Attempted Question)
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Conduct Instructions
Done
Start
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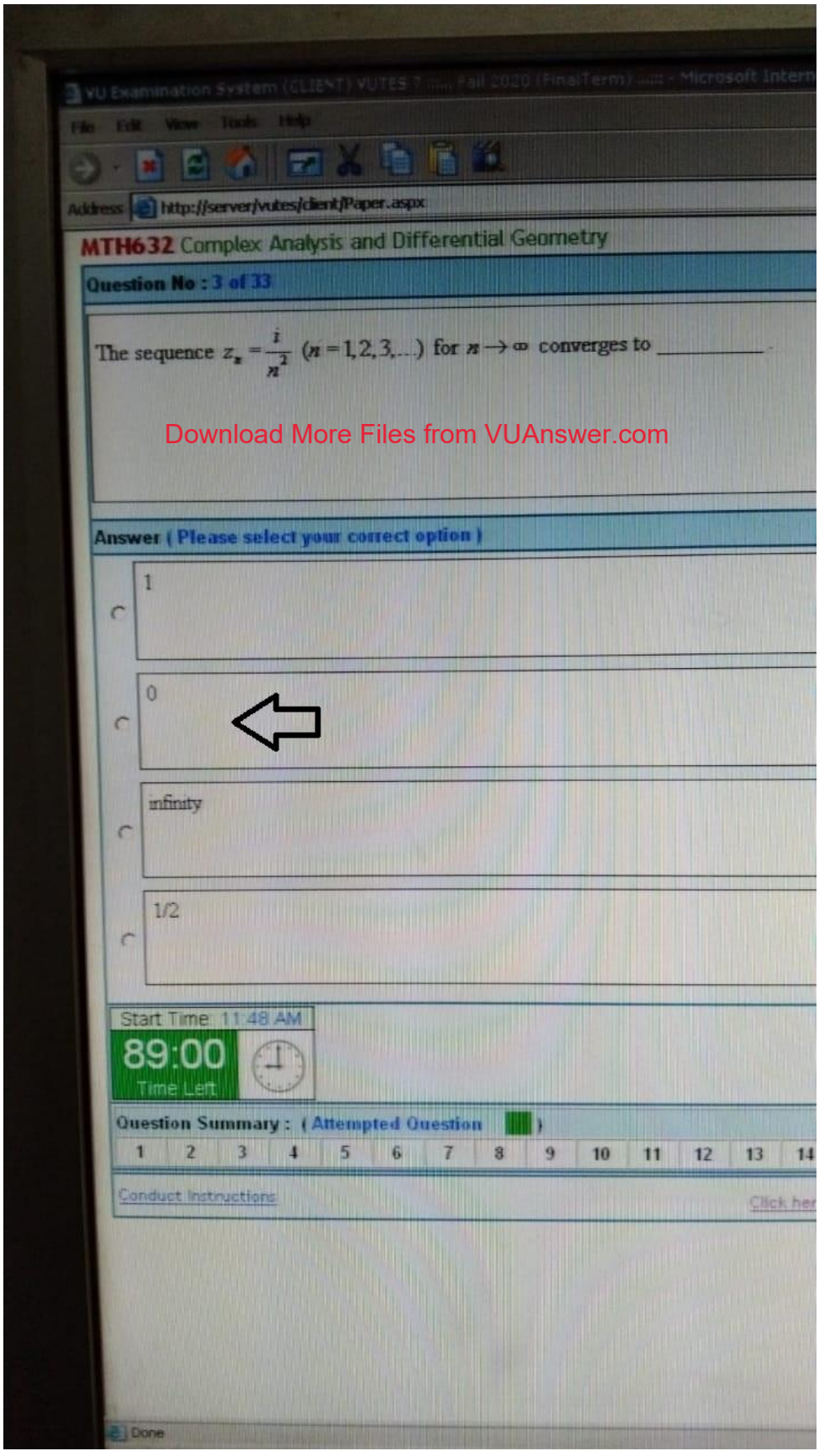

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MTH632 Complex Analysis and Differential Geometry
Question No : 24 of 33
The restriction of a curve $\alpha : I \to R^1$ to a close interval $[a, b]$ is known as a _____.
Answer (Please select your correct option)
curve
segment
line
curve segment
Start Time: 11:39 AM
72:00
Time Left
24
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23
Conduct Instructions
Done
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Address http://server/vutes/client/Paper.aspx
MTH632 Complex Analysis and Differential Geometry
Question No : 25 of 33
Let $\beta : I \to R^3$ be a unit speed curve. The velocity $T = \beta'$, is said to be the _____ on β.
Answer (Please select your correct option)
unit
unit tangent vector field
vector
tangent
Start Time: 11:39 AM
72:00
Time Left
25
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22
Conduct Instructions
Done
Start
VU Examination Syste...
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MTH632 Complex Analysis and Differential Geometry
Marks:
Question No : 22 of 33
An important consequence of the wedge product is that "repeats are zero" due to "alternation rule", that means _____.
Answer (Please select your correct option)
(a) $dx \wedge dx = 0$
(b) $dy \wedge dy = 0$
(c) $dz \wedge dz = 0$
(d) All (a), (b) and (c).
Start Time: 11:39 AM
73:00
Time Left
22
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24
Conduct Instructions
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MTH632 Complex Analysis and Differential Geometry
If a function $f(z) = u(x,y) + iv(x,y)$ is analytic in a domain D, then its component functions u and v are _____.
Answer (Please select your correct option)
harmonic in D.
equal in D.
homogeneous in D.
steady in D.
Start Time: 11:48 AM
89:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20
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MathType (Lite mode) - ...
MathType (Lite mode) - ...
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MTH632 Complex Analysis and Differential Geometry
Question No : 2 of 33
If Z_1 and Z_2 are two nonzero complex numbers then $Log(z_1 z_2) =$ _____.
Answer (Please select your correct option)
$Log\, z_1\, Log\, z_2$
$Log\, z_1 / Log\, z_2$
$Log\, z_1 - Log\, z_2$
$Log\, z_1 + Log\, z_2$
Start Time: 11:48 AM
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Time Left
Question Summary : (Attempted Question)
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MTH632 Complex Analysis and Differential Geometry
Question No : 3 of 33
The sequence $z_n = \frac{i}{n^2}$ $(n = 1,2,3,...)$ for $n \to \infty$ converges to _____.
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Answer (Please select your correct option)
1
0
infinity
1/2
Start Time: 11:48 AM
89:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14
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MTH632 Complex Analysis and Differential Geometry

Question No : 4 of 33

If $f(z) = x + ay + i(bx + cy)$ is analytic, then a, b, c equals to _____

Answer (Please select your correct option)

- $c=1$ and $a=-b$ 
- $a=1$ and $c=-b$
- $b=1$ and $a=-c$
- $a=b=c=1$

Start Time: 11:48 AM

89:00 Time Left

Question Summary : (Attempted Question)

1	2	3	4	5	6	7	8	9	10	11	12	13

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MTH632 Complex Analysis and Differential Geometry

Question No : 5 of 33

The function $f(z) = |z|$ is non-constant _____.

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Answer (Please select your correct option)

- analytic function
- nowhere analytic function
- non-analytic function
- entire function

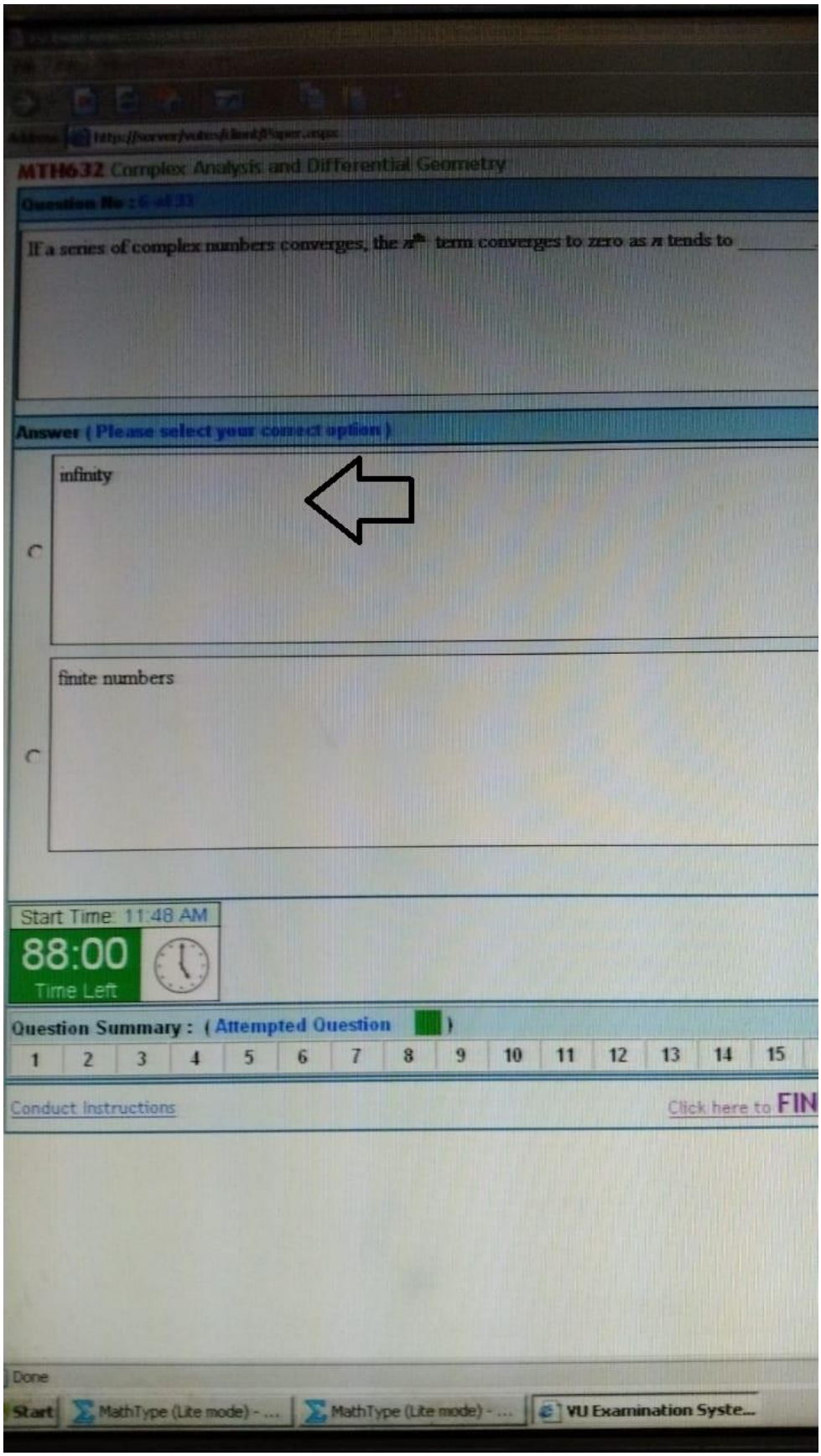
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Question Summary : (Attempted Question)

1	2	3	4	5	6	7	8	9	10	11	12

Conduct Instructions


MTH632 Complex Analysis and Differential Geometry
If a series of complex numbers converges, the nth term converges to zero as n tends to _____
Answer (Please select your correct option)
infinity
finite numbers
Start Time: 11:48 AM
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Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
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MathType (Lite mode) - ...
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MTH632 Complex Analysis and Differential Geometry

Question No : 7 of 33

If $e^{ax}\cos y$ is harmonic then 'a' is _____.

Answer (Please select your correct option)

- 0
- -1
- 2
- i

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Question Summary : (Attempted Question)

1	2	3	4	5	6	7	8	9	10

Conduct Instructions

MTH632 Complex Analysis and Differential Geometry

Question No : 8 of 33

A series converges to a number S if and only if the sequence of remainders tends to _____.

Answer (Please select your correct option)

- infinity
- one
- zero



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Question Summary : (Attempted Question)

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15

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MTH632 Complex Analysis and Differential Geometry

Question No : 9 of 20

The points at which $f(z) = \frac{1}{z^2+1}$ is not analytic are _____.

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Answer (Please select your correct option)

- 1 and -1
- i and $-i$
- 1 and i
- -1 and $-i$

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Question Summary : (Attempted Question)

1	2	3	4	5	6	7	8	9	10	11	12	1

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MTH632 Complex Analysis and Differential Geometry

In the trigonometric functions of complex numbers $\sin z$ is $x =$ _____

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Answer (Please select your correct option)

- $\frac{e^{ix} - e^{-ix}}{2i}$
- $\frac{e^{ix} - e^{-ix}}{2}$
- $\frac{e^{-ix} - e^{ix}}{2i}$
- $\frac{e^{-ix} - e^{ix}}{2}$

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Question Summary : (Attempted Question)

1	2	3	4	5	6	7	8	9	10	11	12	13

Conduct Instructions

Done

MTH632 Complex Analysis and Differential Geometry

Question No : 11 of 20

The principal value of 1^{2i} is _____.

Answer (Please select your correct option)

- 0
- 1
- i
- 2i

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Question Summary : (Attempted Question)

1	2	3	4	5	6	7	8	9	10

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MTH632 Complex Analysis and Differential Geometry

Question No : 12 of 33

Let Z be a complex valued function then $\sinh Z =$ _____

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Answer (Please select your correct option)

- $\frac{e^{z}-e^{-z}}{2}$ 
- $\frac{e^{z}+e^{-z}}{2}$
- $\frac{e^{z}-e^{-z}}{2i}$
- $\frac{e^{z}+e^{-z}}{2i}$

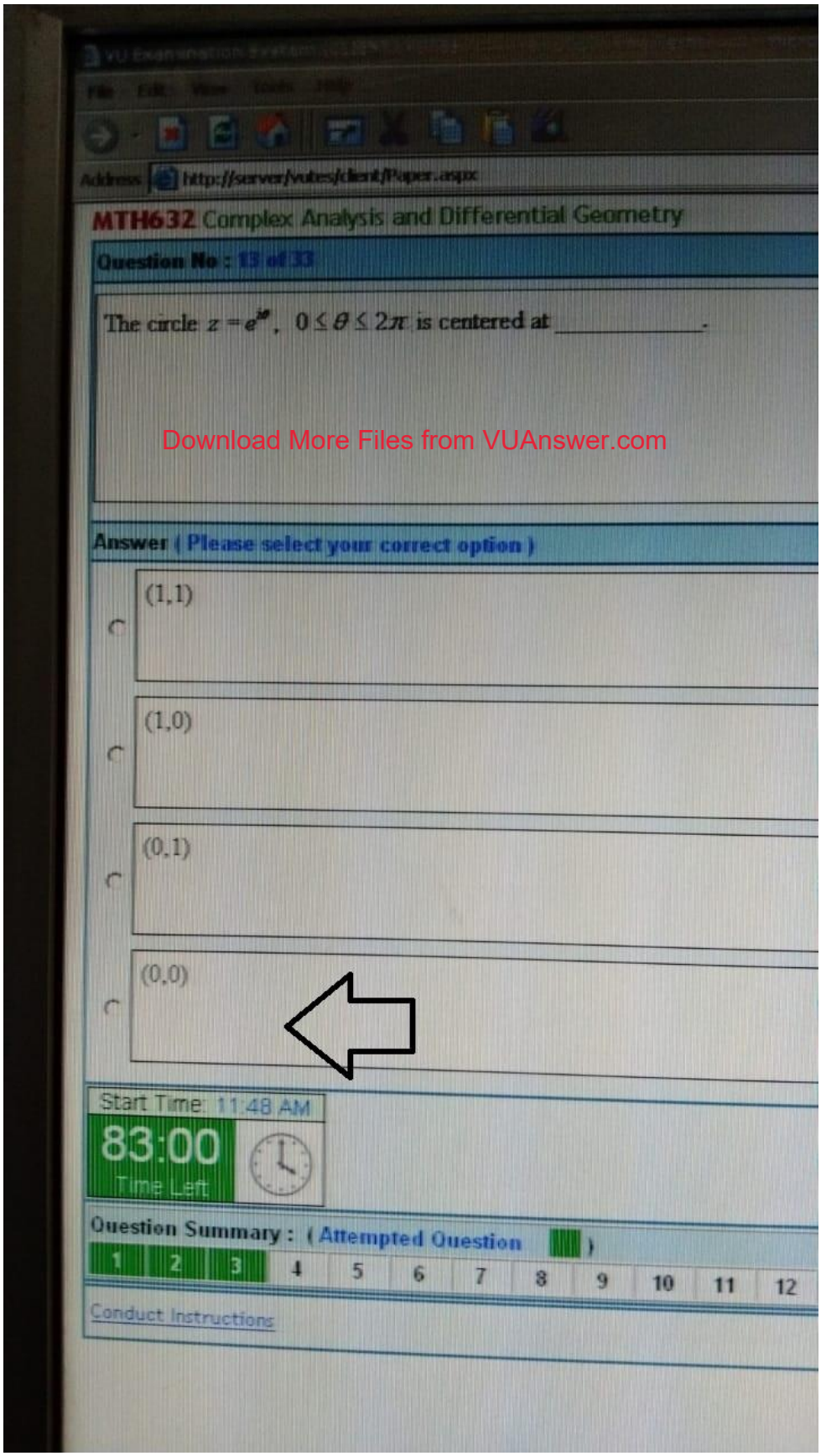
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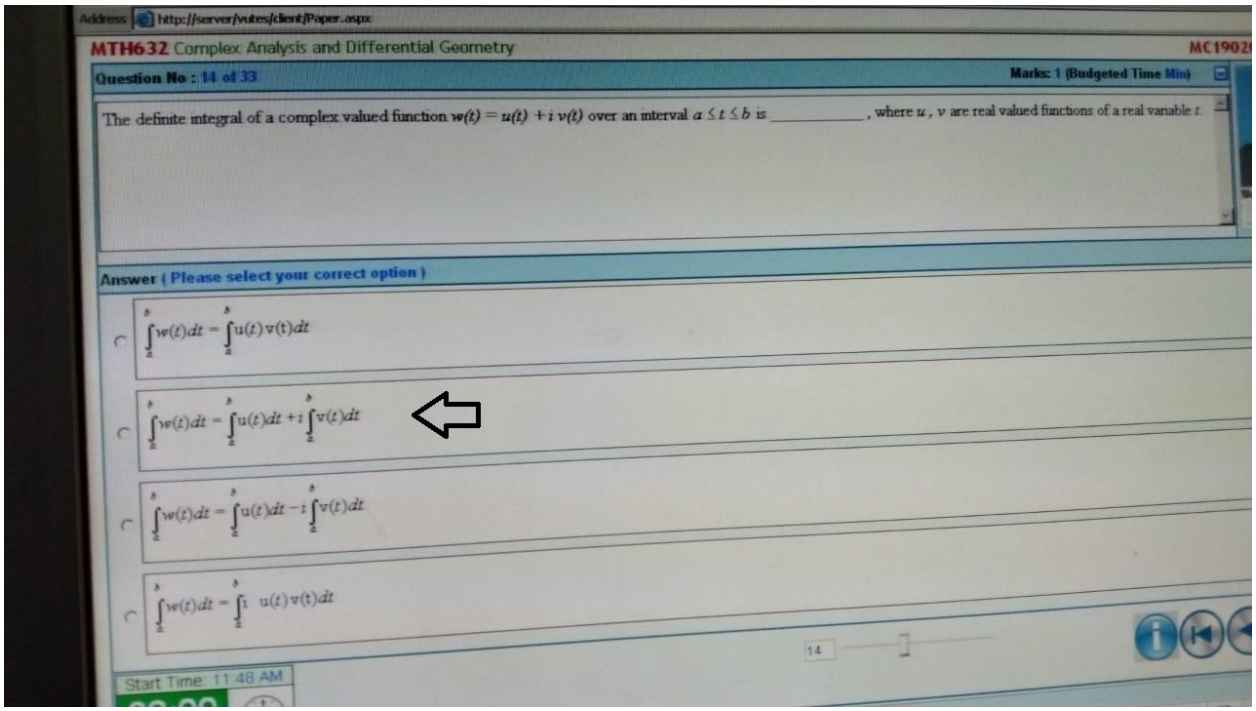
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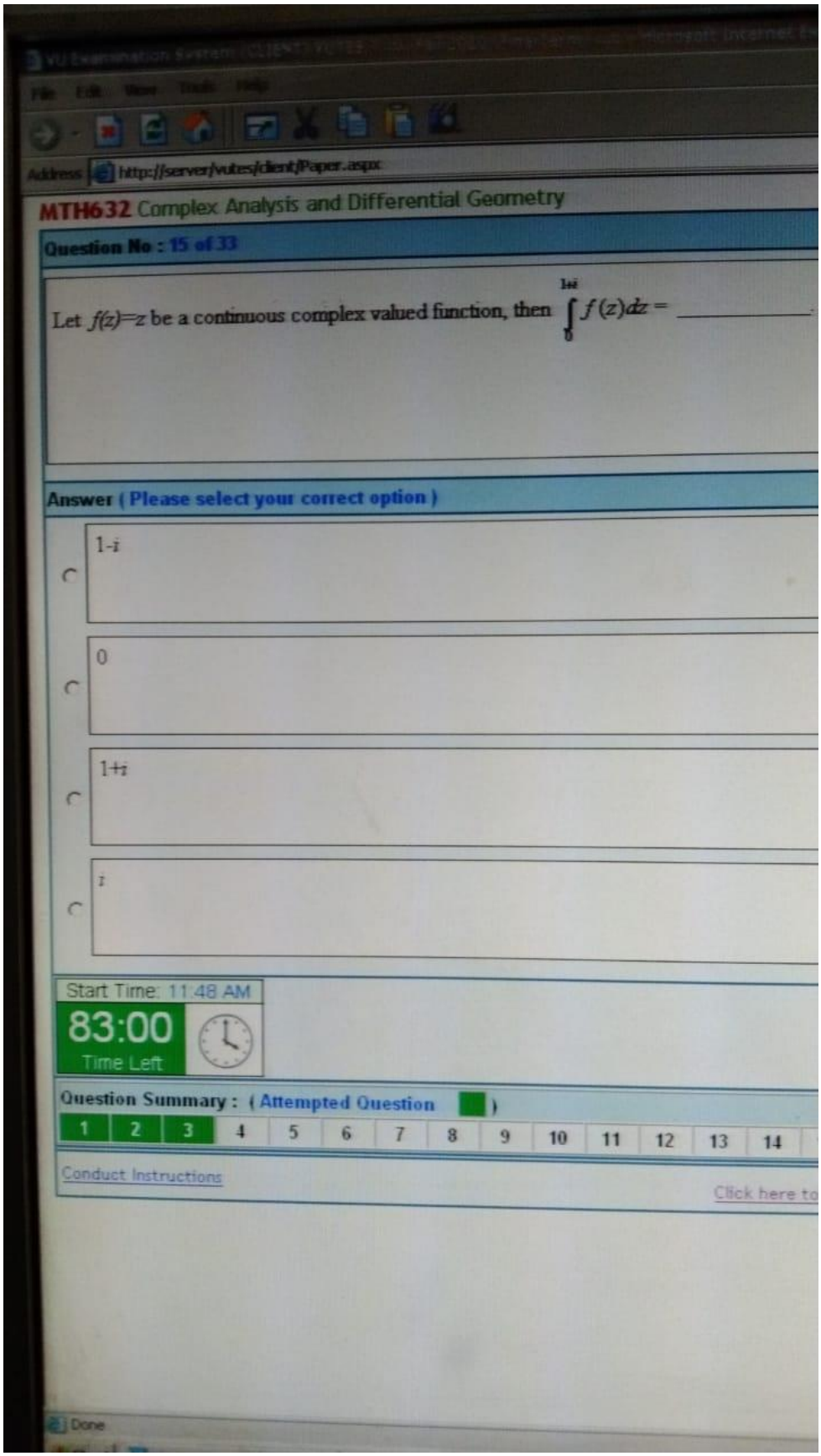
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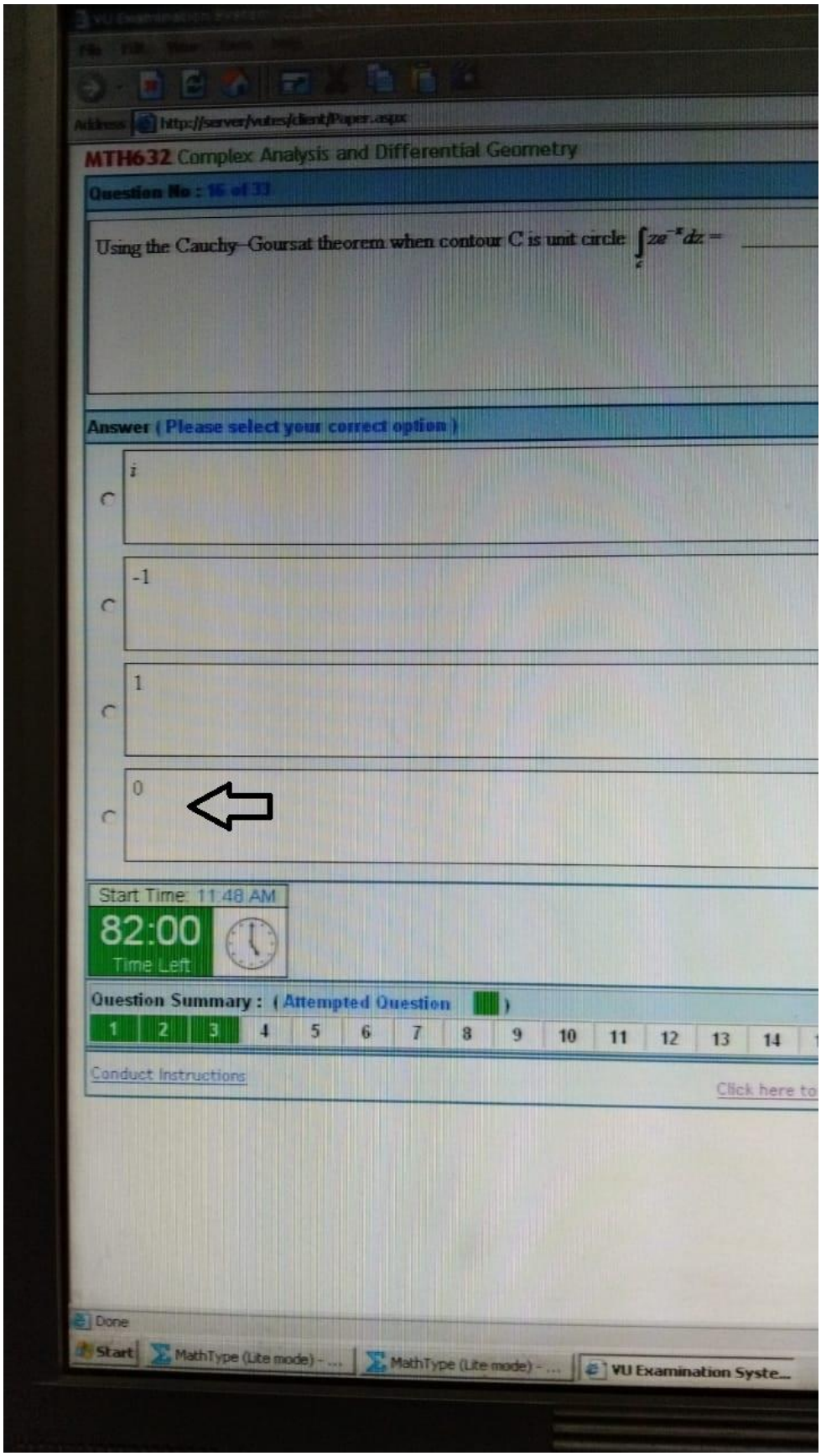

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MTH632 Complex Analysis and Differential Geometry
Question No : 13 of 33
The circle $z = e^{i\theta}$, $0 \le \theta \le 2\pi$ is centered at _____.
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Answer (Please select your correct option)
(1,1)
(1,0)
(0,1)
(0,0)
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Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12
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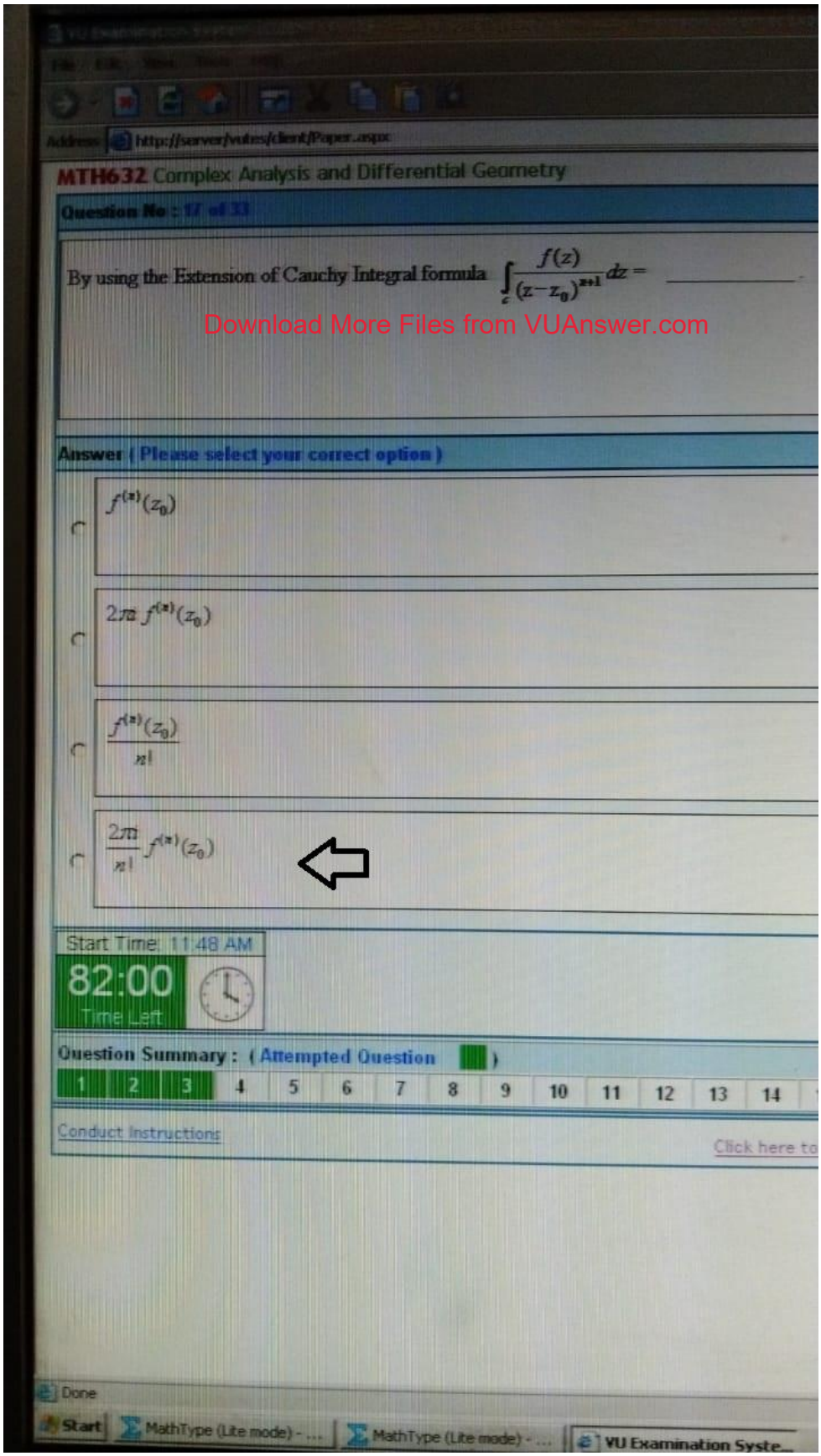


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MTH632 Complex Analysis and Differential Geometry
Question No : 15 of 33
Let $f(z)=z$ be a continuous complex valued function, then $\int_0^{1+i} f(z)dz =$ _____
Answer (Please select your correct option)
1-i
0
1+i
i
Start Time: 11:48 AM
83:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14
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MTH632 Complex Analysis and Differential Geometry
Question No : 17 of 33
By using the Extension of Cauchy Integral formula $\int_c \frac{f(z)}{(z-z_0)^{n+1}} dz =$ _____.
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Answer (Please select your correct option)
$f^{(n)}(z_0)$
$2\pi i\, f^{(n)}(z_0)$
$\frac{f^{(n)}(z_0)}{n!}$
$\frac{2\pi i}{n!} f^{(n)}(z_0)$
Start Time: 11:48 AM
82:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14
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MTH632 Complex Analysis and Differential Geometry

Question No : [illegible]

A triplet $\vec{q} = (q_1, q_2, q_3)$ is called a _____ of Euclidean 3-space E^3.

Answer (Please select your correct option)

line

point

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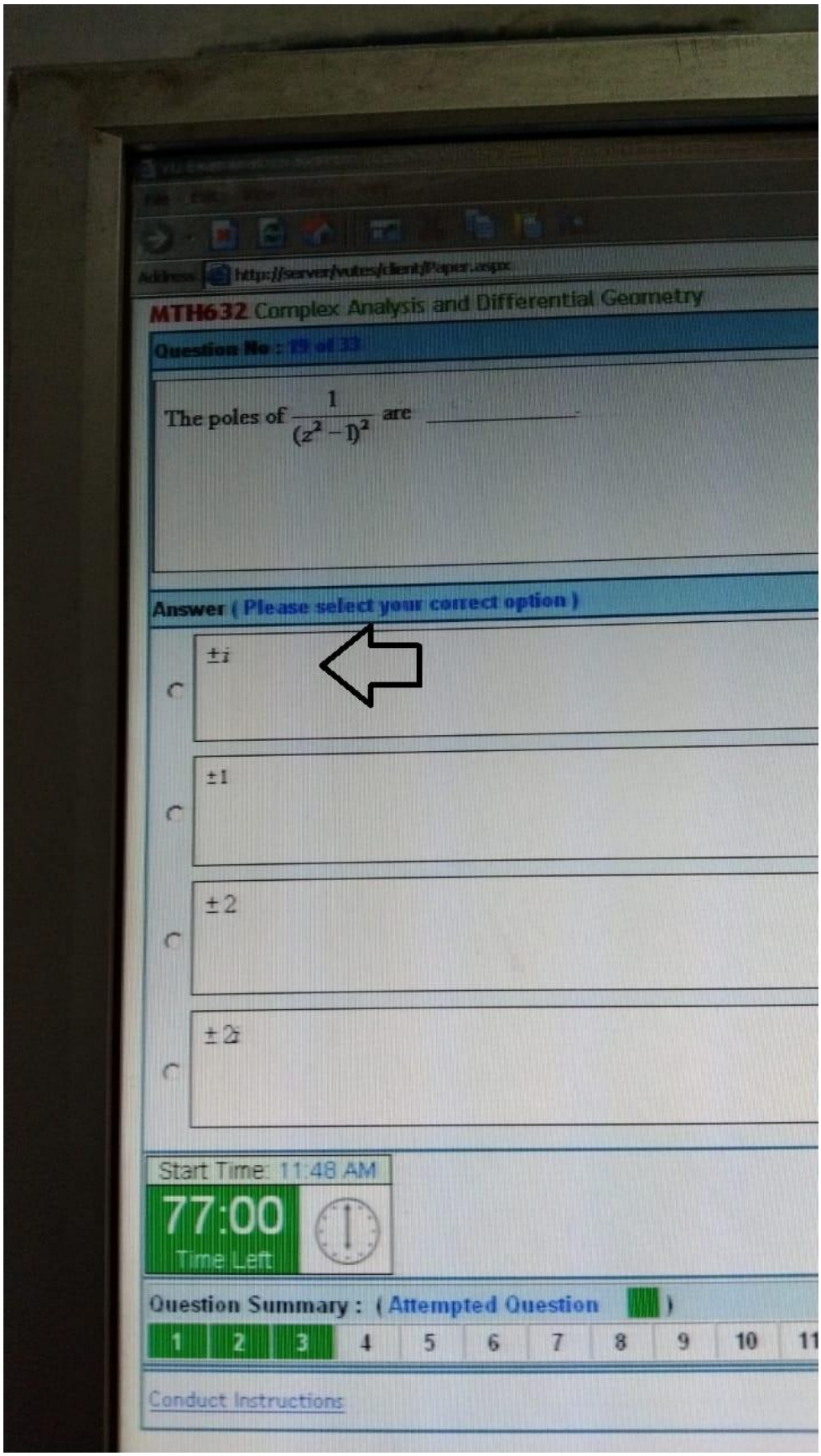
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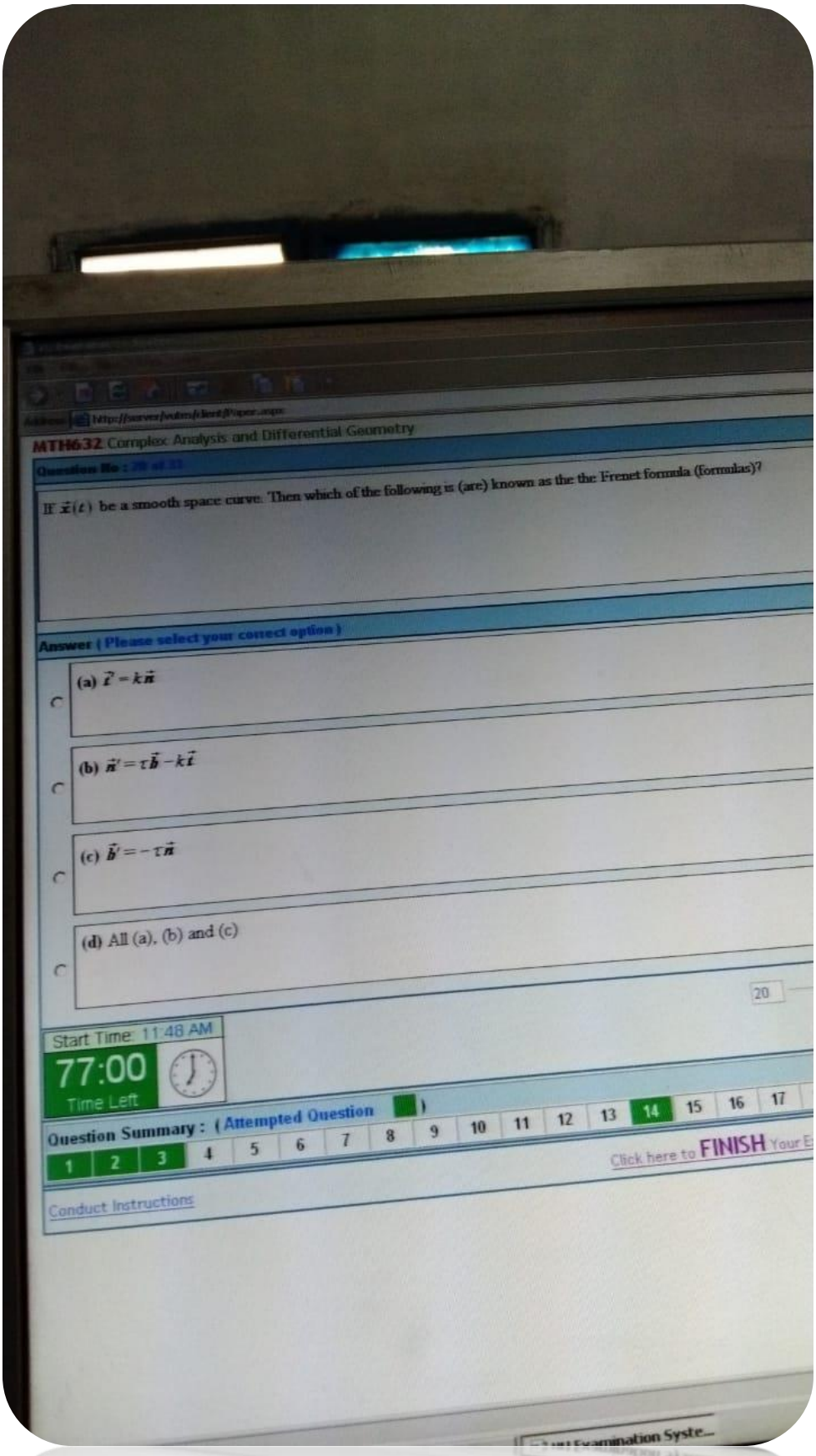
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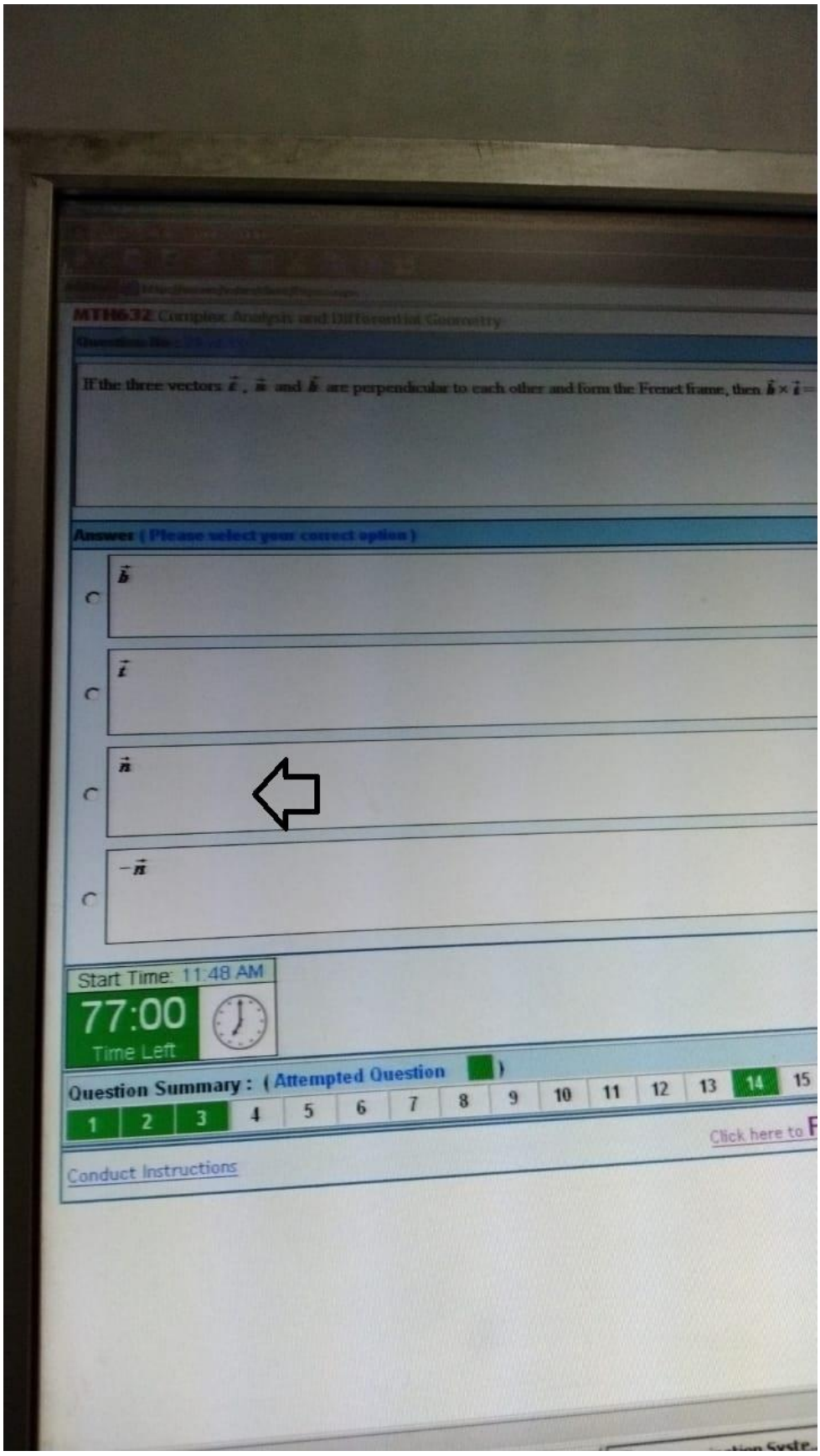
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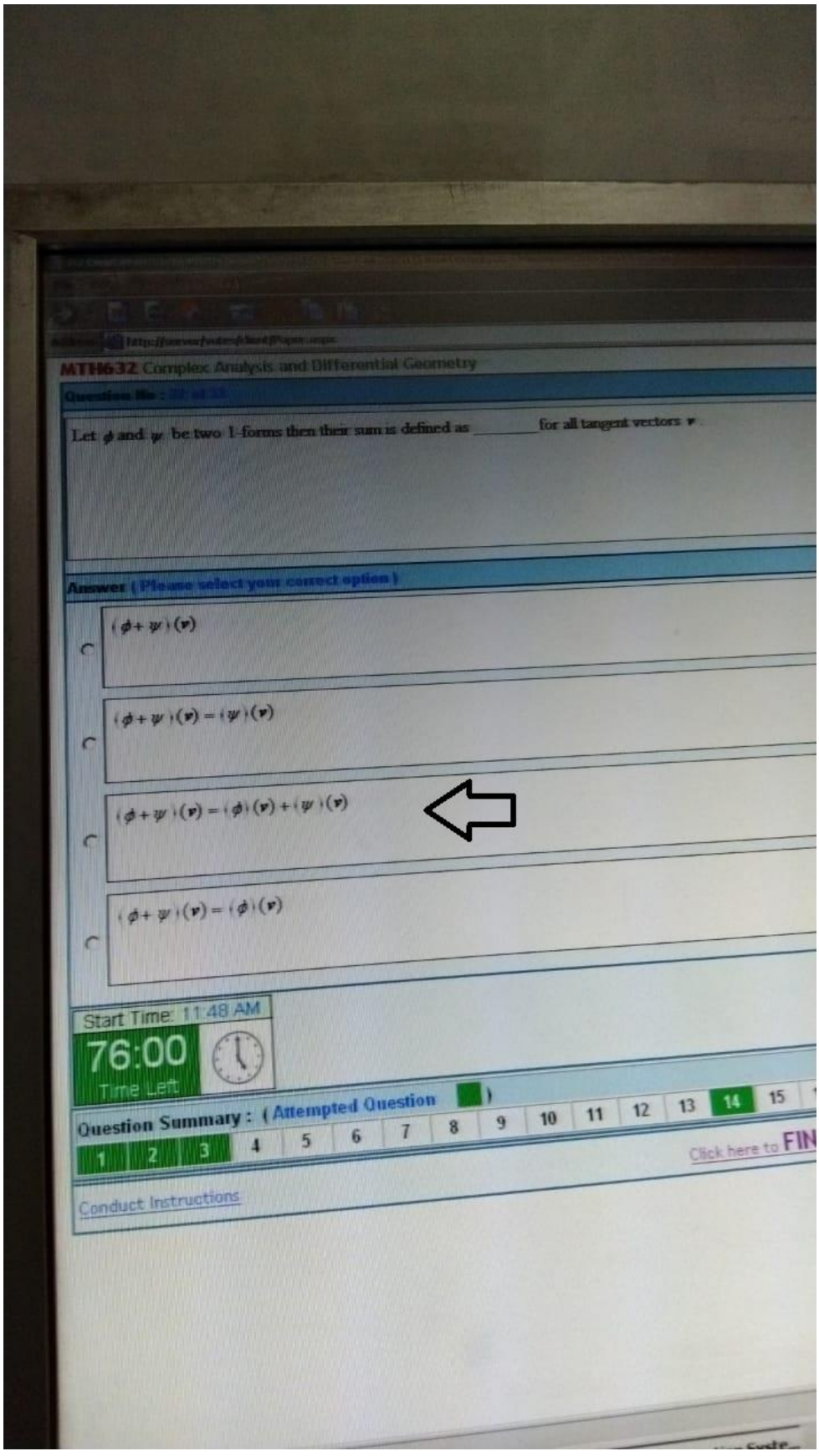
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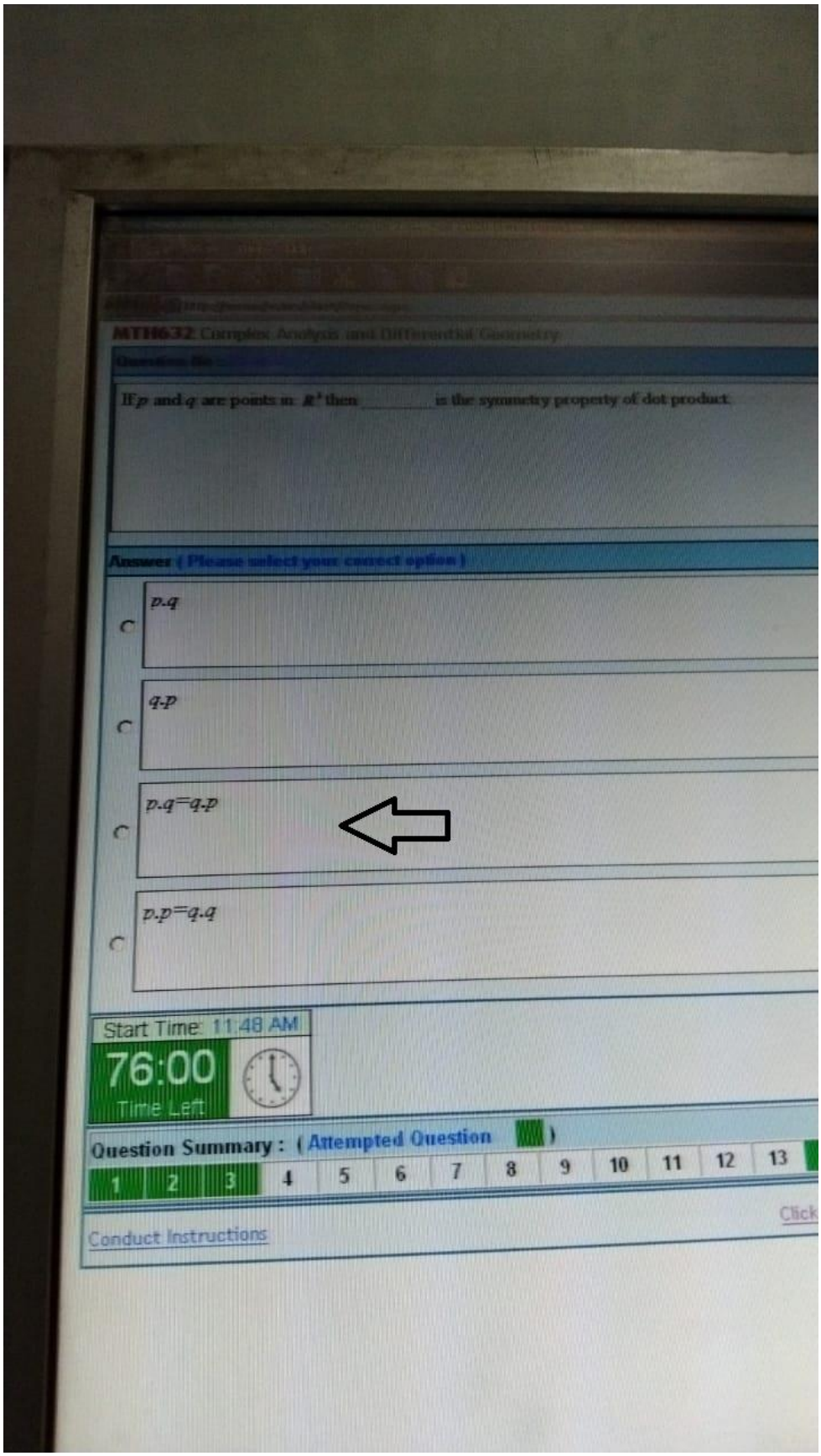
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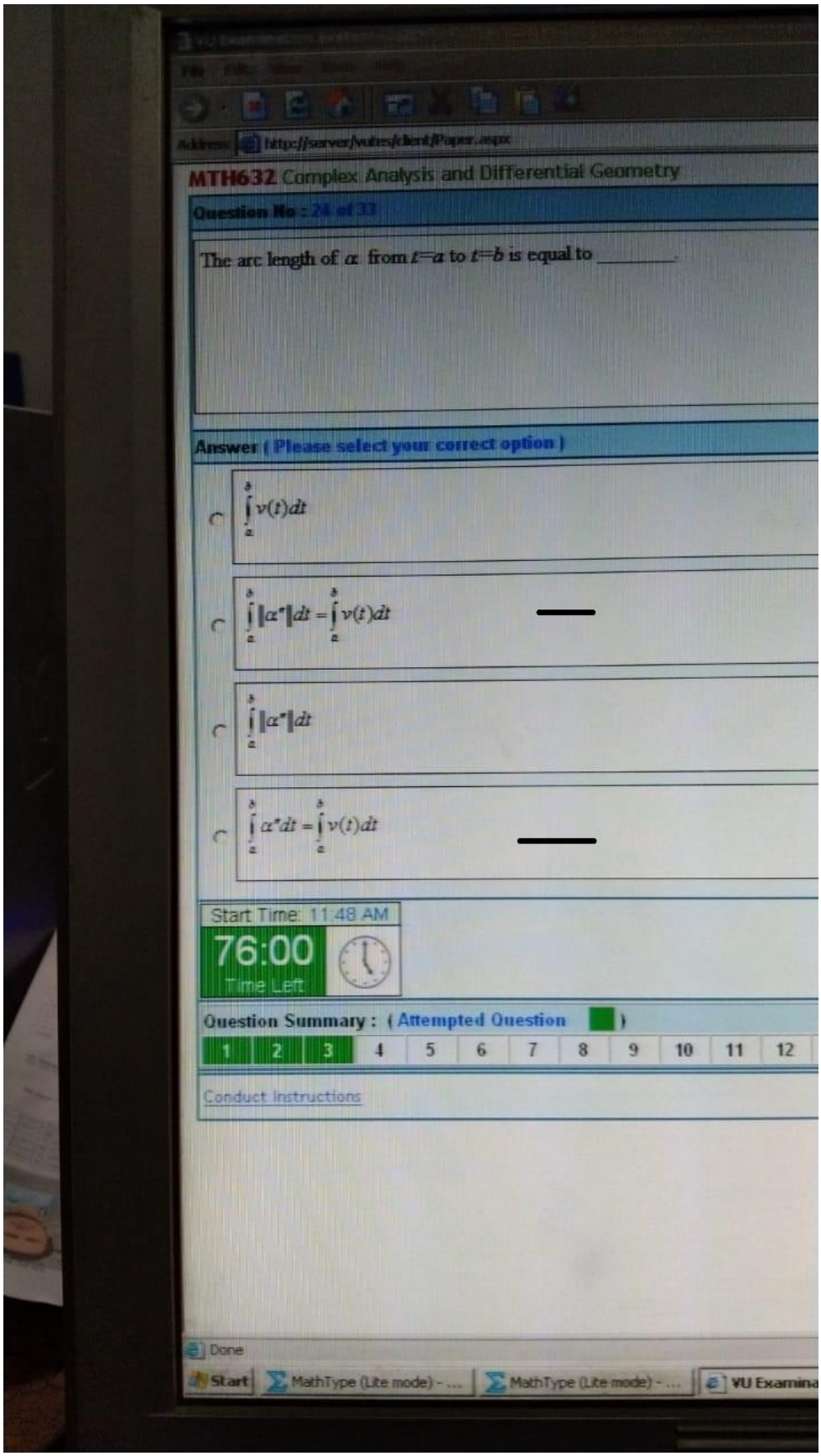

MTH632 Complex Analysis and Differential Geometry
Question No :
The poles of $\frac{1}{(z^2-1)^2}$ are _____
Answer (Please select your correct option)
±i
±1
±2
±2i
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Question Summary : (Attempted Question)
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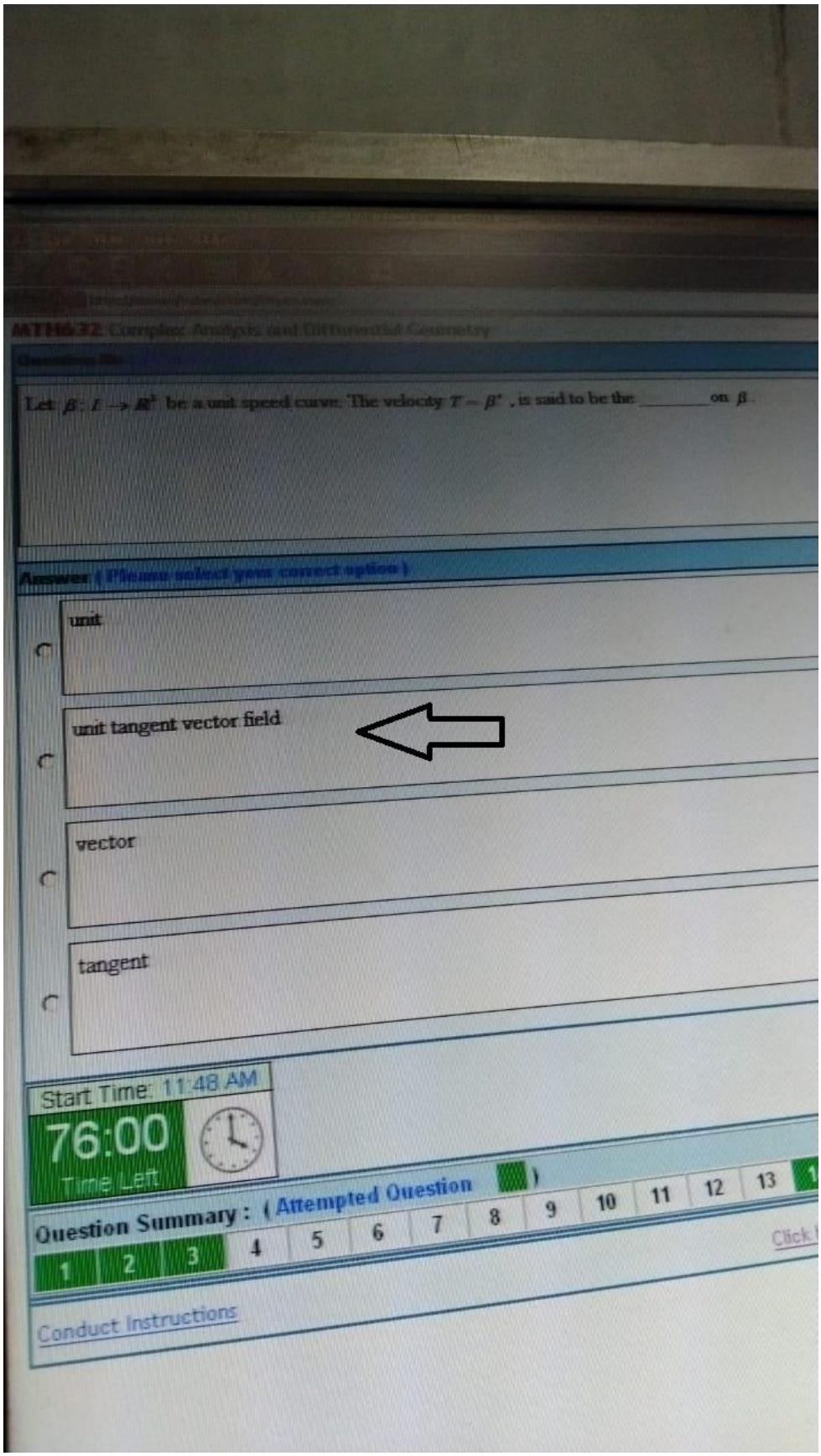

MTH632 Complex Analysis and Differential Geometry
If $\vec{x}(t)$ be a smooth space curve. Then which of the following is (are) known as the the Frenet formula (formulas)?
Answer (Please select your correct option)
(a) $\vec{t}' = k\vec{n}$
(b) $\vec{n}' = \tau\vec{b} - k\vec{t}$
(c) $\vec{b}' = -\tau\vec{n}$
(d) All (a), (b) and (c)
20
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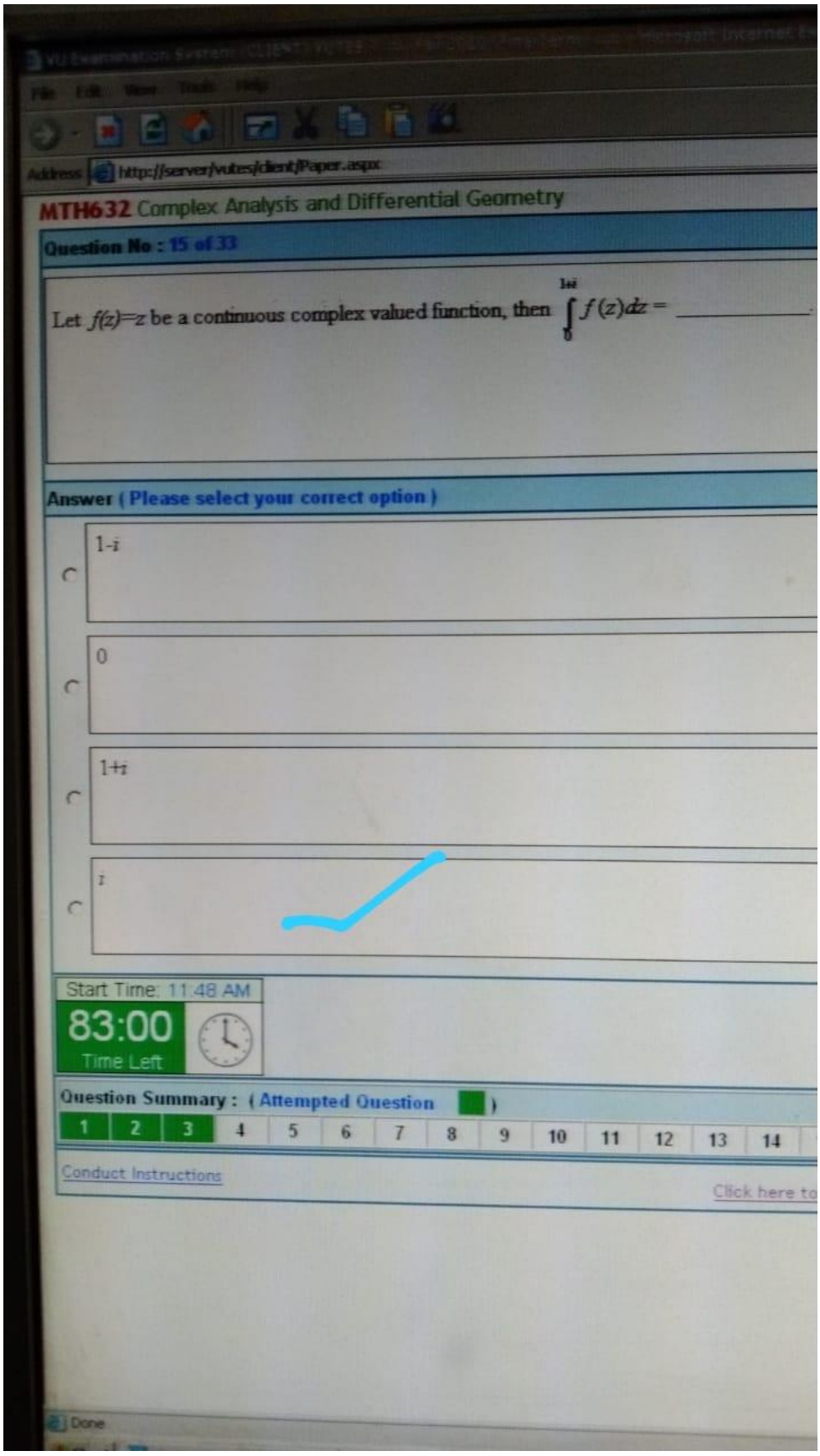

MTH632 Complex Analysis and Differential Geometry
If the three vectors $\vec{t}$, $\vec{n}$ and $\vec{b}$ are perpendicular to each other and form the Frenet frame, then $\vec{b} \times \vec{t} =$
Answer (Please select your correct option)
$\vec{b}$
$\vec{t}$
$\vec{n}$
$-\vec{n}$
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Question Summary : (Attempted Question)
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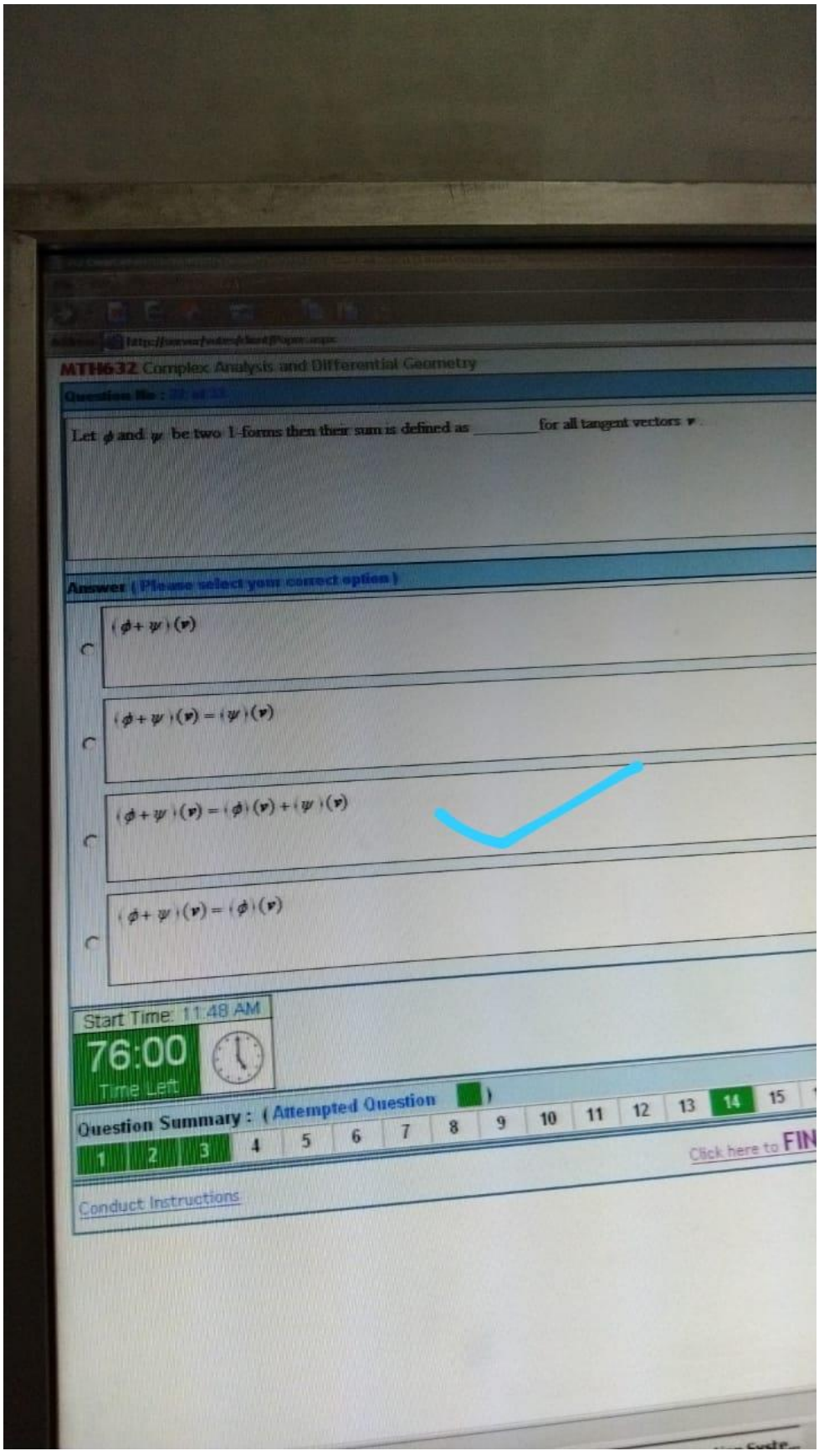

MTH632 Complex Analysis and Differential Geometry
Let ϕ and ψ be two 1-forms then their sum is defined as _____ for all tangent vectors v.
Answer (Please select your correct option)
$(\phi+\psi)(v)$
$(\phi+\psi)(v)=(\psi)(v)$
$(\phi+\psi)(v)=(\phi)(v)+(\psi)(v)$
$(\phi+\psi)(v)=(\phi)(v)$
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Question Summary : (Attempted Question)
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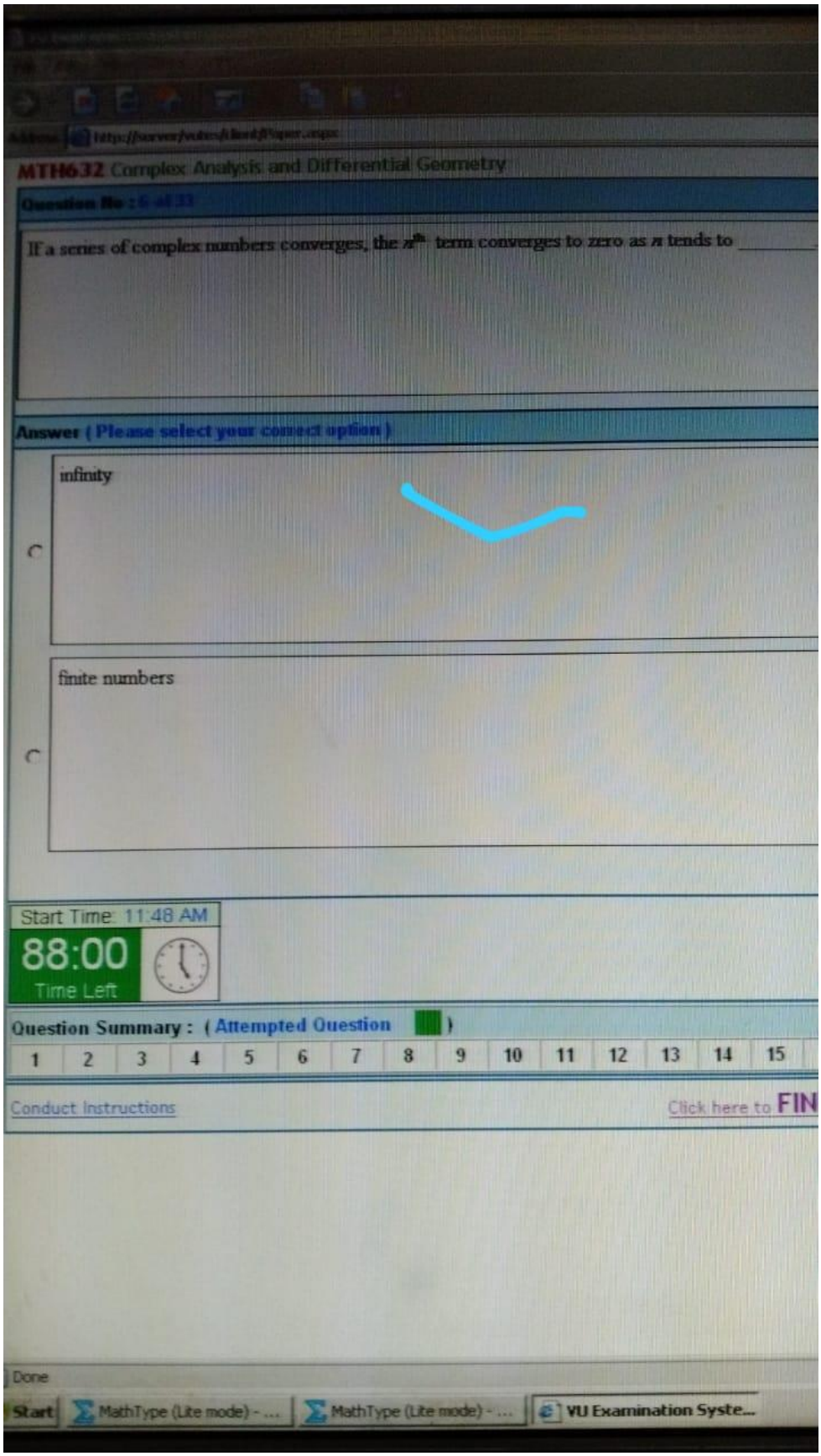

MTH632: Complex Analysis and Differential Geometry
If p and q are points in R^n then _____ is the symmetry property of dot product.
Answer (Please select your correct option)
p.q
q.p
p.q=q.p
p.p=q.q
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Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13
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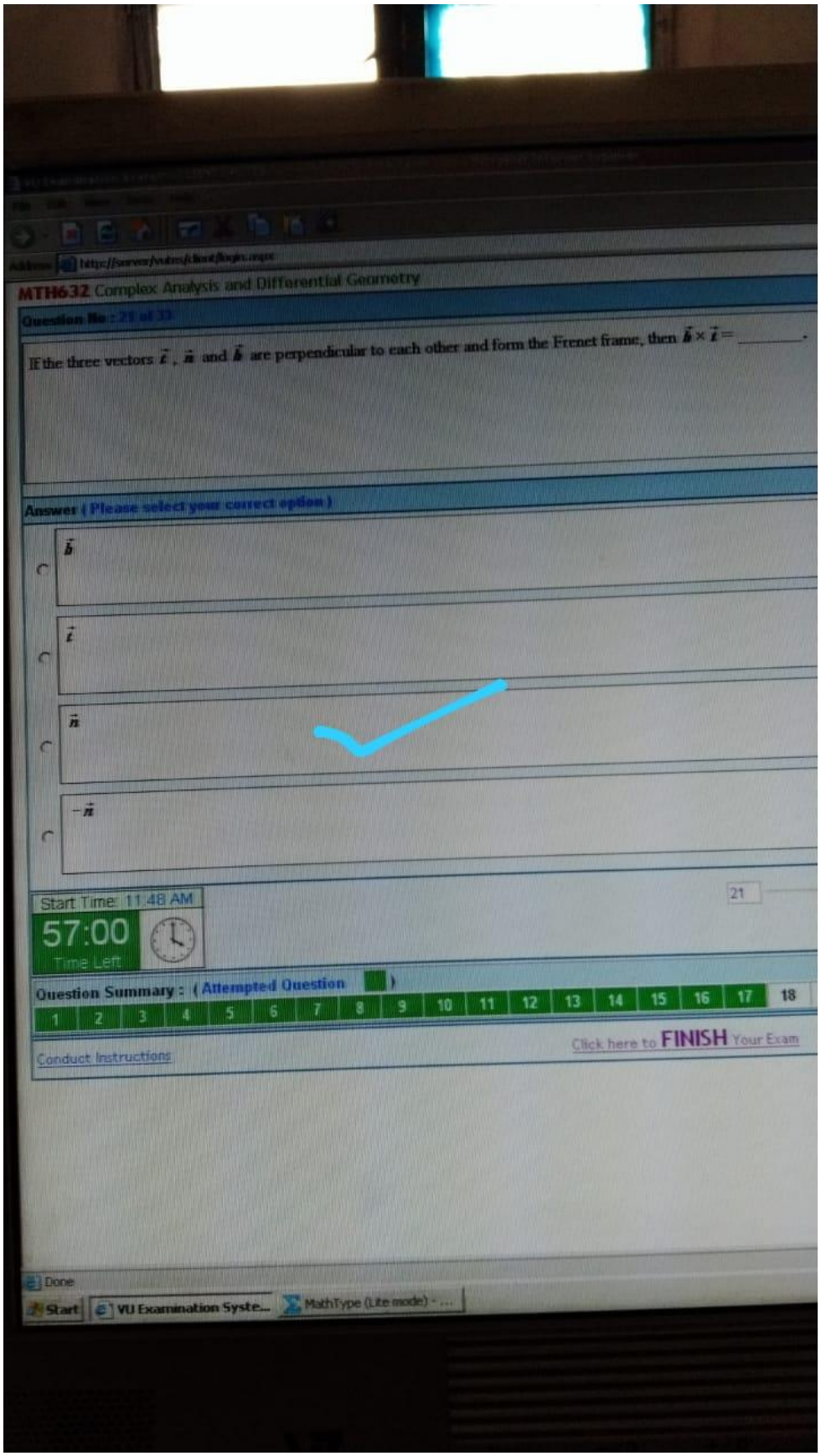

MTH632 Complex Analysis and Differential Geometry
Question No : 24 of 33
The arc length of α from $t=a$ to $t=b$ is equal to _____.
Answer (Please select your correct option)
$\int_a^b v(t)dt$
$\int_a^b \|\alpha'\|dt = \int_a^b v(t)dt$
$\int_a^b \|\alpha'\|dt$
$\int_a^b \alpha' dt = \int_a^b v(t)dt$
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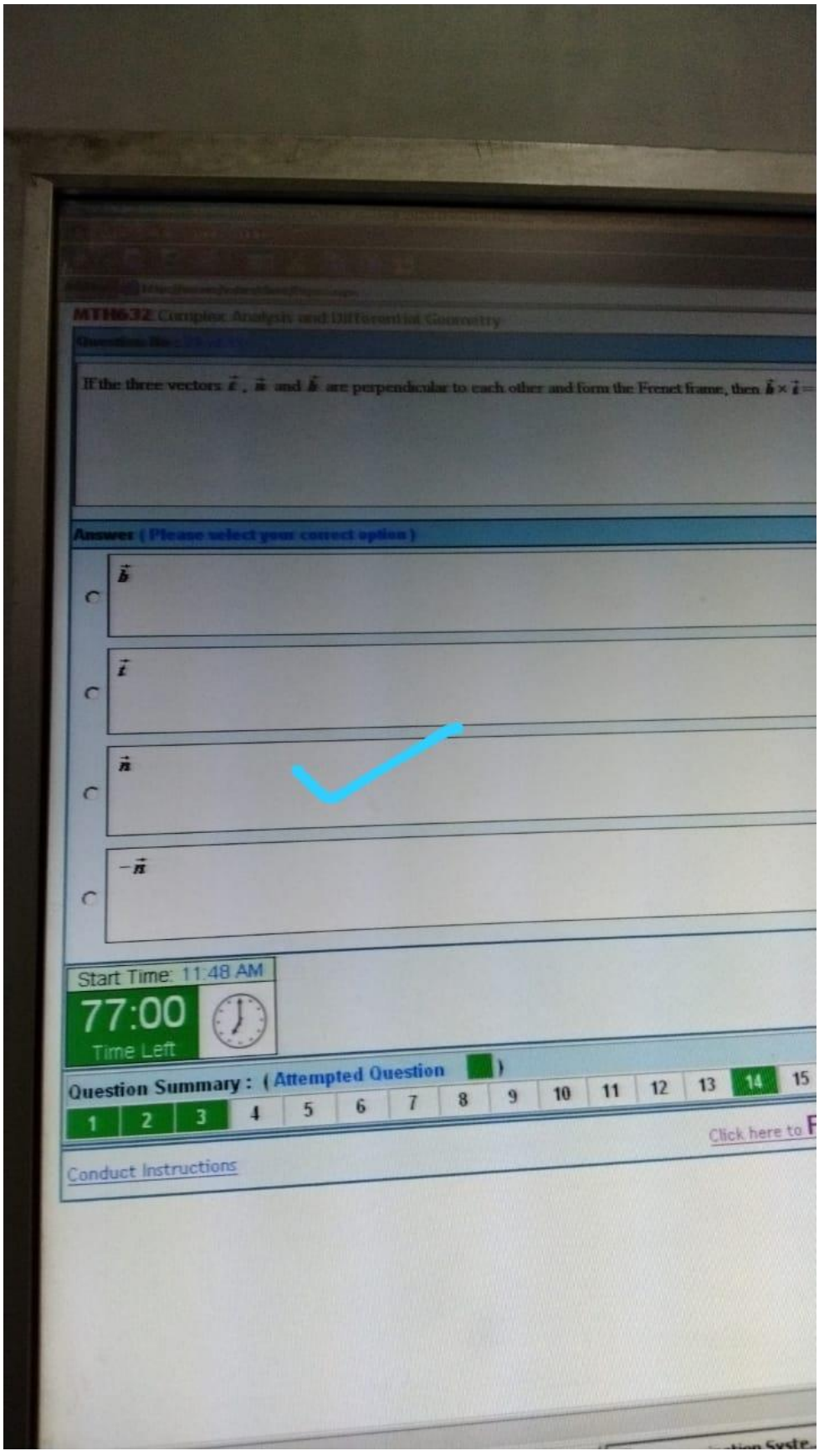

MTH632 Complex Analysis and Differential Geometry
Let $\beta : I \to \mathbb{R}^3$ be a unit speed curve. The velocity $T = \beta'$, is said to be the _____ on β.
Answer (Please select your correct option)
unit
unit tangent vector field
vector
tangent
Start Time: 11:48 AM
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Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13
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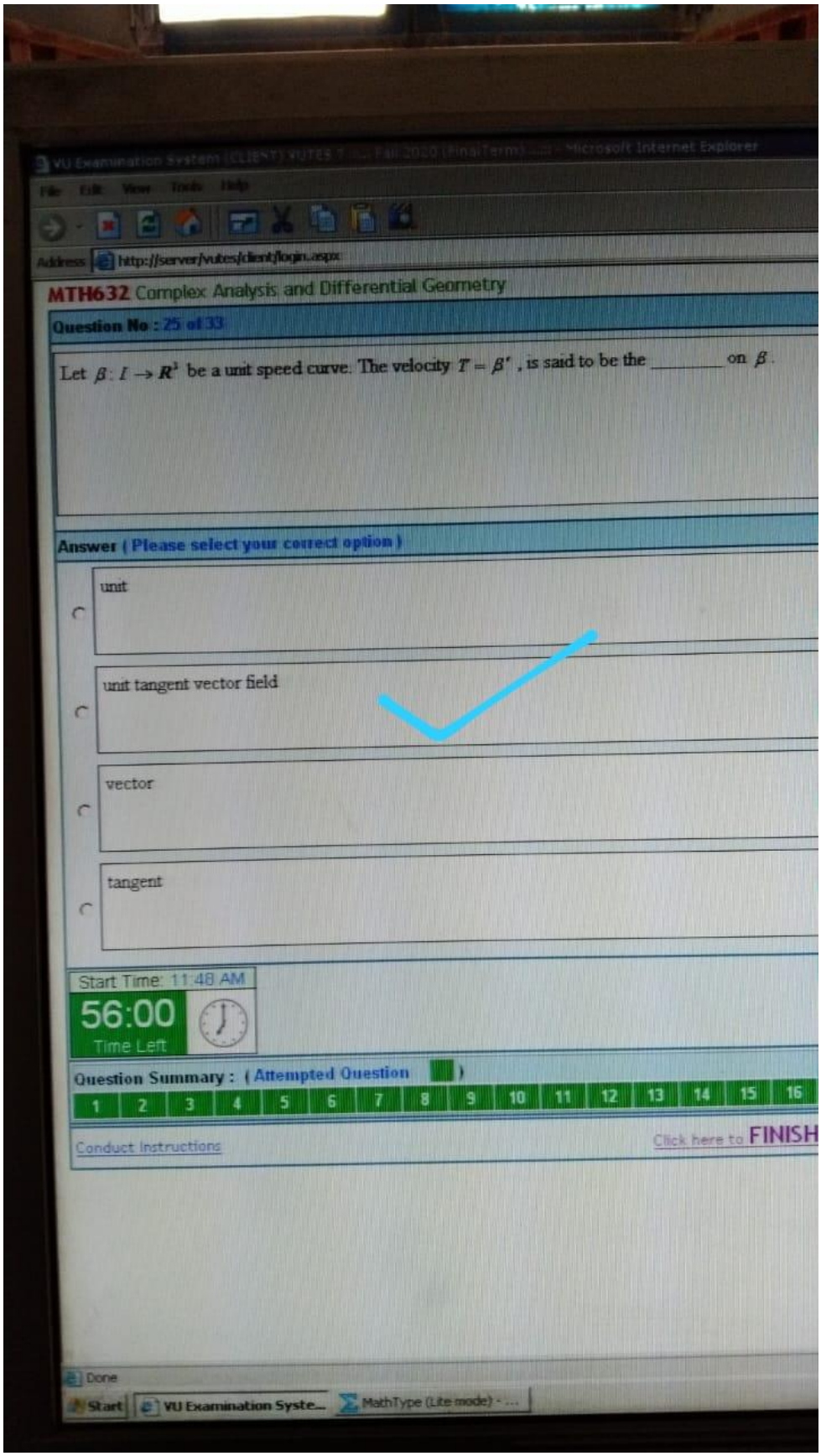

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MTH632 Complex Analysis and Differential Geometry
Question No : 15 of 33
Let f(z)=z be a continuous complex valued function, then $\int_0^{1+i} f(z)dz =$ _____
Answer (Please select your correct option)
1-i
0
1+i
i
Start Time: 11:48 AM
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Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14
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MTH632 Complex Analysis and Differential Geometry
Let ϕ and ψ be two 1-forms then their sum is defined as _____ for all tangent vectors v.
Answer (Please select your correct option)
$(\phi+\psi)(v)$
$(\phi+\psi)(v)=(\psi)(v)$
$(\phi+\psi)(v)=(\phi)(v)+(\psi)(v)$
$(\phi+\psi)(v)=(\phi)(v)$
Start Time: 11:48 AM
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Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
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MTH632 Complex Analysis and Differential Geometry
If a series of complex numbers converges, the nth term converges to zero as n tends to _____
Answer (Please select your correct option)
infinity
finite numbers
Start Time: 11:48 AM
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Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
Conduct Instructions
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MTH632 Complex Analysis and Differential Geometry
If the three vectors $\vec{t}$, $\vec{n}$ and $\vec{b}$ are perpendicular to each other and form the Frenet frame, then $\vec{b} \times \vec{t} =$ _____.
Answer (Please select your correct option)
$\vec{b}$
$\vec{t}$
$\vec{n}$
$-\vec{n}$
Start Time: 11:48 AM
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21
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18
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MTH632 Complex Analysis and Differential Geometry
If the three vectors $\vec{t}$, $\vec{n}$ and $\vec{b}$ are perpendicular to each other and form the Frenet frame, then $\vec{b} \times \vec{t} =$
Answer (Please select your correct option)
$\vec{b}$
$\vec{t}$
$\vec{n}$
$-\vec{n}$
Start Time: 11:48 AM
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Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
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MTH632 Complex Analysis and Differential Geometry
Question No : 25 of 33
Let $\beta : I \to R^3$ be a unit speed curve. The velocity $T = \beta'$, is said to be the _____ on β.
Answer (Please select your correct option)
unit
unit tangent vector field
vector
tangent
Start Time: 11:48 AM
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Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16
Conduct Instructions
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VU Examination Syste...
MathType (Lite mode) - ...

MTH632 Complex Analysis and Differential Geometry

Question No : 11 of 20

The principal value of 1^{2i} is _____.

Answer (Please select your correct option)

- 0
- 1
- i
- 2i

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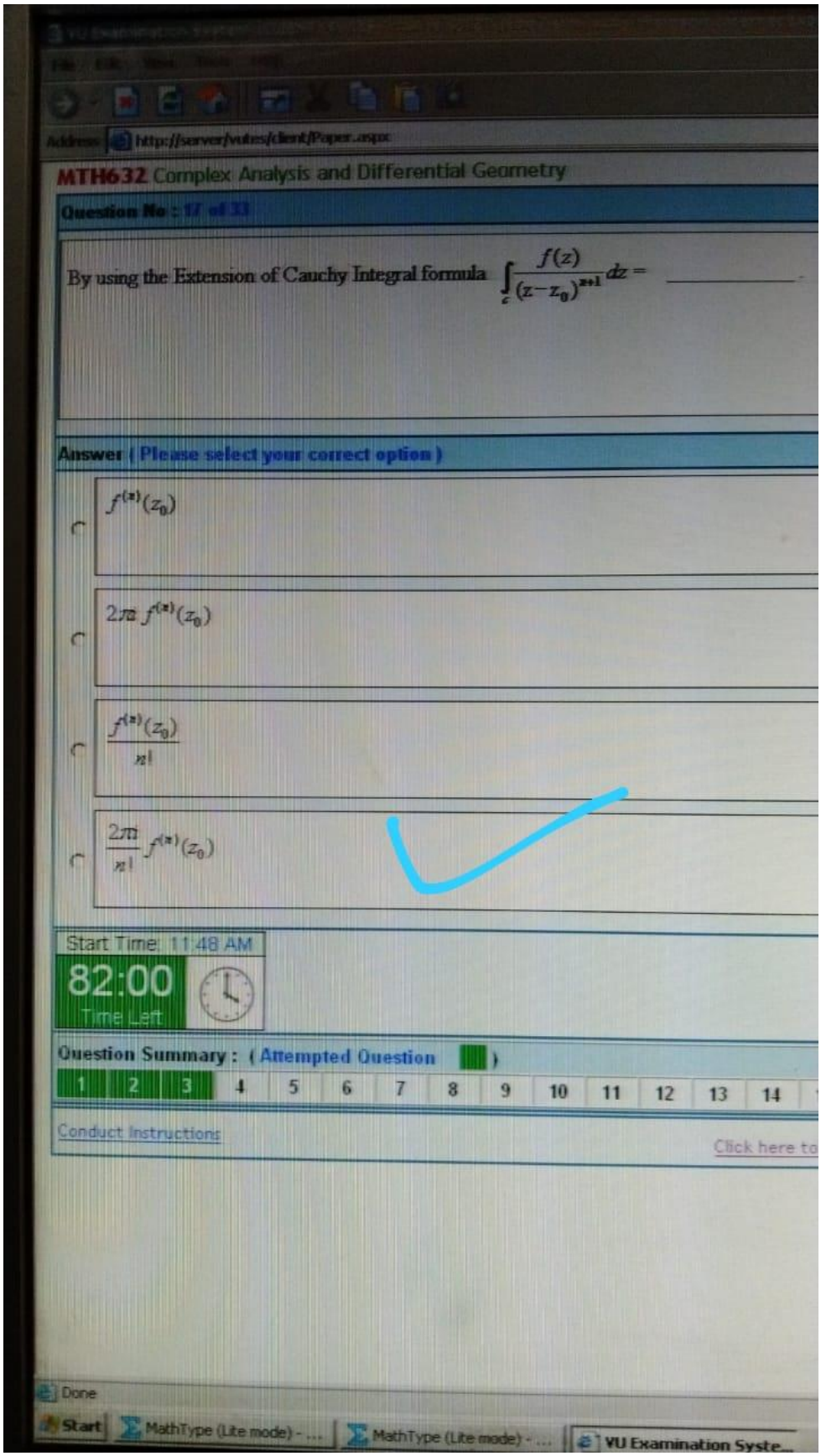
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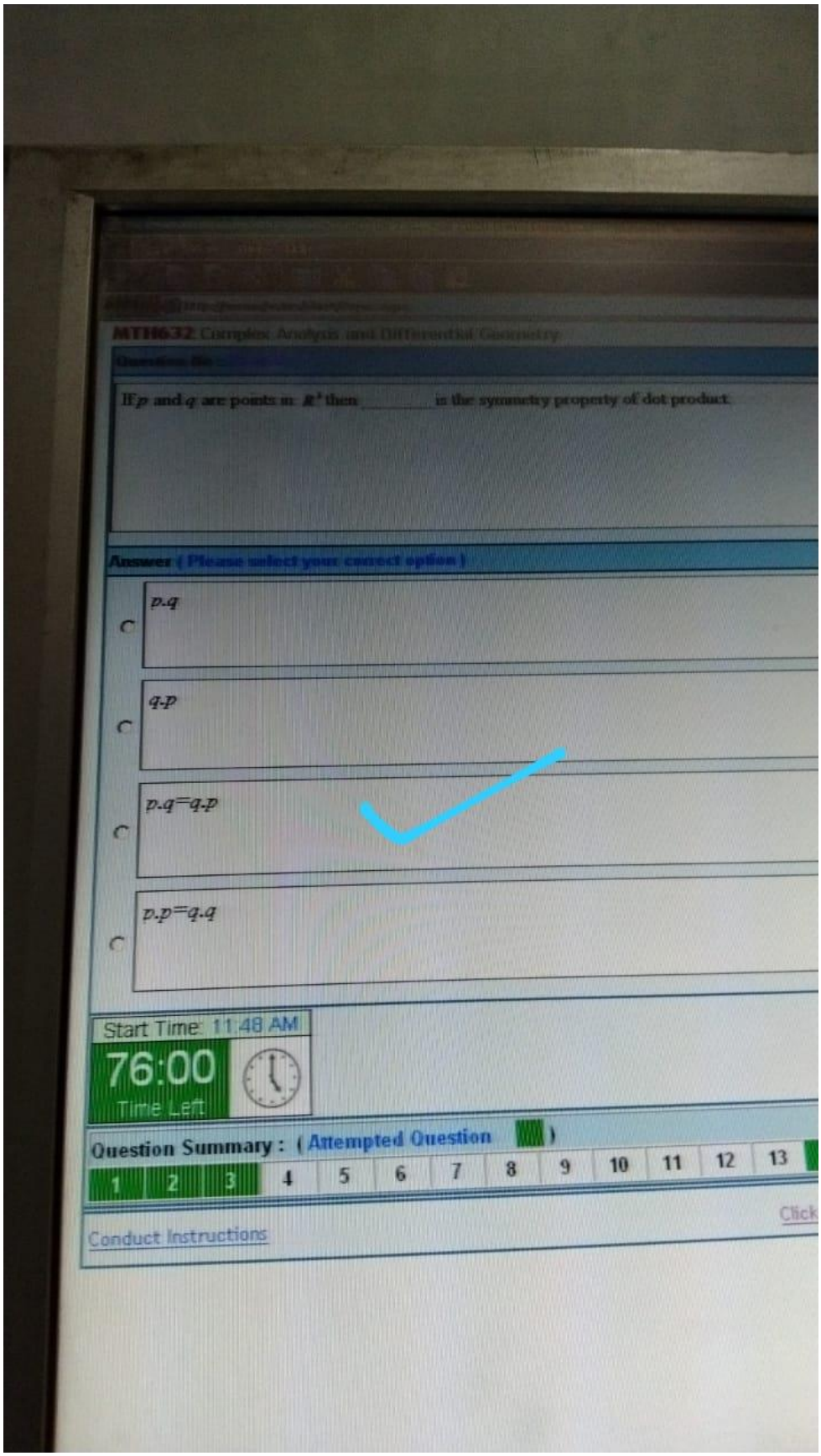
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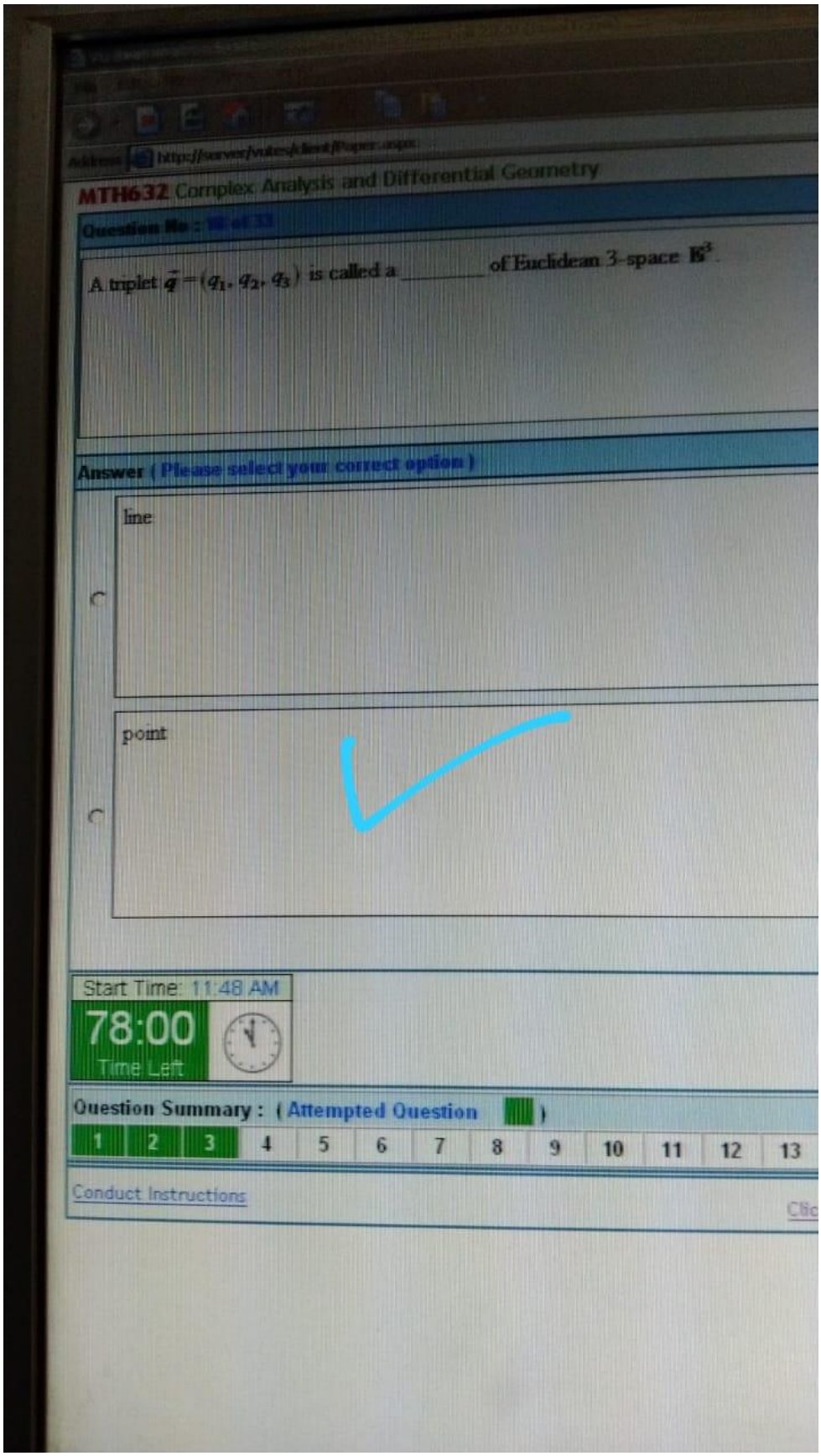
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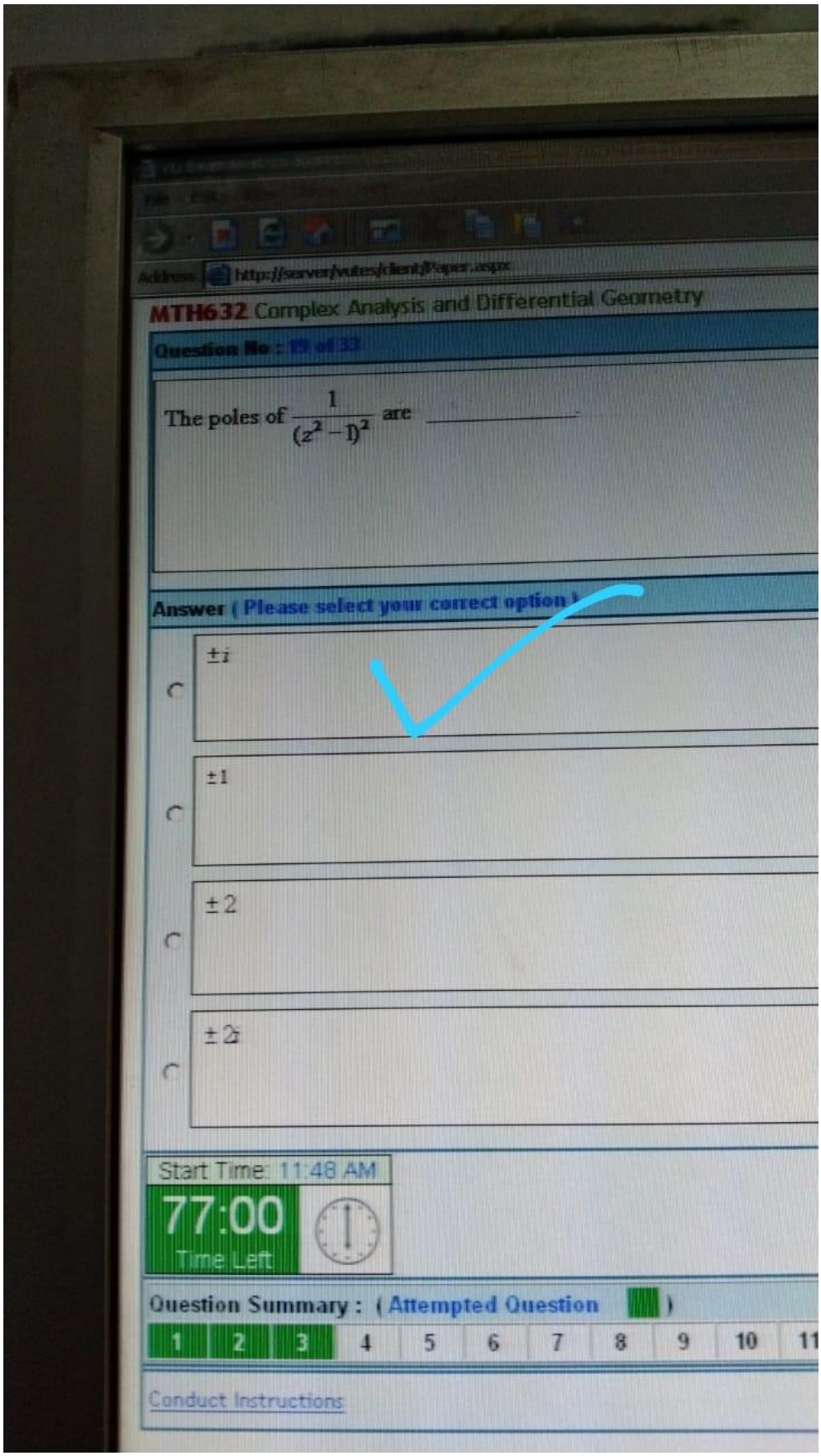
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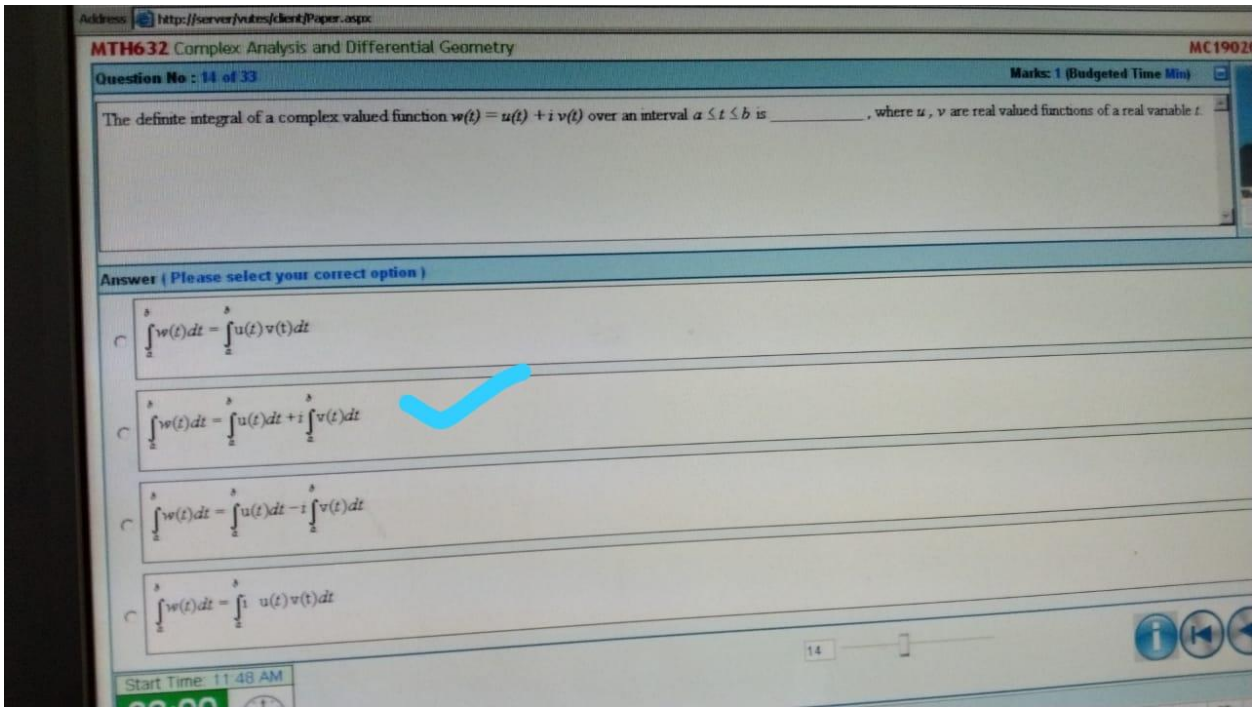
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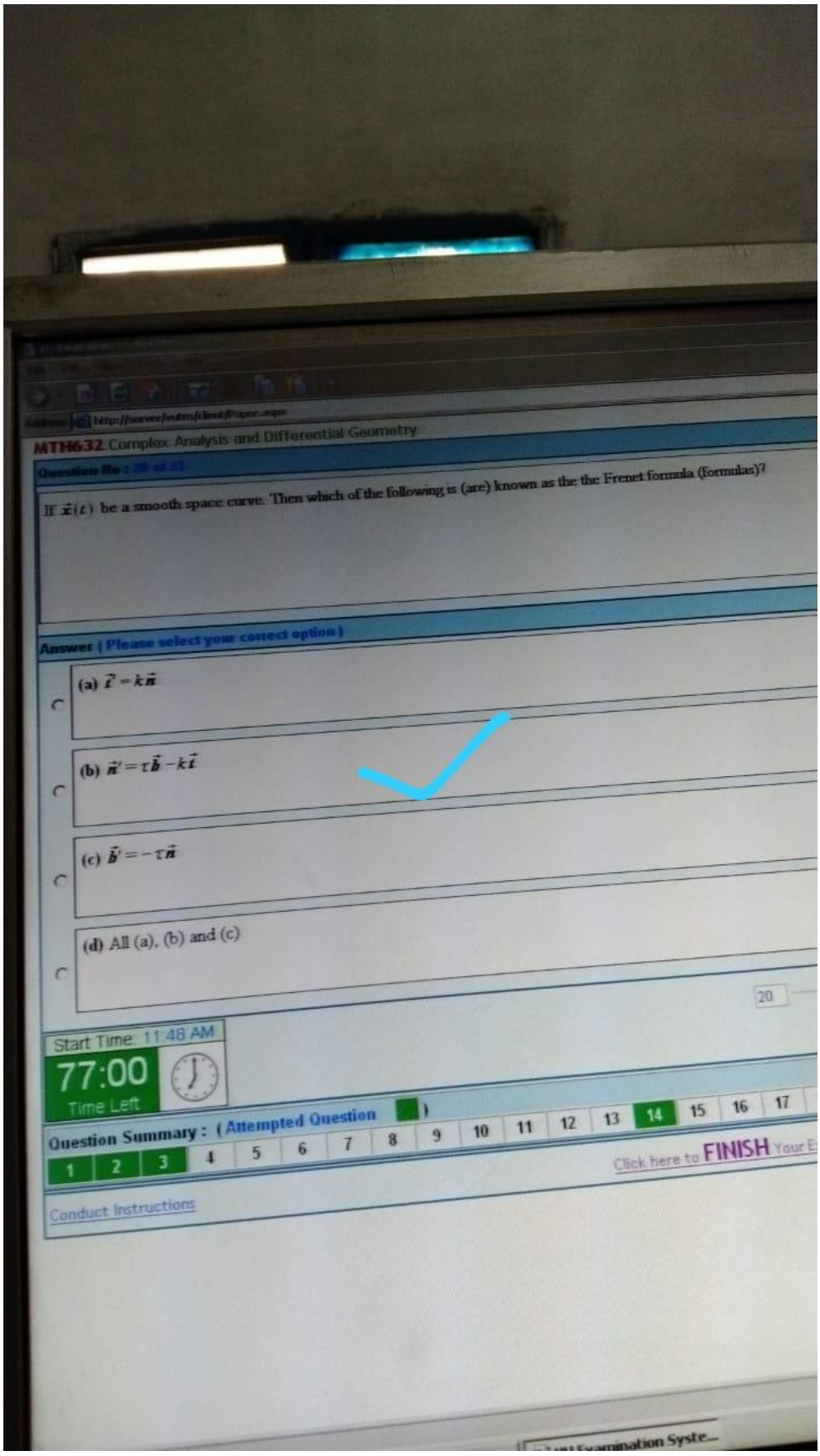

MTH632 Complex Analysis and Differential Geometry
Question No : 17 of 33
By using the Extension of Cauchy Integral formula $\int_c \frac{f(z)}{(z-z_0)^{n+1}} dz =$ _____.
Answer (Please select your correct option)
$f^{(n)}(z_0)$
$2\pi i\, f^{(n)}(z_0)$
$\frac{f^{(n)}(z_0)}{n!}$
$\frac{2\pi i}{n!} f^{(n)}(z_0)$
Start Time: 11:48 AM
82:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14
Conduct Instructions
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Start
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MathType (Lite mode) - ...
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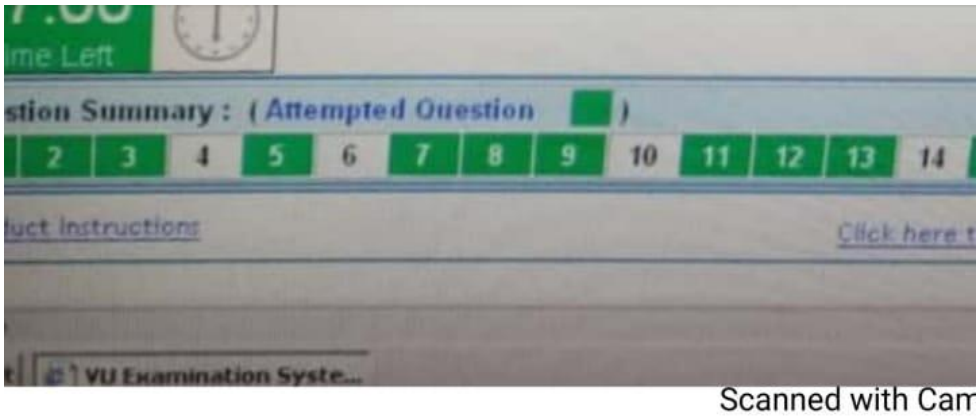

MTH632 Complex Analysis and Differential Geometry
If p and q are points in R³ then _____ is the symmetry property of dot product
Answer (Please select your correct option)
p.q
q.p
p.q=q.p
p.p=q.q
Start Time: 11:48 AM
76:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13
Conduct Instructions


MTH632 Complex Analysis and Differential Geometry
A triplet $\vec{q} = (q_1, q_2, q_3)$ is called a _____ of Euclidean 3-space E^3.
Answer (Please select your correct option)
line
point
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78:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13
Conduct Instructions

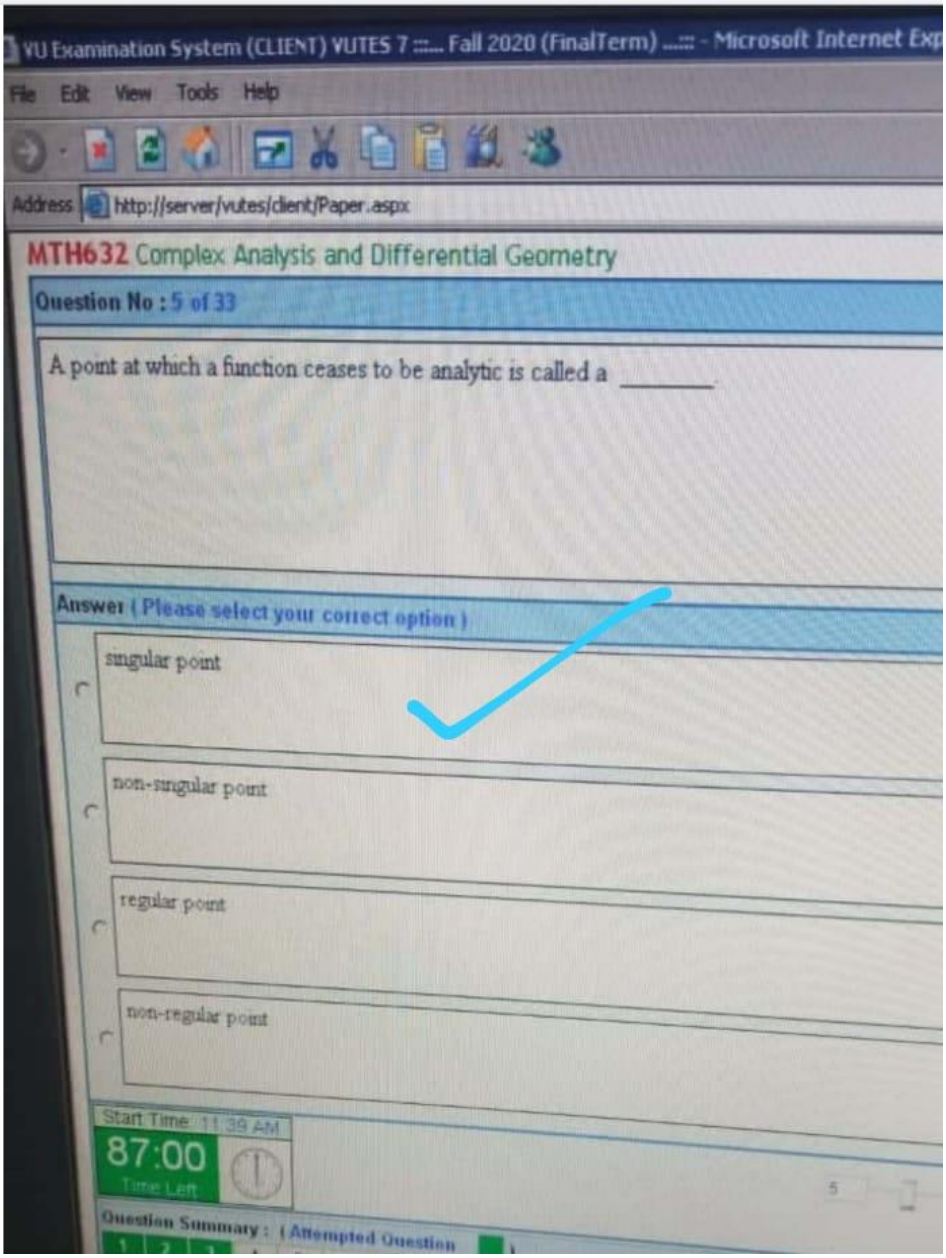

MTH632 Complex Analysis and Differential Geometry
Question No :
The poles of $\frac{1}{(z^2-1)^2}$ are _____.
Answer (Please select your correct option)
±i
±1
±2
±2i
Start Time: 11:48 AM
77:00
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Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11
Conduct Instructions

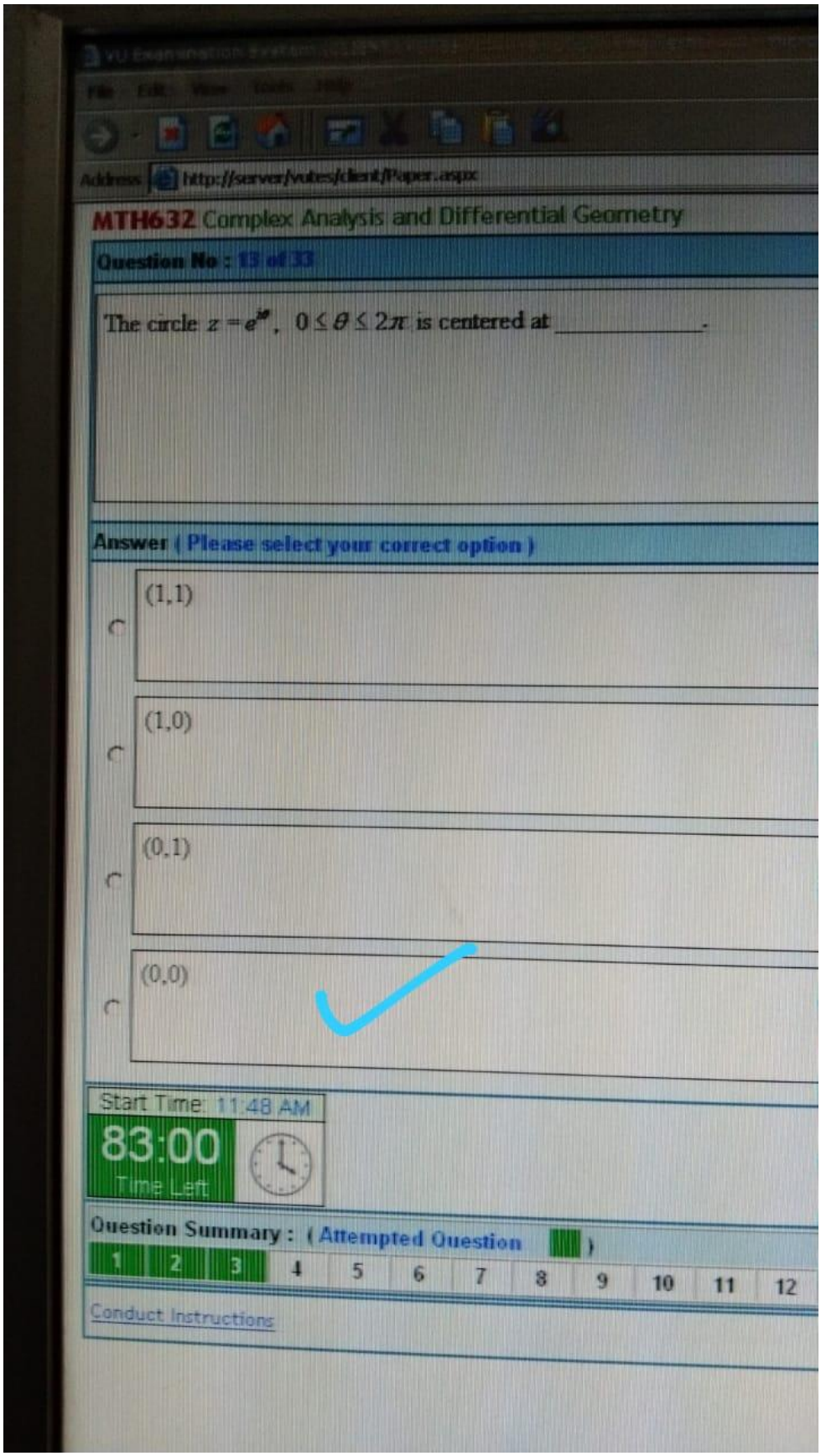

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MTH632 Complex Analysis and Differential Geometry
MC1902
Question No : 14 of 33
Marks: 1 (Budgeted Time Min)
The definite integral of a complex valued function $w(t) = u(t) + i\,v(t)$ over an interval $a \le t \le b$ is _____, where u, v are real valued functions of a real variable t.
Answer (Please select your correct option)
$\int_a^b w(t)dt = \int_a^b u(t)v(t)dt$
$\int_a^b w(t)dt = \int_a^b u(t)dt + i\int_a^b v(t)dt$
$\int_a^b w(t)dt = \int_a^b u(t)dt - i\int_a^b v(t)dt$
$\int_a^b w(t)dt = \int_a^b i\ u(t)v(t)dt$
14
Start Time: 11:48 AM


MTH632 Complex Analysis and Differential Geometry
If $\vec{x}(t)$ be a smooth space curve. Then which of the following is (are) known as the the Frenet formula (formulas)?
Answer (Please select your correct option)
(a) $\vec{t}' = k\vec{n}$
(b) $\vec{n}' = \tau\vec{b} - k\vec{t}$
(c) $\vec{b}' = -\tau\vec{n}$
(d) All (a), (b) and (c)
20
Start Time: 11:48 AM
77:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
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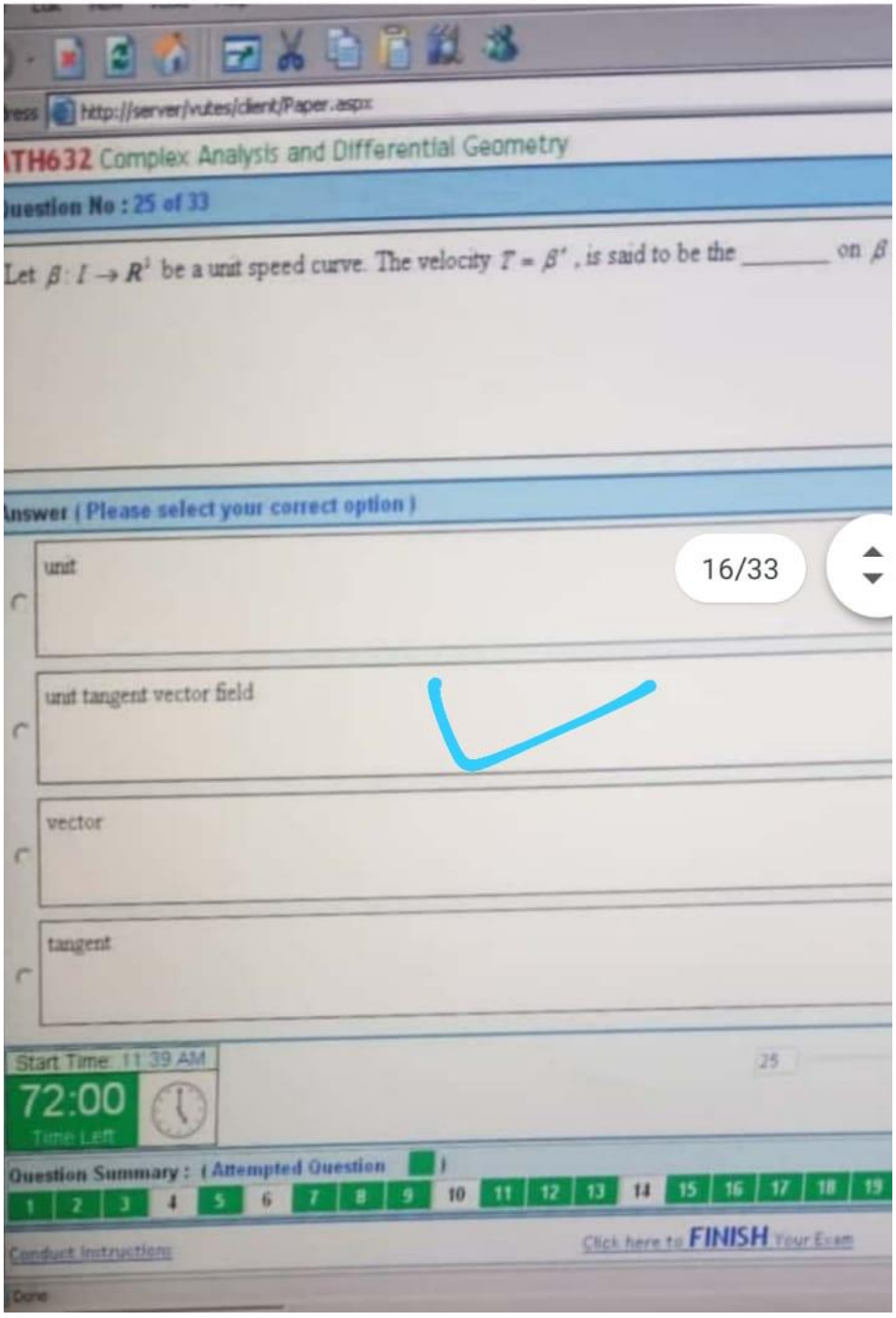


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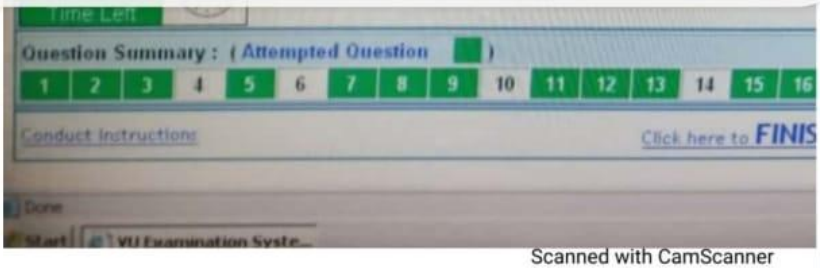



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MTH632 Complex Analysis and Differential Geometry
Question No : 13 of 33
The circle $z = e^{i\theta}$, $0 \le \theta \le 2\pi$ is centered at _____.
Answer (Please select your correct option)
(1,1)
(1,0)
(0,1)
(0,0)
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83:00
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Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12
Conduct Instructions

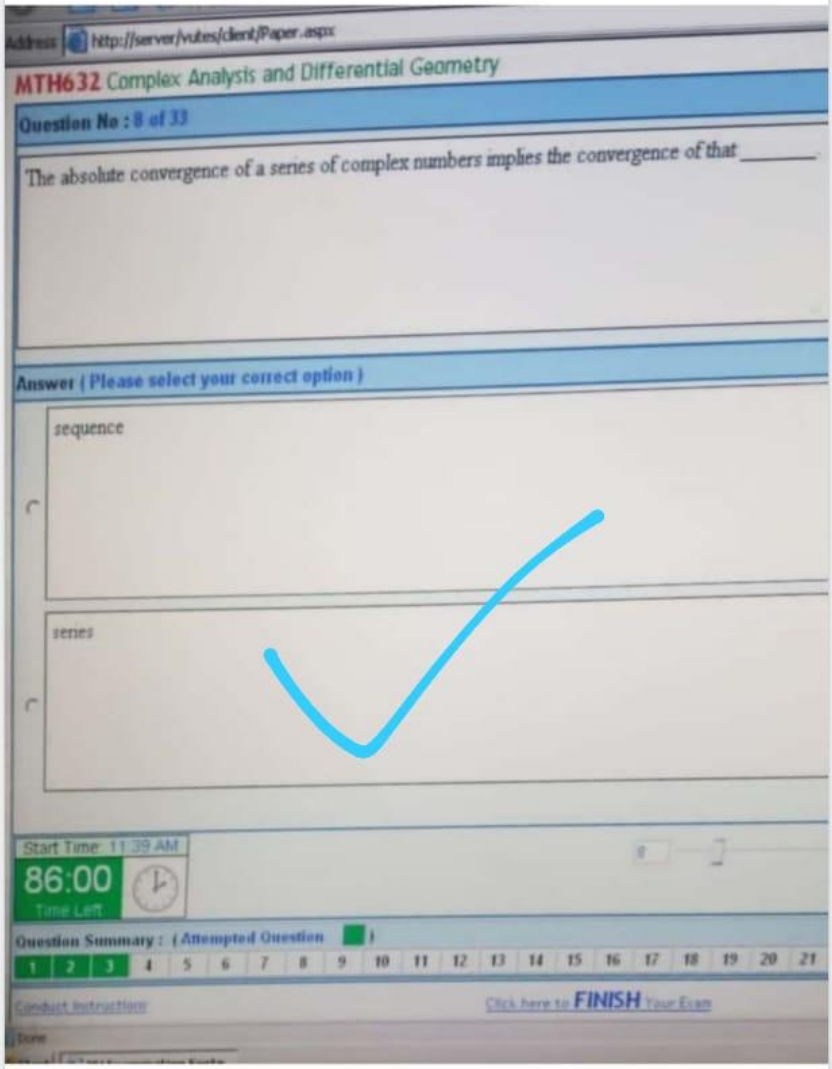

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VU Examination Syste...


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TH632 Complex Analysis and Differential Geometry
uestion No : 25 of 33
Let $\beta : I \to R^3$ be a unit speed curve. The velocity $T = \beta'$, is said to be the _____ on β
nswer (Please select your correct option)
unit
16/33
unit tangent vector field
vector
tangent
Start Time: 11:39 AM
72:00
Time Left
25
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19
Conduct Instructions
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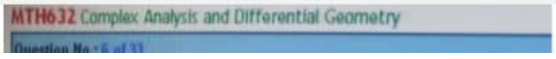
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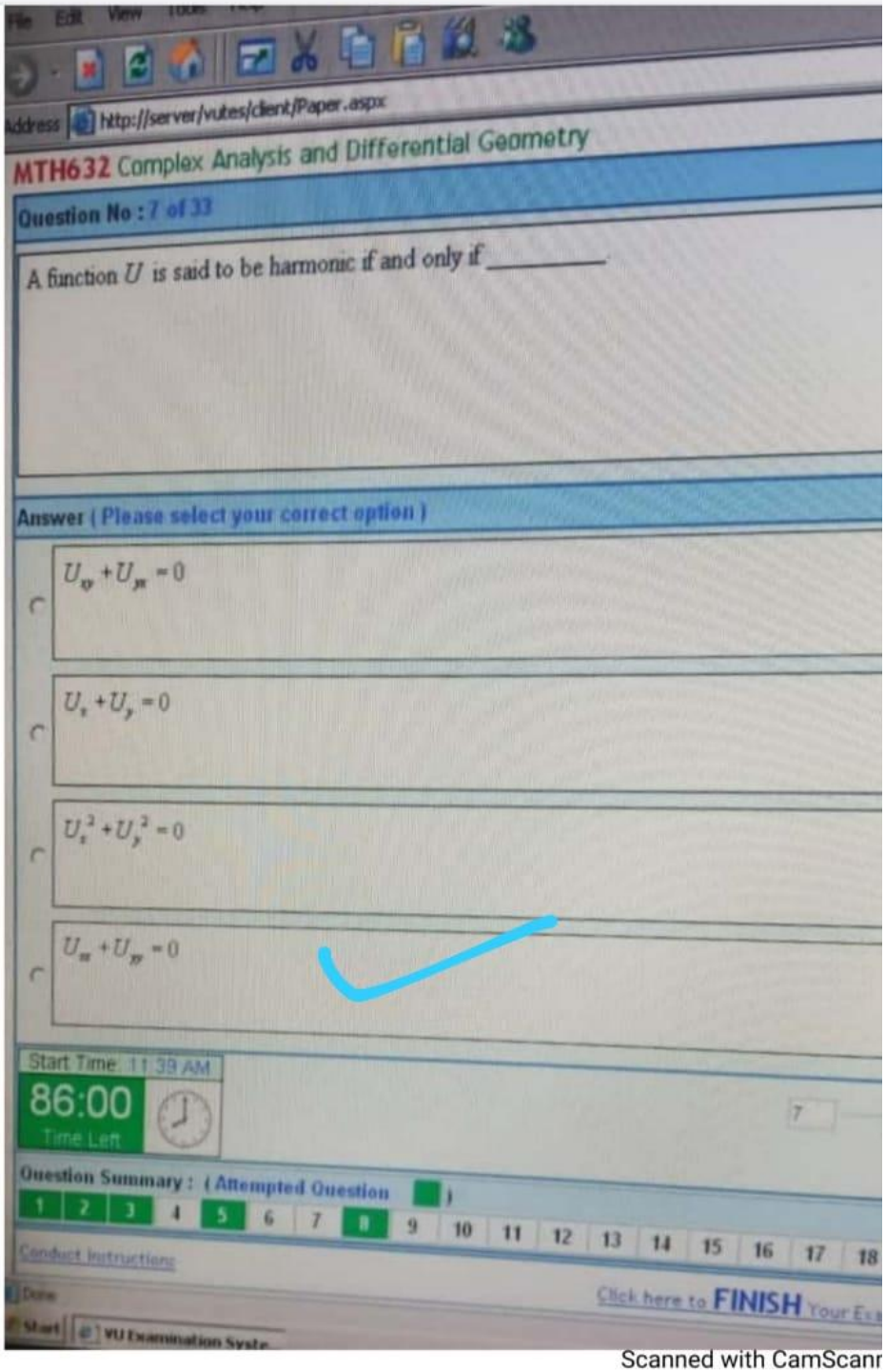
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MTH632 Complex Analysis and Differential Geometr

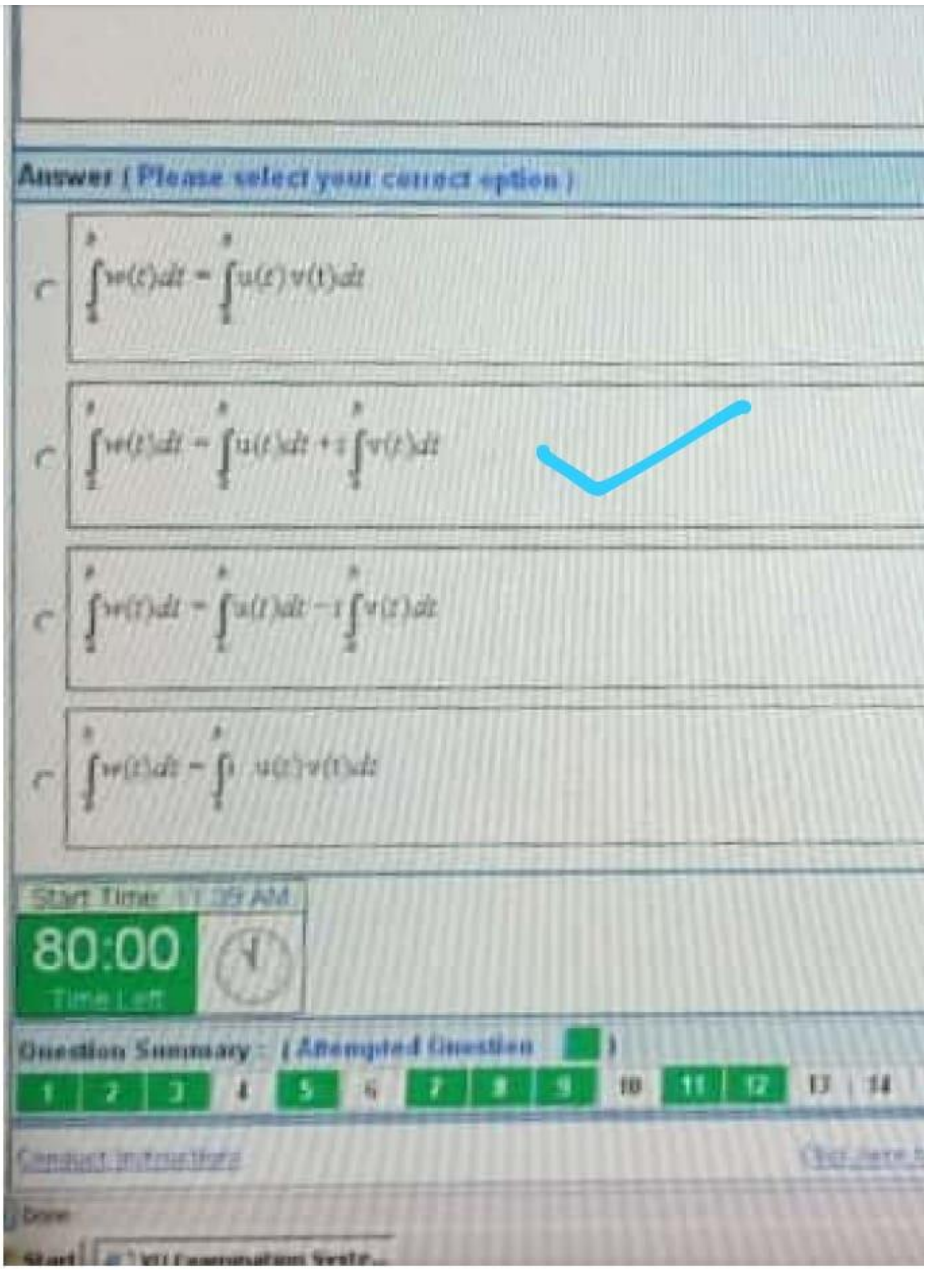
Question No : 11 of 33

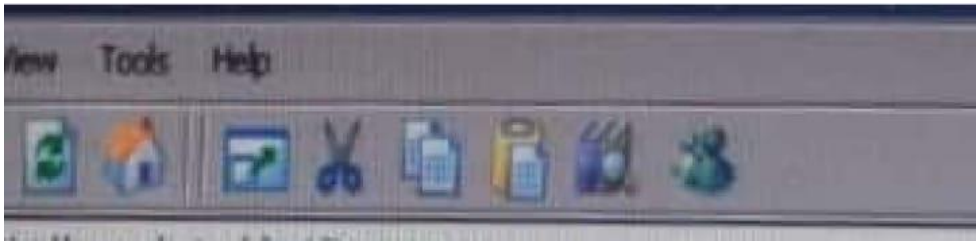
The principal value of 1^{2i} is _____.

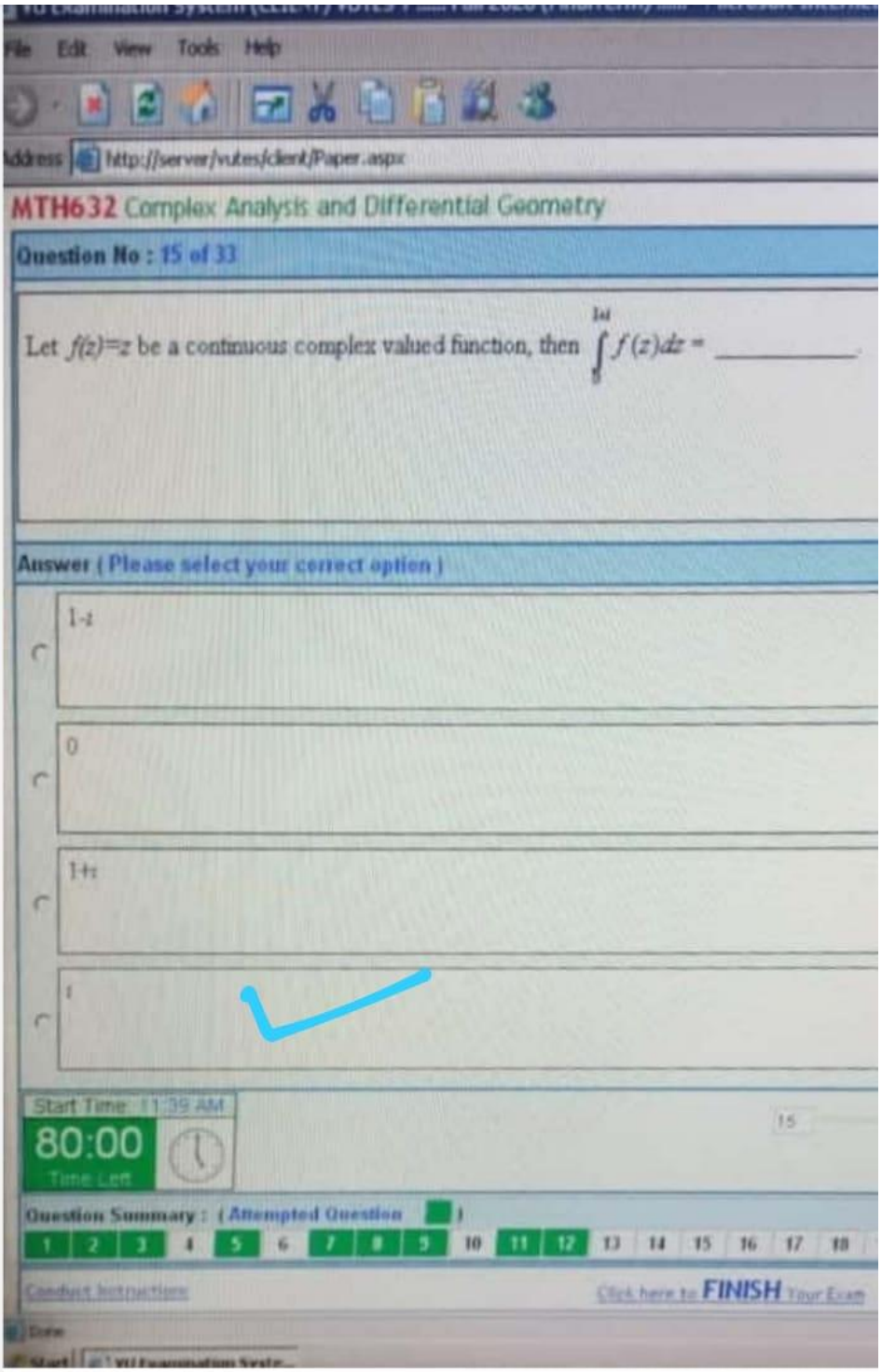
Answer (Please select your correct option)

- 0
- 1
- i
- 2i

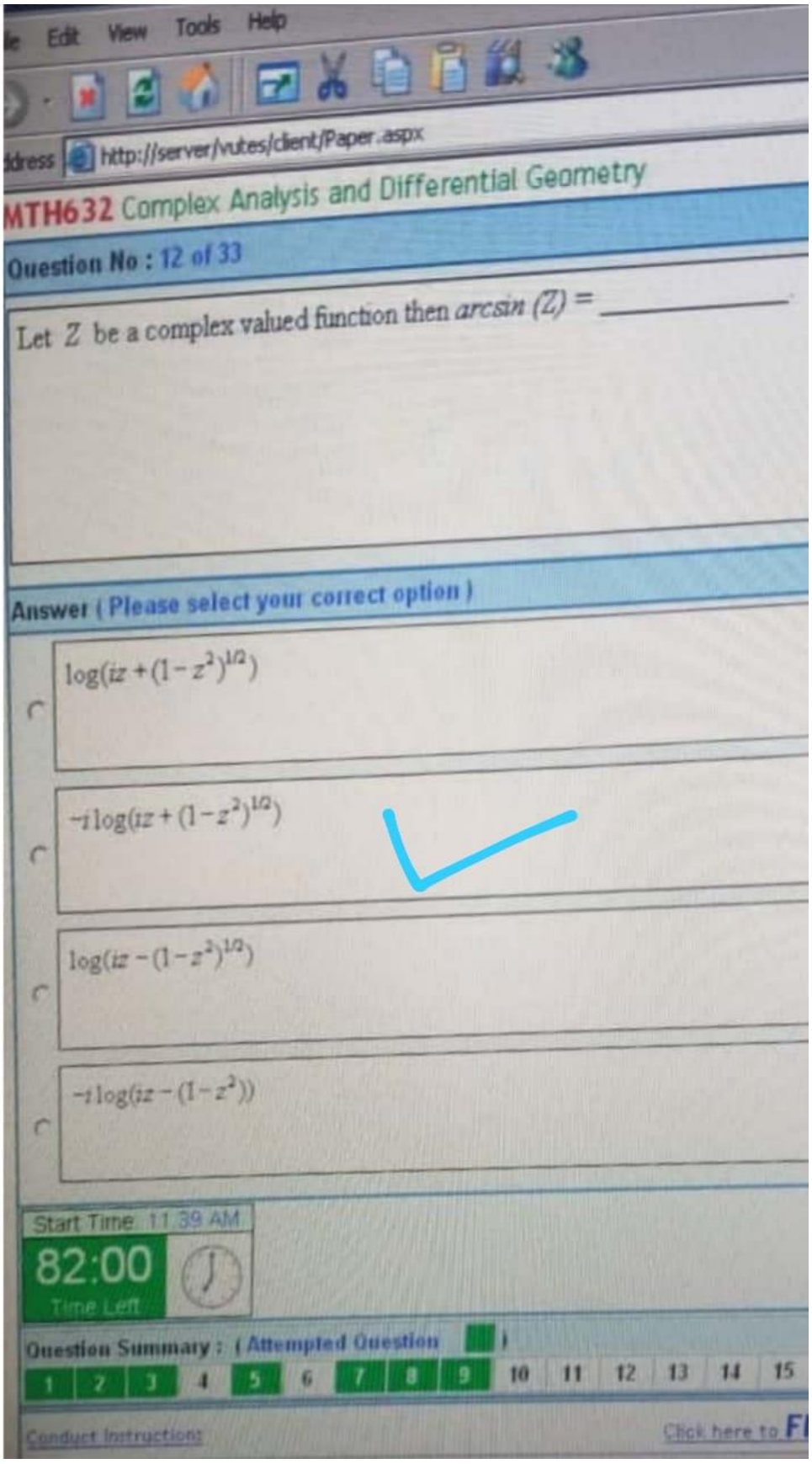
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Answer (Please select your correct option)
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80:00
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Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14


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MTH632 Complex Analysis and Differential Geometry
Question No : 15 of 33
Let f(z)=z be a continuous complex valued function, then $\int_0^{1+i} f(z)dz =$ _____.
Answer (Please select your correct option)
1-i
0
1+i
i
Start Time: 11:39 AM
80:00
Time Left
15
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18
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MTH632 Complex Analysis and Differential Geometry
Question No : 12 of 33
Let Z be a complex valued function then arcsin (Z) = _____
Answer (Please select your correct option)
log(iz + (1 − z²)^1/2)
−i log(iz + (1 − z²)^1/2)
log(iz − (1 − z²)^1/2)
−i log(iz − (1 − z²))
Start Time 11:39 AM
82:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
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